Eclipses by Appointment: 104 Uncatalogued Eclipsing Systems, Catalogue Corrections and a Measured Null Planet Search from Public TESS and Gaia DR3 Data

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The author's role and the use of AI systems are set out in Section 7, Contributions and AI use.

AAVSO keywords = Eclipsing Binaries; Catalogues, Databases, Surveys; Period Analysis, Period Change; Extrasolar Planets; Variable Stars (individual), Observing Targets: HIP 1060, HD 170829, V1192 Cas, TYC 4590-364-1

ADS keywords = binaries: eclipsing; binaries: spectroscopic; catalogs; planetary systems; stars: variables: general; techniques: photometric

ABSTRACT

We report 104 eclipsing systems in public TESS photometry whose eclipses are in no vetted variable-star or eclipsing-binary catalogue: 93 confirmed and 11 probable (the tiers are defined in Section 1). A full-text search of NASA ADS found a prior mention of eclipses for only one (A7); 72 carry unvetted SPOC or TESS-SPOC threshold-crossing events. Fifty-four come from a method we call eclipses by appointment: a Gaia DR3 spectroscopic orbit with a period of 20–60 d predicts when its conjunctions fall, and the TESS light curve is searched only inside those windows. Of 3,749 stars searched ($G < 11$), 70 candidates were re-analysed independently; 50 were confirmed and four left probable. Forty-six more came from transit searches of nearby M dwarfs and from period folds of Gaia orbits shorter than 20 d, 34 of them from 2,559 science targets among 3,022 orbits at G 9–10; four came from edge-on Gaia astrometric orbits. The eclipses contradict twelve Gaia DR3 orbit solutions in period or $e \cos \omega$, and in one system their timing is consistent with the light-travel time predicted by Gaia's astrometric orbit (chance coincidence ~10%). Six eccentric binaries are new heartbeat stars and a seventh is a candidate; we also report three probable blends and 11 catalogue corrections. For 17 of 31 deep systems among A1–A74 and B1–B11, Harvard plates from 1890–1990 show the eclipse at our ephemeris. A transit search of about 3,100 nearby M dwarfs found no new planet where about 1.2 detections were expected under the estimator's assumptions (probability of zero 0.30), because removing known planet candidates removed most stars where a detectable planet is known. We recovered both TIC 4206066 candidates and show TOI-5394.01 to be the occultation of an M-dwarf companion. Next-eclipse predictions are given for every confirmed and probable system.

1. Introduction

Gaia DR3 published about 180,000 spectroscopic orbits (Gaia Collaboration, Arenou et al. 2023), each giving a period, eccentricity, argument of periastron and time of periastron for a binary resolved only in velocity. TESS has, over the same years, photometered most of those stars in its full-frame images. Where the two overlap, an orbit seen within a degree or two of edge-on must eclipse, and TESS records the eclipse whenever a conjunction falls in an observed sector. For periods below about 20 d, a period search at Gaia's period finds such eclipses easily. Between 20 and 60 d it does not. A 27-day sector holds at most one eclipse of each kind, and multi-sector baselines of years can be phased only when the period

is already known to minutes or the single events are deep enough to be found one by one. Catalogued eclipsing binaries are correspondingly scarce in that period range, and the Gaia orbits that would gain most from an eclipse are the ones TESS searches least well. An eclipse fixes the inclination, and with it the companion mass. We present a method that closes this gap and report its results.

The searches reported here began from a planet question rather than a binary one. A public report of two transiting-planet candidates around the K dwarf TIC 4206066 (Rabtsevich 2026) led to a search of about 3,100 nearby M dwarfs in TESS full-frame-image light curves (Section 2.3). That search found no new planet (Section 4 gives the expected yield). It recovered both reported candidates blind, that is, without using the reported periods or epochs (Section 3.7, object C1). It also turned up four uncatalogued eclipsing systems (A1–A4), first seen as planet-like signals, three of them on stars neighbouring the searched dwarf. Two of the four, HIP 83929 (A3) and TIC 70665829 (A4), had Gaia DR3 evidence of a stellar companion that had not previously been combined with the TESS photometry. We therefore inverted the search: starting from Gaia's spectroscopic orbits, we tested each one for eclipses in TESS. For orbits shorter than 20 d this is a period search at Gaia's period, and it found eight more uncatalogued eclipsing systems (A5–A12) among about 2,200 orbits, and 34 more (A75–A109, less A93) among a further 2,559 science targets at G 9–10.

A Gaia orbit also fixes the conjunction times. The period, the time of periastron, the eccentricity and the argument of periastron together determine *when* each conjunction falls. For the median orbit in our sample that prediction is good to about 2 d at TESS epochs. Eclipses can only happen at conjunction, so the search can be restricted to those predicted windows and use a single-event statistic, instead of searching years of data for a period. The number of trials then falls from a four-year period grid to the windows, which hold about half of the usable data for the median star (Section 2.5). An eclipse that lands on the predicted conjunction is also attributed to the star with the Gaia orbit, and attribution is the hardest step in any blended-field TESS search. We call this *eclipses by appointment*. It produced 54 of the 104 systems in this paper, 50 confirmed and 4 probable (Section 3.1).

The conjunction prediction itself is not new. Tarrants & Mendes (2023) applied the same formula to Gaia DR3 SB1 elements to vet already-known TESS and K2 planet candidates, comparing the predicted conjunctions with the transit epochs, and called it their least reliable test. Yamaguchi et al. (2024) stated the principle for self-lensing pulses from white-dwarf companions and applied it to one 358-d Gaia orbit. Eclipses in Gaia orbits have been found by other routes. Callahan et al. (2025) crossed all Gaia SB1/SB2 orbits with TESS in a search for heartbeat stars and found eclipses in 18 of their 112 systems; they cut SB1 orbits at $P < 13$ d and used a periodogram rather than the ephemeris. El-Badry et al. (2026) found 25 eclipsing binaries among the Gaia orbits of compact-object candidates. In 14 of them the eclipse period is much shorter than the Gaia period: hierarchical triples of the same kind as our A54 and A73. Czavalinga et al. (2023) used Gaia outer periods to find tertiary eclipses in known eclipsing binaries. Beck et al. (2024) searched for eclipses in the TESS full-frame-image light curves of the six oscillating red giants whose Gaia astrometric inclinations are within 3° of edge-on. This was an earlier form of the edge-on astrometric search of Section 2.6. Their sample includes the seismic SB1 that is our A60, which they do not flag as eclipsing. We are not aware of any previous use of the conjunction prediction as the search itself, with the Gaia spectroscopic-orbit catalogue as the parent sample and TESS searched only inside the predicted conjunction windows, at 20–60 d where a period search is far less effective. That statement rests on a web and arXiv search; the ADS full text was searched for the objects (Section 2.7), not for the method. Tarrants & Mendes (2023) and Yamaguchi et al. (2024) used the prediction for vetting; using it as the search is the step this paper adds.

What is new here is therefore (i) the method of Section 2.5 and its yield, about 1.4 new eclipsing binaries per hundred Gaia orbits searched at G < 11; (ii) the 104 systems, with ephemerides and next-eclipse predictions (Appendix A); (iii) eleven catalogue corrections and twelve Gaia DR3 orbit solutions that the eclipses contradict; and (iv) a null planet result with a measured lower bound on the expected yield, and the reason it was null. Most of the brighter systems had been flagged by the SPOC or TESS-SPOC pipelines (Jenkins et al. 2016; Caldwell et al. 2020) as threshold-crossing events without being followed up. “Uncatalogued” here means absent from the vetted catalogues and the literature, not unseen by any pipeline (Section 2.7). The conventions, including the status tiers, are set out at the end of this section. The object labels (A1–A109, B1–B11, C1–C3) were assigned in this paper and are used as identifiers throughout.

Literature check. On 2026 October 12 we queried the NASA ADS full-text index for 248 identifiers of the 74 A-objects (the TIC number, the Gaia DR3 source identifier and every catalogue name) and read the title of every document returned. The query found one prior report of eclipses. Bryant et al. (2023) identified A7 as an eclipsing binary while vetting giant-planet candidates and excluded it from their list, without publishing an ephemeris or entering it in any catalogue. A7 is kept in the table with that note. For the other 69 systems, no document returned has a title indicating a variability or eclipse report. On October 13 the same query, extended to every reference SIMBAD lists for each star, covered 242 identifiers of the 35 systems from the G 9–10 fold (Section 2.4). It showed that one of them, BD–16 52, is the published eclipsing binary EBLM J0021–16, which is therefore listed among the other objects (Table 3) and not counted here. Of the other 34 (A75–A108), one has a prior literature record (four others, A81, A94, A100 and A105, carry only the unvetted table-2 flag of Kostov et al. 2025): the SuperWASP disposition catalogue (Schanche et al. 2019) lists A93 as an eclipsing binary, with a follow-up disposition of blended eclipsing binary and a note that one spectrum showed three line systems, but no period. SIMBAD carries that flag as the object type EB*, so A93 is kept as an entry, with the first ephemeris for it, but is counted as previously catalogued. For A109 the three references SIMBAD lists, all catalogue compilations, were read, with a web search on its names; none reports variability. A title screen cannot exclude a mention in the body or a table of an unrelated paper, and the full-text index does not cover every table in every paper. A prior report under a name we did not query could also be missed. The novelty claim is therefore conditional on this check and on the catalogues of Section 2.7. A separate scan of the SPOC and TESS-SPOC data-validation products (Section 2.7) found threshold-crossing events for 46 of those 74 objects (44 of the 70 counted), and the re-analyses of A75–A109 found them for 29 of those 35 (28 of the 34 counted), so 72 of the 104 counted systems carry one; Table A1 lists them all. A threshold-crossing event is an automated detection inside a MAST pipeline product, and none of these was vetted or promoted to a TESS Object of Interest. The catalogue terms used in the entries are defined under Conventions.

1.1 Conventions

Time. TESS times are given as BTJD = BJD_{TDB} − 2457000. Epochs are also given as BJD_{TDB}. Gaia DR3 times of periastron are relative to J2016.0, which is BTJD 389.0.

Eclipse names. We define conjunction A as the conjunction at true anomaly $\nu = 90° - \omega$ and conjunction B as that at $\nu = 270° - \omega$, with ω the argument of periastron of the primary in Gaia's solution. "Primary" eclipse means the deeper one unless an entry says otherwise. A *transit* is the smaller star crossing the larger, an *occultation* the smaller star hidden.

Units. ppt and ppm are parts per thousand and per million of the aperture flux. P is the orbital period in days, e the eccentricity. $R_\odot$, $M_\odot$ and $R_\oplus$ are solar radius, solar mass and Earth radius. Tmag is the TESS-band magnitude from the TESS Input Catalog (TIC 8.2), G the Gaia magnitude.

Labels and order. The A, B and C labels were assigned in this paper and follow the order of discovery; the Results sections are ordered by significance, so labels within a section are not consecutive. The next-eclipse table in Appendix A is in label order.

Status words. *Confirmed* means that the system passed every check of the protocol of Section 2.5 that its data allow: repeated events re-measured from raw flux, a period from the eclipses themselves, a difference image placing the deficit on the target against every neighbour bright enough to produce it, at least one independent pipeline, and a geometry consistent with Gaia's orbit. *Probable* means that the eclipses are real and repeat but one of those checks is missing, or failed in the independent re-analysis (one pipeline, no pixel test, a host not separated from a neighbour, a second event only in flagged data). The entry says which, and next-eclipse predictions are given all the same. *Plausible* is a single event, or two events with the period untested. *Open* and *refuted* keep their plain meanings. *By-catch* means a real variable on the target whose signal does not follow the Gaia orbit that brought the star into the sample. The tiers were assigned after the internal adversarial review (Appendix B), and every status word in an entry, in Table A1 and in `tables/objects.csv` follows them. The counts in this paper are 93 confirmed and 11 probable. The probable systems are A3, A5, A6, A7, A10, A19, A37, A40, A53, A68 and A74. A46 is a by-catch variable, kept as an entry but not counted. A93, a confirmed eclipsing binary that SIMBAD already types as one, is likewise kept but not counted. Labels are never reassigned when a status changes.

Catalogue status. *Uncatalogued* means that the object's eclipses were absent, at the time of the check (2026 October 11–12), from the AAVSO VSX, SIMBAD variability types, the Gaia DR3 eclipsing-binary solutions (`vari_eclipsing_binary` and the eclipsing solution types of the non-single-star tables), the TESS eclipsing-binary catalogue of Prša et al. (2022), IJspeert et al. (2024), the vetted tables of Kostov et al. (2025) and the ASAS-SN variable-star catalogue. Individual entries list further catalogues where they were checked; the checks are described in Section 2.7. “Kostov table 2” means that the object appears in the ~900,000-row machine-learning flag list of Kostov et al. (2025). That is an unvetted automated flag, so we treat such objects as uncatalogued; Table A1 records the flag. “Known SB” means that the star was already in SIMBAD as a spectroscopic binary; the eclipses are what is new. Depths are as measured in the named pipeline's flux and are not intrinsic: they are diluted by any light from neighbours that the pipeline does not correct for (Section 2.8).

2. Data and methods

2.1 TESS photometry

All photometry is from the Transiting Exoplanet Survey Satellite (TESS; Ricker et al. 2015), retrieved from MAST. The primary source was the MIT Quick-Look Pipeline (QLP; Huang et al. 2020; Kunimoto et al. 2021) full-frame-image light curves, at 30-min (sectors 1–26), 10-min (27–55) and 200-s (56 onward) cadence. For the long-period work we used the raw aperture flux `SAP_FLUX`, normalised per sector, rather than QLP's detrended `KSPSAP_FLUX`/`DET_FLUX`, because in post-sector-56 QLP files the detrending suppresses long eclipses (Section 2.8). Wherever independent reductions of the same pixels existed, we used them as cross-checks. These were SPOC 2-min light curves (Jenkins et al. 2016), TESS-SPOC full-frame-image light curves (Caldwell et al. 2020), TGLC (Han & Brandt 2023), eleanor (Feinstein et al. 2019) and GSFC-ELEANOR-LITE (Powell et al. 2022), TASOC (Handberg et al. 2021), CDIPS (Bouma et al. 2019), and the TARS (Boyle et al. 2026), T16 (Hartman et al. 2025) and TEQUILA (Ogunwale et al. 2025) high-level science products in MAST. Where an entry says that *n* pipelines agree, these are the pipelines meant. Target-pixel files and TESScut full-frame-image cutouts were used for difference images and centroids.

2.2 Gaia DR3 orbits

Spectroscopic and astrometric orbits are from the Gaia DR3 (Gaia Collaboration, Vallenari et al. 2023) non-single-star table `gaiadr3.nss_two_body_orbit` (Gaia Collaboration, Arenou et al. 2023; Halbwachs et al. 2023; Gosset et al. 2025), joined to `gaia_source` for magnitudes, RUWE and the radial-velocity amplitude statistic `rv_amplitude_robust`. The solution types used were SB1, SB2, SB1C, SB2C and AstroSpectroSB1 for the spectroscopic work and the pure astrometric `Orbital` type for Section 2.6. Stars were matched to the TESS Input Catalog (TIC 8.2; Stassun et al. 2019; Paegert et al. 2021) by Gaia source identifier, through the MAST API or the VizieR copy `IV/39/tic82`.

Gaia DR3 gives the time of periastron T_p relative to J2016.0 = BTJD 389.0. The conjunction times follow from Kepler's equation. Conjunction A is at true anomaly $\nu = 90° - \omega$ and B at $\nu = 270° - \omega$. From ν follow the eccentric anomaly E, the mean anomaly $M = E - e \sin E$, and

$$t_c = T_p + 389.0 + M\,P / 2\pi \ \ (\mathrm{BTJD}),\ \ \text{plus integer multiples of } P.$$

Uncertainties were propagated by Monte Carlo through Gaia's errors on P, T_p, e and ω, with 4,000 draws in the appointments search and 20,000 in the fold of Section 2.4. The appointments search drew the four parameters independently, which is adequate for setting window widths. The conjunction test of the fold and of the independent re-analyses propagates Gaia's full covariance matrix, rebuilt from the published correlation vector (`corr_vec`). The draws were re-centred on their median modulo P, so that a conjunction close to periastron does not wrap a whole cycle. For circular solutions (SB1C, SB2C) $\omega = 0$ was assumed and flagged.

2.3 Transit searches of nearby M dwarfs

Sample. The parent sample was drawn from TIC 8.2 through the MAST API: dwarfs with T_{eff} 3200–4300 K, Tmag 8.5–11 and distance < 40 pc, 3,490 stars. Every star with a TESS Object of Interest (TOI) or community TOI (CTOI) in the ExoFOP lists fetched on 2026 October 11 was removed (3,412 left), and every star whose TIC contamination ratio was 0.05 or more (2,057 left). Of those, 1,299 had 3–8 QLP sectors. Other searches extended this sample to crowded fields, fainter and smaller stars, and the known planet hosts the first cut had removed. Table 1 (Section 4) lists every search with its completeness.

Preparation. Per sector: cadences with QUALITY = 0, 0.5 d trimmed after every gap, 10-min bins, a 1-d running-median flatten, and clipping. The low clip was at 8σ, which removes transits deeper than about 8σ per point.

Searches. The first pass ran a box least-squares (BLS; Kovács, Zucker & Mazeh 2002) search per sector on a shared log-period grid (0.5–15 d, $\Delta \ln P = 5 \times 10^{-4}$, boxes 0.72–3 h), summed the log-likelihood across sectors (incoherently, so phase could float between years), and refined the top five peaks with a coherent BLS. The second pass ran one coherent BLS over all sectors on the full grid (876,000 periods over a four-year baseline) and measured its completeness with 282 injected box transits. Further passes searched:

- periods of 0.17–0.5 d (about 690,000 periods, boxes 17–77 min), with the grid spacing set by a timing-slip budget;
- fainter late-M dwarfs (Tmag 11–13.5);
- the smallest stars at fixed brightness (radius 0.10–0.25 $R_\odot$, Tmag < 11);
- single transits at periods of 24 d and beyond;
- the 1,009 crowded-field stars with contamination ratio ≥ 0.05;
- 78 known planet hosts, with their known signals masked.

Vetting. Each peak was ranked by how far its coherent signal-to-noise ratio (SNR) stood above the star's other peaks, and flagged for odd/even depth differences, a dip at phase 0.5, sector-to-sector depth changes, a duration too long for the star's density, and boxes at the longest trial duration. Survivors then went through three further tests. The first was a cyclic-roll null: each sector's flux was rolled in time, which keeps red noise and destroys any transit, and then searched identically (36–100 trials). The second was per-pixel deficit maps and difference-image centroids in at least two field orientations. The third was archive queries: Gaia RUWE and `rv_amplitude_robust`, VSX and the eclipsing-binary tables. The order of these tests matters, as A3 showed (Section 3.4). A candidate that passes every photometric test has only passed the photometry, whereas Gaia RUWE and a VSX cone search are 30-second queries that can reject it at once. Every candidate from these searches that reached the pixel stage was an eclipsing binary, a blend, a rotator, a pulsator or a flare.

2.4 Period folds of Gaia orbits shorter than 20 days

Nearby sample (parallax > 10 mas). The sample was every Gaia DR3 SB1, SB2, SB1C, SB2C or AstroSpectroSB1 orbit with P < 20 d and parallax > 10 mas, 824 orbits in all. Of these, 14 were already in Gaia's `vari_eclipsing_binary` table and became positive controls, leaving 810, of which 743 matched TIC and 737 had QLP light curves (4,107 files). The QLP flux was flattened with a 2-d running median and clipped upward only. A coherent BLS over all sectors then searched a window of $\pm\max(5\sigma_P, 0.1\%)$ around P_{Gaia} and around $P_{Gaia}/2$ (for twin eclipses), with boxes of 0.7–6.3 h. The score was a z value against 300 random periods on the same star. All 14 controls were recovered, 13 at $z \geq 6$ and one at 4.6 (very deep eclipses inflate a star's own null). Of 736 searched, 63 reached $z \geq 5$. Triage removed VSX matches within 30″. It then separated eclipses from rotation and ellipsoidal variation by removing the first two Fourier harmonics and searching again. The survivors' TESS eclipse epochs were compared with Gaia's predicted conjunctions (Section 2.2), and every VizieR table with a row within 5″ was listed.

Distant sample (parallax ≤ 10 mas or missing). The same method and code, run on the complement in brightness order. The first subsample, G < 9, held 1,523 orbits; 1,445 were searched (1,338 science targets and 107 Gaia-eclipsing controls). Of the 107 controls, 95 reached $z \geq 5$. Ten of the twelve misses were deep eclipsers whose own eclipses inflate the null, so triage added a supplementary acceptance criterion: z < 5 with depth > 5,000 ppm and SNR more than three times the null median. Triage took 279 stars, of which 168 were eclipse-like and 115 were in no vetted catalogue. All 115 folds were inspected visually, and 11 showed clean, sharp eclipses. These were re-checked independently with separately written code. The re-check refolded the light curves, measured per-sector and odd/even depths, recomputed the conjunctions independently and repeated the catalogue cone searches with Algol as a positive control. It also re-measured A8 and the three shallowest claims in raw SAP flux with per-event local baselines, fetched TESS-SPOC photometry where it existed, and ran a centroid test on the one object with a conjunction tension. Seven of the 11 survived (A8–A12, B6, B7); four were demoted (Section 3.8). The attribution rule in both samples is the same: an eccentric system's eclipse must land on Gaia's predicted conjunction.

Distant sample, G 9–10. The same search was then run on the next subsample, the 3,022 orbits with $9 \leq G < 10$. Of these, 2,777 had QLP light curves and were searched: 2,559 science targets and 218 controls already in Gaia's eclipsing-binary table, of which 199 (91%) reached $z \geq 5$ (208 with the supplementary criterion); this recovery of known eclipsers measures the search's completeness. Triage took 562 science targets. The 238 with any VSX entry within 30″ were set aside. Of the other 324, the 257 that were eclipse-like after the harmonic-removal test went to the catalogue cone searches, which found 61 of them in a vetted eclipsing-binary table. The folds of the other 196 were inspected visually: 38 showed eclipse-shaped dips and nine more showed marginal ones. Per-event depths in the raw SAP flux removed one, which showed no discrete event in any sector. Each of the other 37 was re-analysed

independently to the full protocol of Section 2.5: per-event timing fits in the raw SAP flux, difference images, a second pipeline, and the conjunction test through Gaia's full covariance (Section 2.2) at Gaia's own epoch, to which the eclipse ephemeris is carried with the TESS period. Thirty-six are confirmed eclipsing systems. One is the published eclipsing binary EBLM J0021–16 (Table 3); the other 35 are A75–A109 (Section 3.3). Of these, A93 was already typed as an eclipsing binary in SIMBAD, from a SuperWASP flag without a period, so 34 are counted as new. The remaining star, BD+30 9 (TIC 22503559), is an ellipsoidal variable with no eclipse: its two minima fall on Gaia's conjunctions (Table 3). Two groups were not pursued: about 175 eclipse-like stars with a VSX entry within 30″ (about 100 of them with a non-eclipsing type for the nearest entry), and the nine marginal dips. The yield of this subsample is therefore a lower bound, and the type corrections it may hold were not searched for. The score z ranks candidates and is biased upward, because the search keeps the best of a window of trial periods around Gaia's. The controls measure completeness; purity comes from the inspection and the re-analyses.

2.5 Eclipses by appointment: Gaia orbits with P = 20–60 days

Sample. `gaiadr3.nss_two_body_orbit` joined to `gaia_source`, solution types SB1, SB2, SB1C, SB2C and AstroSpectroSB1, 20 ≤ P < 60 d and G < 12: 11,446 orbits (10,713 SB1, 593 SB2, 120 AstroSpectroSB1, 19 SB2C, 1 SB1C), 46 of them in Gaia's eclipsing-binary table. The quartiles of the period uncertainty are 0.010, 0.021 and 0.044 d. Two subsamples were searched. The G < 10 subsample: 1,528 orbits, 1,444 matched to TIC, 1,422 with QLP light curves (8,716 files), 40 Gaia-eclipsing controls. The G 10–11 subsample brought the number of stars searched, both subsamples together, to 3,749.

Windows. For each orbit the conjunction times and their uncertainty σ_T were computed at every TESS cycle as in Section 2.2. The uncertainty grows with the number of cycles n since 2016 roughly as

$$\sigma_T(n) = [\, (n\,\sigma_P)^2 + {\sigma_{Tp}}^2 + (\text{terms in } \sigma_e, \sigma_\omega)\,]^{1/2},$$

which for the median orbit in the sample is about 2 d by 2021, and under 3 d for two-thirds of the orbits. Each predicted window had a half-width of $\max(3\sigma_T, 0.25\ \text{d})$, capped at 7 d.

Statistic. Each sector was prepared the same way: QLP `SAP_FLUX`, QUALITY = 0, 15-min bins. The noise was taken from the point-to-point scatter, which is insensitive to the star's slow variability. It is also little affected by the eclipses themselves, which occupy a small fraction of the cadences. In each window, a single-event box with durations of 1–14 h was compared with the mean of two symmetric side windows 0.75–2.5 durations out, which cancels linear trends. Two shape tests were added. The *far ratio* is the depth against a baseline 3–6 durations out, divided by the near depth; it is about 1 for an eclipse and not 1 for a ramp or a pair of flares. The second test requires both halves of the box to be in deficit. The null was the same maximum-SNR-in-window statistic measured in 40 random windows of the same width on the same star, outside every predicted window, giving z = (SNR − median)/MAD.

This z is a rank statistic against the star's own null, not a Gaussian significance. With 40 null windows, the smallest empirical false-alarm rate that can be expressed per star is about 1 in 40. The z ≥ 8 cut of the triage is therefore a ranking threshold. Its false-alarm rate was controlled by the later stages (the shape tests, inspection and the independent re-analysis), not by z alone.

The windows are wide. Over the 3,708 science stars of the search (the 40 eclipsing controls excluded; `appointments/coverage/` in the deposit), they hold a median 51% of the usable cadences (interquartile range 29–72%). They hold more than 80% for 18% of the stars, and every cadence for 165 stars (4.5%), where no null exists. For a further 591 stars (16%), z is undefined because all 40 null windows land on the same feature (MAD = 0). Seven of the 54 reported systems never faced the z test for these reasons; the triage code records how each was admitted (`appointments/triage.py`). The null and z cuts apply only

where a null could be built. A star without one passes on the remaining pre-set cuts alone: in-window SNR ≥ 8, a depth ratio against the far baseline within 0.6–1.6, the deficit present in both halves of the box, and the event within 3σ of Gaia's conjunction. A37, A40 and A42 were admitted that way, with no null at all. A27, A54 and A64 (40 nulls each) and A51 (five nulls) had MAD = 0, and still passed the cut that no random window reach the in-window SNR.

One reported system failed a triage cut. A18's far-baseline ratio is 2.19, because its 3-h search box sat inside a 17-h total occultation and the far baseline fell within the eclipse. It was taken from the search's raw SNR ranking (SNR 120, z = 79) before the triage cuts were written, and rests on six eclipses in three sectors and on the independent re-analysis. It should not be counted toward the automated selection's performance. Apart from A18, inspection decided among triage passers and did not admit stars the cuts had rejected.

As the coverage figures above show, the windows remove about half of the data rather than most of it. The method's gain is as much in attribution, an event at the predicted phase of an orbit already known, as in the number of trials. On a Tmag 7 star the nominal SNR of a 14-h box reaches 300 from scattered light alone, so the per-star null and the far ratio are the statistic, not the SNR.

Cuts. Candidates were retained on the following criteria:

1. SNR ≥ 8;
2. far ratio in [0.6, 1.6];
3. both halves in deficit;
4. event within $3\sigma_T$ (or 0.03 in phase) of a conjunction;
5. no null window reaching it;
6. z ≥ 8 where a null exists;
7. the 14-h box accepted only with a far ratio near 1.

A per-sector common-mode light curve (the median of all stars in the sector) was built, and a box was flagged when the common-mode deficit exceeded 300 ppm and 30% of the box depth. In the G < 10 subsample the common-mode flag removed 34 of 181 candidates and none of the eclipsers later confirmed.

Controls. Of the 40 Gaia-flagged eclipsers in the G < 10 subsample, 32 show a 5–30% eclipse at the predicted conjunction, with $|\Delta\varphi|$ mostly < 0.03 and within $3\sigma_T$. Two further known long-period eclipsers were also recovered blind: HD 181068, the triply eclipsing triple found by *Kepler* (P 45.52 d), and HD 196022 (IJspeert et al. 2024). Twenty catalogued eclipsing binaries were recovered blind in each subsample.

Triage and classification. Candidates passing the cuts were matched against VSX, ASAS-SN, Prša et al. (2022), IJspeert et al. (2024), Kostov et al. (2025, tables 2–4) and SIMBAD. Stars with a VSX entry of a non-eclipsing type (rotation, RS CVn, MISC, VAR) were kept, because an eclipse at conjunction is still uncatalogued and makes a type revision. Every remaining candidate was plotted (the light curve in each predicted window) and classified visually. Five classifiers inspected the 161 stars of the G < 10 subsample that passed the cuts and were not known eclipsers (134 in no catalogue, 27 with a VSX entry of another type). Seven inspected the 280 G 10–11 candidates that no eclipsing-binary catalogue lists. Most false positives were orbit-edge scattered-light ramps, flux level steps across data gaps (especially around the sector 99 gaps), flux drifts before reaction-wheel momentum dumps, and a sector-wide common mode. The common mode appears as pre-gap dips of 600–4,400 ppm on fixed dates, seen in the median of 35–124 unrelated stars. In the G 10–11 subsample the yield fell steeply down the triage ranking: 44 stars classified as eclipsers among the first 168 candidates, 2 among the next 104, and none among 8 neighbour-contaminated ones.

Independent re-analysis. Every star classified as an eclipser, and every unusual non-eclipse signal, was re-analysed independently by a separate agent instance with separately written code, briefed to refute the claim (see Contributions). There is one report per candidate, kept with the paper's sources (Appendix B), and every report has the same sections: archives; eclipse measurements; ephemerides; independence checks; physics; verdict and next eclipses. Each re-analysis:

1. re-measured every eclipse from raw SAP flux, including cadences with QUALITY ≠ 0 where an eclipse fell in flagged data, and looked for eclipses not used by the search;
2. fitted a linear ephemeris to all events of both kinds and reported the observed-minus-calculated (O−C) residuals;
3. computed difference images (in-eclipse minus out-of-eclipse) and located the flux deficit relative to the target and to every Gaia neighbour bright enough to produce the dip;
4. re-measured the eclipses in at least one independent pipeline (SPOC, TESS-SPOC, TGLC, eleanor, TASOC, TARS, T16, TEQUILA or CDIPS, as available) and tested for a common-mode or background (`SAP_BKG`) origin;
5. fitted the eclipse geometry (a two-sphere or limb-darkened model where the events allowed) and compared the eccentricity and argument of periastron from the eclipse spacing and durations with Gaia's orbit, and the depths and durations with what Gaia's mass function allows;
6. predicted the next eclipses, and where a sector in the archive held a predicted event the search had not used, looked for it blind.

The search trimmed 0.5 d after every data gap. QLP's 200-s sectors have many flag-induced gaps, so this trim removed whole eclipses; it was reduced to 0.1 d and the G < 10 subsample was searched again. Of the 18 G < 10 candidates that passed visual classification, two (HD 201073, HD 7423) were dismissed on inspection before re-analysis. The other 16 were re-analysed, and one was refuted. Of the 55 G 10–11 candidates, one (TYC 5976-120-1, already in VSX as an eclipser) was not re-analysed. The other 54 were, and three were refuted. Section 3.1 lists them.

2.6 Edge-on Gaia astrometric orbits

We selected the 6,652 Gaia DR3 pure astrometric orbits (solution type `Orbital`) with P < 300 d and G < 13, recovered each orbit's inclination from its Thiele–Innes coefficients A, B, F and G, and kept the 191 within 2° of edge-on. The inclination recovered this way is only a ranking statistic, because the transformation has a documented inclination bias. Of the 191, 182 matched TIC and 177 had 1,339 QLP light curves. The search used raw `SAP_FLUX` with a five-day local median and upward-only clipping, in two period windows, one around Gaia's period and one around half of it, each with a width set by Gaia's period uncertainty. Candidates had to show repeated events, period agreement with Gaia, recovery in an independent pipeline where one existed, and attribution to the target in difference images. Both survivors were re-analysed independently; one (TIC 316045876) is withdrawn (Section 3.2). The ω convention was first tested on 400 AstroSpectroSB1 orbits. For these, the astrometric ω from A, B, F, G agrees with the spectroscopic ω from C, H to a median 4°; the opposite sign gives 45°. Conjunction times were propagated by Monte Carlo through Gaia's full covariance matrix.

A complementary search of the same 177 light curves had two parts. The first was a per-sector box search over 0.3–13 d on raw QLP SAP, requiring two sectors to agree on a period; it recovered the 5.854-d period of TIC 316045876 and found A73. The second was an inspection of the deepest single events. It revealed eclipses at Gaia's own period that the window search had scored below the detection threshold (A72, A74 and the plausible case of Table 3). Each was then re-analysed independently with the protocol of Section 2.5.

2.7 Catalogue checks

The catalogues that define “uncatalogued” (Conventions) were checked as follows. Each object was cone-searched (typically 30″, wider for bright stars) in VSX (Watson et al. 2006), SIMBAD (Wenger et al. 2000), the Gaia DR3 `vari_eclipsing_binary` (Mowlavi et al. 2023), `vari_classifier_result` (Rimoldini et al. 2023; Eyer et al. 2023) and NSS tables, Prša et al. (2022; VizieR J/ApJS/258/16), IJspeert et al. (2024), Kostov et al. (2025; J/ApJS/279/50, tables 2–4) and the ASAS-SN variable-star catalogue (Shappee et al. 2014; Kochanek et al. 2017; Jayasinghe et al. 2018; VizieR II/366). Catalogues without a service of their own were queried through VizieR (Ochsenbein et al. 2000), and each catalogue was tested with a known eclipsing binary (usually Algol) as a positive control. Many entries list further catalogues where they were checked: the GCVS (Samus' et al. 2017), ASAS-3 (Pojmański 2002), SB9 (Pourbaix et al. 2004), the WDS (Mason et al. 2001), the CCDM (Dommanget & Nys 2002), Hipparcos (ESA 1997), ATLAS (Heinze et al. 2018), the WISE variables of Chen et al. (2018), the Gaia DR2 variables (Holl et al. 2018), RAVE (Steinmetz et al. 2020), OGLE-IV, ExoFOP TOIs and CTOIs, and a web search on the star's names. The NASA ADS full-text index was queried on 2026 October 12 for 248 identifiers of the 74 A-objects (TIC number, Gaia DR3 source identifier and every catalogue name), and every hit was read by title. The one prior report it found is A7 (Section 1). A second query on October 13 covered 242 identifiers of the 35 systems from the G 9–10 fold and the titles of every SIMBAD reference for them (Section 1); it skipped survey designations (2MASS, WISE, UCAC, GSC, PPM, SAO and Gaia DR2). A star discussed in a paper only under a name we did not query, or only in a table the index does not reach, would still be missed.

Live check. After the manuscript was complete, on 2026 October 13, a separate session with its own code cone-searched all 74 A-objects live in 14 VizieR catalogue tables and SIMBAD (1,414 VizieR cones at 20″ and 101 SIMBAD cones at 10″, with no query errors): VSX, ASAS-SN, the Gaia DR3 classifier and eclipsing-binary tables, Prša et al. (2022), IJspeert et al. (2024) and SIMBAD, which define “uncatalogued” here, and six catalogues this paper had not claimed to check for every object: the GCVS, the NSV (Samus’ et al. 2017), ASAS-3 in V and in I, ATLAS, the ZTF periodic and suspected variables (Chen et al. 2020) and the Catalina periodic variables (Drake et al. 2014). The B-objects served as a positive control: the paper says they are catalogued, and they came back catalogued (VSX 9 of 12; SIMBAD has 11 of the 12, six of them with a variability type). No A-object has an eclipsing-type entry for itself in any of the 15. The one eclipsing-type row within 20″ of any A-object belongs to the star 11.3″ from A20, which A20's entry names and excludes. The six added catalogues hold rows for five A-objects, all already stated in their entries (A31's GCVS type, the VSX MISC entries of A25 and A52, A4's ATLAS “dubious” flag, two ZTF suspected variables 13–18″ from A48), and none is an eclipsing type. The same check found the VSX rotational entry on A3 that its catalogue-status line now records (Section 3.4). The same code was then run on A75–A108, again with the B-objects as positive control (644 VizieR cones and 46 SIMBAD cones, no query errors; the controls came back catalogued as before). SIMBAD types A93 as an eclipsing binary, from a SuperWASP flag (Section 1). None of the other 33 has an eclipsing-type entry in any of the 15; the only other row is a ZTF suspected variable, untyped, 15″ from A98. A109, added after that run, went through the same code (14 VizieR cones, none with a row within 20″; SIMBAD types it only as a spectroscopic binary), and its re-analysis searched to 60″ in VSX, ASAS-SN, the Gaia DR3 variability tables, the TESS eclipsing-binary catalogues, SB9 and the WDS: the only entry is a Gaia δ Scuti-type classification of a G 13.7 neighbour 40″ away.

Pipeline detections. Subsequently, the SPOC and TESS-SPOC data-validation products in MAST (Jenkins et al. 2016; Caldwell et al. 2020) were scanned for every object in this paper. Sixty-six of the 101 objects scanned have at least one threshold-crossing event (TCE), among them 46 of the 74 A-objects (44 of the 70 counted). The re-analyses of A75–A109 searched the same products and found TCEs for 29 of those 35 (28 of the 34 counted), none promoted to a TOI; in all, 72 of the 104 counted

systems carry a TCE. In many the TCE period is the eclipse period or an alias of it (A6, A7, A11, A12, A16, A49, A61, A73 and others); in others it is unrelated or spurious. A TCE is an automated event inside a pipeline product, never vetted and never promoted to a TESS Object of Interest. It is not a catalogue entry. None of these systems reached VSX, the Gaia eclipsing-binary table, the vetted TESS eclipsing-binary catalogues or the literature, so they remain uncatalogued in the sense defined under Conventions. Table A1 lists the TCE for each system; the entries that had already noted one keep the note.

2.8 Known systematics

The following families of false signal were identified:

- BLS boxes at the longest trial duration: spot rotation not removed by the flattening.
- Peaks at 12–14.8 d, near the 13.7-d TESS orbit. Spurious peaks probably begin at about 10.4 d, but excluding that range would also exclude real planets (Section 4.3).
- Events anchored at BTJD 1355–1395 (early-sector ramps) and at BTJD ≈ 1413.3 (sector 4), each identified by coincident events on two unrelated stars.
- Gap-edge ramp points, which produced a 3,300-ppm "transit" on TIC 86232609 and a spurious 1% dip on TIC 4206066 at BTJD 4003.1.
- QLP's detrended flux (`DET_FLUX`/`KSPSAP_FLUX`), which suppresses long eclipses in post-sector-56 files: it removed the 20-h primary of A16 entirely, halved one eclipse of A17, and reduced A8's 18-ppt eclipse to 0.8 ppt in some sectors.
- The sector-wide common mode of pre-gap dips (Section 2.5), flux level steps across the gaps of sectors 99 and 101–104, and dips coinciding with peaks in the background flux `SAP_BKG`.
- Orbit-edge scattered light, which repeats in orbits with no conjunction; checking the neighbouring orbit reveals it.
- At $P < 0.5$ d, contact and ellipsoidal binaries in the aperture, and pulsators. A dip at phase 0.5 means the true signal is faster than the period found.
- QLP timestamps. The QLP sector-80 light curves run 2.7 min late against SPOC 2-min and TESS-SPOC in the two systems where this was measured (A11 and A98), and the two sector-80 events of A99 are late by the same amount. Whether the offset applies to the whole sector has not been tested; the entries of A11, A98 and A99 state how their sector-80 timings enter the fits. At 30-min cadence (sectors 1–26) QLP mid-times can be off by minutes: in A92, those of sectors 14 and 15 are about 6 min late against three independent pipelines, while in its 10-min and 200-s sectors QLP agrees with them to 0.1 min. This was seen in that star and has not been tested in others.
- QLP depths. For A98 the depth in QLP's `SAP_FLUX` matches TESS-SPOC's crowding-corrected PDCSAP flux, so in that star QLP's aperture flux already carries the TIC contamination correction, although the file header does not say so. We have not established this for other stars, and depths are quoted as measured in the named pipeline. The QLP aperture changes from sector to sector, so a neighbour can make one sector's eclipse 9–27% shallower than the others, most often in sectors 99–102 (A85, A87, A91, A94, A96). In A81 the depth instead declines gradually, by about 9% from sector 7 to sector 88. Depths are therefore taken from the sectors that agree with difference images or a second pipeline.
- Flare pairs: two flares hours apart raise both side windows of a single-event box, and in injections manufacture a candidate 16–17% of the time (TIC 189882802, Section 3.8).

3. Results

The 54 eclipsing systems found by appointment come first. They are followed by the edge-on astrometric cases, the Gaia-orbit folds, the M-dwarf searches, the probable blends, the catalogue corrections, the planet candidates and the other objects. The labels (A1–A109, B, C) follow the order of discovery, not this arrangement, so the numbering within a section is not continuous. Nearly every entry has a figure, a list of key facts and a short account. Ephemerides and next-eclipse predictions for all confirmed and probable systems are collected in Appendix A.

3.1 Long-period eclipsing binaries found by appointment (A16–A18, A20–A71)

The appointments search (Section 2.5) produced 54 eclipsing systems, of which 50 are confirmed and four (A37, A40, A53, A68) are probable. The G < 10 subsample contributed 13 of them (16 stars re-analysed independently) and the G 10–11 subsample 41 (54 stars re-analysed independently). Fifty-two are long-period eclipsing binaries found at Gaia's conjunctions, with periods from 20.29 d (A35) to 57.24 d (A63). Two were found incidentally in the same light curves, at periods other than Gaia's: A45 (31.15 d, against a Gaia double-lined period of 21.68 d) and A54 (a 3.02-d eclipsing pair whose 39-d Gaia orbit is spurious). A third by-catch object, A46, is a 0.38-d contact or ellipsoidal variable on an unresolved 1.87″ pair. It is a new variable and is kept as an entry, but it is not shown to be an eclipsing system and is not counted among the 104. Each object was re-analysed independently with separately written code (Section 2.5). Most have both eclipse kinds, an ephemeris good to minutes, a geometry fitted against Gaia's orbit, at least one second pipeline and a difference image on the target. Where the re-analysis left the physics or the host open, the entry says so. Where a check of the protocol is missing, the status is probable. In the G < 10 subsample, one candidate was refuted, two were dismissed on inspection before re-analysis, one (BD+16 2639) remains a plausible single event, and known eclipsers including HD 181068 were recovered blind. In the G 10–11 subsample, three candidates were refuted, three stars were recovered as already known, and two remain plausible but unresolved. All three refutations were contamination by neighbours 36–60″ away, two of which are themselves uncatalogued eclipsers. The entries, the yield table and the list of refuted, plausible and recovered candidates follow.

3.1.1 Sample and yield

The method is set out in Section 2.5; this section gives its outcome. In the $G < 10$ subsample, 181 stars passed the cuts. Of these, 20 were already-catalogued eclipsing binaries, 27 had a variable-star entry of a non-eclipsing type and 134 were in no catalogue. Visual inspection of the 161 stars that were uncatalogued or typed otherwise found 10 with repeating eclipse-shaped dips on Gaia's ephemeris and three more marginal ones. Most of the rest showed orbit-edge scattered-light ramps, level jumps across data gaps, flagged slides before reaction-wheel desaturations, the sector-wide common mode of pre-gap dips, pulsation or spots. In the G 10–11 subsample, 447 of the 2,327 stars passed the cuts, and we inspected the 280 that no eclipsing-binary catalogue lists. There were 44 eclipser calls among the first 168 by triage rank, 2 among the next 104 and none among 8 neighbour-contaminated stars. Most false positives were level steps across the sector-99 gaps and the pre-gap common-mode dips. Every star classified as an eclipser, or as showing an unusual non-eclipse signal, was then re-analysed independently with separately written code (Section 2.5).

Table 3.1.1. Yield of the appointments search.

	***G* < 10 subsample (1,422 stars)**	***G* 10–11 subsample (2,327 stars)**

	***G* < 10 subsample (1,422 stars)**	***G* 10–11 subsample (2,327 stars)**
Sent to independent re-analysis	16 stars (two more, HD 201073 and HD 7423, were dismissed on inspection before any check)	54 stars (TYC 5976-120-1, already in VSX as an eclipser, was not sent)
Eclipsing systems, uncatalogued	13 confirmed	35 confirmed + 4 probable (A37, A40, A53, A68) long-period; by-catch: TYC 447-247-1 (A54, 3.02 d) and TYC 7897-48-1 (A45, a 31.15-d eclipsing binary); BD+30 2277 (A46) is a contact or ellipsoidal variable and is not counted
Confirmed heartbeat stars, new	1 (HD 193389)	2 (TYC 8203-301-1, with tidally excited oscillations at 22, 23 and 24 times the orbital frequency; BD+59 2405)
Recovered as already known	20 catalogued eclipsing binaries recovered blind (not sent for re-analysis)	20 catalogued eclipsing binaries recovered blind; 3 on target (TOI-1897 and TOI-5149, planet candidates that are M-dwarf eclipsing binaries; TYC 1194-680-1, a SuperWASP EA with the wrong period); 2 known eclipsing binaries on neighbours 63″ away
Refuted	1 (systematics)	3 (all neighbour contamination at 36–60″; two of the neighbours are uncatalogued eclipsers themselves)
Plausible, unresolved	1 single event (BD+16 2639)	2 (TYC 7687-990-1, whose eclipses appear in 2025 but not 2019–2023; Wolf 73, a single event)

The 54 eclipsing systems are catalogue entries A16–A18 and A20–A29 (the G < 10 subsample, all confirmed) and A30–A45 and A47–A71 (the G 10–11 subsample; A37, A40, A53 and A68 probable). Two of them (A45, A54) are by-catch: real eclipsing binaries on the target whose eclipses do not follow the Gaia orbit that brought the star into the sample. A46 (BD+30 2277) is also by-catch. It is a contact or ellipsoidal variable with its host component unresolved, and its entry is kept below but not counted.

Eclipses missed by the automated search.

1. **The post-gap trim.** The search first discarded 0.5 d after every data gap longer than 0.2 d, to suppress ramps. QLP sectors have many flag-induced gaps, so whole eclipses were lost: five of six in TYC 2819-1027-1 (A18), one of five in HD 26958 (A17) and one of seven in TYC 2173-965-1 (A16). The trim was reduced to 0.1 d and the G < 10 subsample was searched again; the far ratio and the side windows already reject the ramps that the trim was meant to suppress.
2. **Eclipses under QLP quality flags.** The search used QUALITY = 0 cadences only. In many systems whole eclipses lie in flagged cadences, mostly flagged for scattered light, although the flux itself is usable. These eclipses were recovered from the flagged QLP cadences, from other pipelines, or from custom cutout photometry (this work). They are: five of the ten eclipses of BD+29 2845 (A30); two of the four primaries of TYC 3747-2187-1 (A71); whole eclipses in BD-14 1559 (A58), TYC 3187-1132-1 (A59), TYC 4094-1571-1 (A70), TYC 243-1377-1 (A69), BD+27 4098 (A65), HD 279736 (A56), TYC 3729-162-1 (A50), BD+28 4244 (A51), BD+60 397 (A55, timing only) and TYC 986-148-1 (A40); partial eclipses in TYC 7121-1393-1 (A57); the sector-87 eclipse of TYC 5406-1905-1 (A52), under a scattered-light surge where QLP's background fails; and the second event of TYC 8355-1052-1 (A53), on which that confirmation rests.

Two points affect the values below. First, the widest box that fitted is not the eclipse duration. The search box stopped at 14 h, while confirmed eclipses run to 47 h, so every duration below is the fitted value. Second, a variable-star entry of another type (rotation, "MISC", "VAR:") does not make an eclipse known, so such stars appear below as type revisions.

3.1.2 Confirmed and probable eclipsing binaries

Each entry gives the key facts as measured in this work. Depths are fractional losses of the system's light in the photometric aperture. Epochs are mid-eclipse times in BTJD (BJD_{TDB} − 2457000). “A” is the conjunction at true anomaly 90° − ω (the star Gaia tracks is behind) and “B” the one at 270° − ω (it is in front). “Primary” denotes the deeper eclipse (Conventions) except where an entry says otherwise. Positions are the TIC J2000 coordinates, and *G* and Tmag are from Gaia DR3 and the TIC. “Not listed in” gives the catalogues in which the object's eclipses are absent (the definition of uncatalogued is in the Conventions). The ADS full-text check (Section 1) found no prior report of eclipses for any of these systems, and 31 of them carry unvetted SPOC or TESS-SPOC threshold-crossing events, listed in Table A1 (Section 2.7). A “Kostov table 2” flag is an entry in an unvetted machine-learning list of about 900,000 rows, which we count as uncatalogued. Next-eclipse times are those predicted in the re-analyses, with geocentric UTC where available. We recomputed the UTC times for this paper from the BTJD with TT − UTC = 69.184 s and no barycentric correction, so they may be off by up to ±8.3 min.

Entries are in catalogue order except that four are placed first: A47 (HIP 1060, with a total eclipse and eleven events over eight years), A20 (HD 170829, at *G* 6.3 the brightest star in this paper), A17 (HD 26958, *G* 8.2) and A31 (V1192 Cas, catalogued as a semiregular pulsator but shown here to be an eclipsing binary).

Per-object entries (55: A47, A20, A17 … A71) are in the Supplement, Section S3.1.2, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

3.1.3 Heartbeat stars

Three Gaia orbits produced a signal at periastron that is not an eclipse. It is a brightening and dimming locked to the closest approach of the two stars, caused by the tidal distortion of the primary seen at an angle: a “heartbeat”. For each star we fitted the equilibrium-tide model of Kumar et al. (1995), $\Delta F/F = S[1 - 3\sin^2 i \sin^2(\nu+\omega)]/R^3$, to the light curve alone. We then compared the eccentricity and periastron angle it returns with Gaia's independent radial-velocity orbit. That comparison is the test: a heartbeat at the right shape and phase is attributed to the star whose orbit it reproduces. None of the three eclipses at the fitted inclination.

The three are HD 193389 (TIC 51223058; Gaia SB1 P 20.7966 d, e 0.36), an F5.5 bright giant whose model recovers Gaia's ω to 0.3° and *e* to 0.016 (3.2σ by Gaia's error); BD+59 2405 (TIC 405590521; P 25.90 d, e 0.59), an A2 star in Trumpler 37; and TYC 8203-301-1 (TIC 147413927; P 22.04 d, e 0.68). The last also shows a coherent 842-ppm oscillation at 23.0009 ± 0.0006 times the orbital frequency, with weaker peaks at exactly 22 and 24 times it. An oscillation locked to an exact harmonic of the orbit is the signature of a tidally excited oscillation, and we read this one as such. The periods of all three are beyond the reach of the TESS heartbeat searches (Li et al. 2024a,b; Solanki et al. 2025) and of the Gaia–TESS cross-match of Callahan et al. (2025), which cut SB1 orbits at $P < 13$ d, so none of the three was in those samples.

Four more come from outside the appointments search. The continuation of the Gaia-orbit fold of Section 2.4 to G 9–10 set aside nine light curves whose feature was a brightening and a dip at one phase rather than an eclipse. Folded on Gaia's own periastron time, extrapolated over 300–950 cycles, seven of the nine put the feature at periastron, about 3 in 10,000 by chance. Three of those seven are published heartbeat stars (two in Callahan et al. 2025, one in Li et al. 2024a), which the test recovered without knowing it. The other four are in no variable-star catalogue and no heartbeat search. Their periods, 4.7–8.2 d, lie inside the range those searches covered, so their absence was checked by position and in the papers' own tables. Each was re-analysed independently with separately written code, to the

standard of the first three. BD+61 2067, BD+37 4854 and CD-66 2078 are confirmed: the light curve alone returns Gaia's e and ω. Two of the four Gaia solutions are double-lined (SB2) pairs of near-twin F stars (CD-66 2078 and TYC 9045-989-1). For BD+37 4854, whose ω is near 90°, the pure tidal model returned the mirror-image angle 180° − ω, because near that angle the two give time-reversed copies of almost the same pulse. Adding Doppler beaming of the primary, at the amplitude its velocity predicts, removes the ambiguity and returns Gaia's ω directly. TYC 9045-989-1 is a candidate. Its pulse is on the target and locked to Gaia's conjunctions, and its e agrees, but its ω comes back 20° (2.4σ) from Gaia's, and at e = 0.12 its signal is largely ellipsoidal.

Per-object entries (7: HD 193389, BD+59 2405, TYC 8203-301-1, BD+61 2067, BD+37 4854, CD-66 2078, TYC 9045-989-1) are in the Supplement, Section S3.1.3, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

3.1.4 Recovered, refuted and unresolved

Table 3.1.2 lists the candidates that are not new confirmed eclipsing binaries: systems the search recovered that were already known (positive controls in effect), candidates that were refuted, and three that remain open. Three of the four refutations share one cause: the dips belong to a neighbour 36–60″ away. Difference images show this, and so does the depth shrinking as the aperture tightens. The fourth (TIC 71350905) was orbit-edge scattered light. Two of those neighbours are themselves uncatalogued eclipsing binaries, recorded as leads rather than as results.

Table 3.1.2. Appointment candidates that are not new confirmed eclipsing binaries.

Object	Outcome	What the check found
Recovered, already known		
TOI-1897 = TIC 420645189	Known: retired planet candidate	TOI-1897.01 and .02 are this system's primary and secondary eclipses, retired by TFOPWG as FP/EB on 2024-09-17; P = 48.533998 ± 0.000022 d. Recovered blind. New: the Gaia SB1 gives f(M) = 0.0088 $M_\odot$, a 0.25 $M_\odot$ M-dwarf companion (K_1 12.4 km/s against about 50 m/s for the putative planet), and a sixth A eclipse in s86
TOI-5149.01 = TIC 286094277 = TYC 3578-1416-1	Known TOI; reinterpreted as an EBLM	A planet candidate since 2022-02-08; the 1.00–1.11%, 9.4–9.8-h transit is the TOI signal, P = 27.37163 ± 0.00006 d. New: a 602 ± 57 ppm secondary eclipse at phase 0.697 (z = 7.1; also in SPOC, TESS-SPOC and TARS); e cos ω = 0.3128 and e sin ω = −0.093 match Gaia; f(m) = 0.0014 gives m_2 = 0.11–0.13 $M_\odot$, R_2 = 0.15 $R_\odot$, T_2 ≈ 3100 K, so the companion is a low-mass star (an EBLM) rather than a planet. Speckle imaging shows a companion at 0.086″ (Δm 1.97)
TYC 1194-680-1 = TIC 435855781 = 1SWASP J004527.74+210427.9	Known eclipser; period and type revision	Listed as EA/EB with P = 1.66 d in Thiemann et al. 2021 (SuperWASP citizen science; not in VSX). TESS gives P = 22.450574 ± 0.000054 d (Gaia SB2 22.44398 ± 0.0033, 2.0σ off); 45% and 39% eclipses; joint SB2 fit: M = 1.144 + 1.161 $M_\odot$, R = 1.93 + 2.13 $R_\odot$, e = 0.1826, ω = 57.4°. This is a revision of the period and type, not a discovery
HD 181068 and seven catalogued EBs	Known; recovered blind	HD 181068, the Kepler triply eclipsing triple, recovered blind, and seven catalogued eclipsing binaries among the first 711 stars triaged: LW Pup, TIC 188625601, four IJspeert et al. 2024 OBAF eclipsers and one Prša et al. 2022 TESS EB. The full rerun of the G < 10 subsample recovered 20 known eclipsing binaries in all
Refuted		
TIC 71350905 = TYC 4361-615-1	Refuted: systematics	The claimed long periastron eclipse was orbit-edge scattered-light ramps: sector 40's first orbit, with no conjunction within 14 d, shows the same 1% drop. The mass function (0.0025 $M_\odot$) makes the companion a ~0.2 $M_\odot$ M dwarf whose transit could be at most 0.14% deep

Object	Outcome	What the check found
TIC 212431508 = TYC 5162-1158-1	Refuted: neighbour, itself an uncatalogued eclipser	Both dips (13% and 4.3%) are on Gaia DR3 4230027841458799744 (G 11.02, 35.9″, same parallax, probably a wide companion), which loses 16.1% and 5.3% while the target loses 0.6% and −0.3%; depth falls 13 → 2.2 → 0.7% as the aperture tightens. The neighbour is itself a Gaia SB1 (P 33.294 d) whose A–B spacing is 2.9σ from its own orbit: a separate lead
TIC 17238861 = TYC 2508-895-1	Refuted: neighbour, itself an uncatalogued eclipser	The 1.9% dips are 10.5% and 9.6% V-shaped eclipses of TIC 17238865 = TYC 2508-1098-1 (G 12.20, 60.4″, same parallax but not bound); the target's core pixels change by no more than 0.1–0.4%. The neighbour's period is any of 732.0556/N d (N = 1–28); 48.80 d is one of these 28 aliases, the one tied to the target's orbit
TIC 265677843 = TYC 3335-389-1	Refuted: background eclipsing binary	The 7.37% dip is a 32.4% eclipse of Gaia DR3 250601205600534272 (G 12.06, 37″, parallax 0.85 mas, a background star); the target loses 0.01 ± 0.04%; TARS 0.83% against QLP 7.37%. Attribution to the target would also require a period 3.9σ from Gaia's
Plausible single events		
BD+16 2639 = TIC 334634917 = HIP 70251	Plausible, unconfirmed	One 0.52%, 2.50-h grazing eclipse at BTJD 2676.6639, 0.4σ from Gaia's A conjunction, identical in five pipelines and on the target's pixels, in the only TESS sector (s50). The geometry (i ≈ 86.3°) predicts the 2.5–3.0-h duration and no secondary. An unrelated dip has a chance of a few percent of falling inside a ±0.5 d window. Next A: BTJD 4345.49, 4365.60 (±0.17)
Wolf 73 = HIP 6972 = TIC 365156806	Plausible, unconfirmed	One 2.98 ± 0.10%, 3.72-h flat-bottomed dip at BTJD 2481.9196 in SPOC 2-min data, 0.29σ from Gaia's B conjunction, on target; no other conjunction covered. A grazing occultation of a companion of at least 0.50 $M_{\odot}$. SPOC's own pipeline flagged the same event in 2021 at a spurious 32.139-d period. Next B: BTJD 4327.91 ± 0.47 (Gaia's period only)
Unresolved		
TYC 7687-990-1 = TIC 271053559	Plausible; eclipses only in 2025	In s89: a 31.33% V-shaped eclipse and two flat 5.19% and 5.51% eclipses, all on the target (difference images, TARS, clean common mode), at Gaia's 24.65-d period (P = 24.6548 ± 0.0018 d). However, the same ephemeris falls on covered, flat data in s8, s9 and s62, and a scan of every period from 24.58 to 24.76 d (the s89 value at ±50σ, Gaia's at ±5σ) finds, for each one, at least one of s8, s9, s35 and s62 covering the A conjunction with unflagged data and nothing deeper than 0.63% there, so the absence before 2025 is not an artefact of the one-sector ephemeris. Gaia also has a 300.8-d astrometric orbit on the star; our hypothesis, which is not proven, is a compact triple whose inner orbit precessed into eclipse between 2023 and 2025; the implied nodal-precession period of a few decades allows it. Next A: BTJD 4327.29, 4351.95 (±0.05; the third body can add hours)

3.2 Eclipses on edge-on Gaia astrometric orbits (A19, A72–A74)

Of 177 stars with nearly edge-on Gaia astrometric orbits and TESS coverage (Section 2.6), two passed these tests: repeat events, agreement with Gaia's independently measured period, independent SPOC photometry, the catalogue checks and pixel attribution. Both also passed independent re-analysis (Section 2.5). One of them, TIC 316045876, is withdrawn as a long-period eclipser, because its dips recur every 5.854 d (Section 3.8). Both the search and the re-analysis had looked for repeats only near Gaia's period and its half. The other, A19, is listed as probable. Its three eclipses repeat on a single ephemeris at a period that agrees with Gaia's to 0.7σ. Against that, they fall about 2.5σ from the nearer predicted conjunction, two of the three events exist only in QLP-flagged cadences or in QLP alone, and the one TESS-SPOC depth (85 ppt) is 30% below QLP's. Spectroscopy would be decisive. The same protocol had passed TIC 316045876. Two further stars were not retained as entries: TIC 299833082, with a single 40-ppt event, and TIC 333783808, with two events 298.57 d apart, 4.3σ from Gaia's 286.80-d period, and an irregular phased light curve.

The short-period search of the same light curves (Section 2.6) added two confirmed systems, one probable and one plausible. A72 (TIC 279636729) eclipses on its Gaia orbit itself, with three 38% primaries and a 24% secondary over 12 cycles of 215.01 d. The third primary was predicted blind and recovered in flagged data within 10 min of the predicted time. A73 (TYC 4590-364-1) is a 17.54-d eclipsing binary inside a 125-d Gaia orbit. Its eclipses have faded from 33% to 13% in four years, and their timings vary by hours. A third body is needed to explain this, though the data do not show that it is Gaia's. A74 (TIC 427345302) has two matching 4.6% eclipses in TESS, three cycles apart. An archival check after the review found a ZTF night at the predicted time of the cycle between them, on which the star is faint by the predicted amount, and two further ZTF nights at one secondary phase. These raise it to probable. TIC 271520202 has two matching eclipses at a period consistent with Gaia's and is kept in Table 3 (Section 3.8) until a third event. All four were re-analysed independently with separately written code (Section 2.5).

Per-object entries (4: A19, A72, A73, A74) are in the Supplement, Section S3.2, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

3.3 Eclipsing systems from the Gaia-orbit period folds (A5–A12, A75–A109)

The first eight of these systems, A5–A12, all have Gaia DR3 spectroscopic orbits shorter than 20 d, and TESS shows eclipses at Gaia's period. For the eccentric orbits the eclipse also lands on the conjunction Gaia predicts, which attributes the eclipse to the star with the orbit. A5–A7 come from the fold of 736 orbits with parallax > 10 mas, and A8–A12 from the fold of 1,445 orbits with G < 9 and parallax ≤ 10 mas or missing (Section 2.4). All eight were already known spectroscopic binaries (SIMBAD SB* or "**"); the eclipses are new. Positions and magnitudes in the entries are TIC 8.2 values. Periods, epochs, depths and durations come from timing fits made for this paper on the raw QLP SAP flux: trapezoid fits to every fully covered event and a weighted linear ephemeris (Appendix A). Where the search's own box values differ, they are given in brackets. The figures for A5–A7 were redrawn for this paper from the QLP light curves at the search period. Those for A8–A12 are the discovery folds from the search output, folded at the search period (for A10, at half of it). They show detrended flux where the search used it, so their depths differ from the raw-flux values quoted. Four of the eight (A5, A6, A7, A10) are listed as probable. The fold searched only within $\pm\max(5\sigma_P, 0.1\%)$ of Gaia's period, so period agreement is guaranteed by construction, and none of the four has a pixel test or a second pipeline. A5 and A10 are circular orbits with no conjunction-phase test. A8, A9, A11 and A12 are kept as confirmed. A9 rests on its pixels, ASAS-3 photometry and a 27.08-d ephemeris, and A12 on two pipelines and a centroid test. A8 and A11 rest on their event counts, in ten and eight sectors, and their conjunction-phase matches. After the pre-submission review, both gained difference images that place the deficit on the target, and light curves from TESS-SPOC (A8) and SPOC 2-min (A11) that reproduce QLP's depths and mid-times; their entries give the numbers. Two catalogue corrections from the same runs (B6, B7) are in Section 3.6, and the four demoted candidates are in Section 3.8.

Per-object entries (8: A5, A6, A7, A8, A9, A10, A11, A12) are in the Supplement, Section S3.3, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

A75–A109 come from the fold of the 3,022 orbits with G 9–10 (Section 2.4). Each was re-analysed independently to the full protocol of Section 2.5 before it entered the paper, and all 35 are confirmed. A93 was already typed as an eclipsing binary in SIMBAD (Section 1), so 34 of them are counted as new. In every one, difference images place the deficit on the target and exclude every Gaia neighbour within 60″ that is bright enough to produce it, at least one second pipeline reproduces QLP's depths and mid-times, and per-event timing fits in the raw SAP flux give the period without reference to Gaia's, except

that for A76 and A92 the choice of twice the fold period rests on Gaia's double-lined orbit. All 35 were already known spectroscopic binaries: Gaia gives 20 single-lined orbits, 11 double-lined and four double-lined circular. The periods run from 1.039 d (A109) to 19.605 d (A82). In nine systems (A76, A77, A84, A92, A93, A95, A98, A106, A107) the fold ran at half the orbital period, because the two eclipses are similar enough to fold together, and the true period is Gaia's; in A77 and A84 the two depths still differ, by factors of 1.6 and 1.9. Five show only one eclipse per orbit, the second conjunction missing the stars in an eccentric orbit or the companion contributing too little light (A78, A88, A89, A97, A99). The entries are in label order, which for A75–A108 is right-ascension order. Their figures were drawn for this paper from the raw QLP SAP flux, folded on the fitted ephemeris, with dashed lines at the measured eclipse phases (a few also mark Gaia's predicted conjunctions). Depths and durations are from the timing fits, except for A85, whose adopted depth comes from the pixels and SPOC. Where the eclipses test Gaia's orbit and fail, the entry says so; the cases are collected in Section 5.3.

Per-object entries (35: A75, A76, A77 … A109) are in the Supplement, Section S3.3, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

3.4 Eclipsing systems from the M-dwarf transit searches (A1–A4)

All four were detected as planet-like signals on nearby M or K dwarfs. Three of them (A1, A2, A4) are eclipses on a neighbouring star whose light contaminates the dwarf's aperture. The fourth (A3) is on the target. It is listed as probable: Gaia shows that its companion is most likely a star, but the companion's orbital period is not known.

Per-object entries (4: A1, A2, A3, A4) are in the Supplement, Section S3.4, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

3.5 Probable eclipsing binaries with unresolved hosts (A13–A15)

Three signals from the crowded-field and known-host searches are eclipses on faint stars near the targets. The pixels locate them, but not to a single star, or not with enough sectors to quantify the noise. They are listed as probable, not confirmed.

A13 Near TIC 58553437

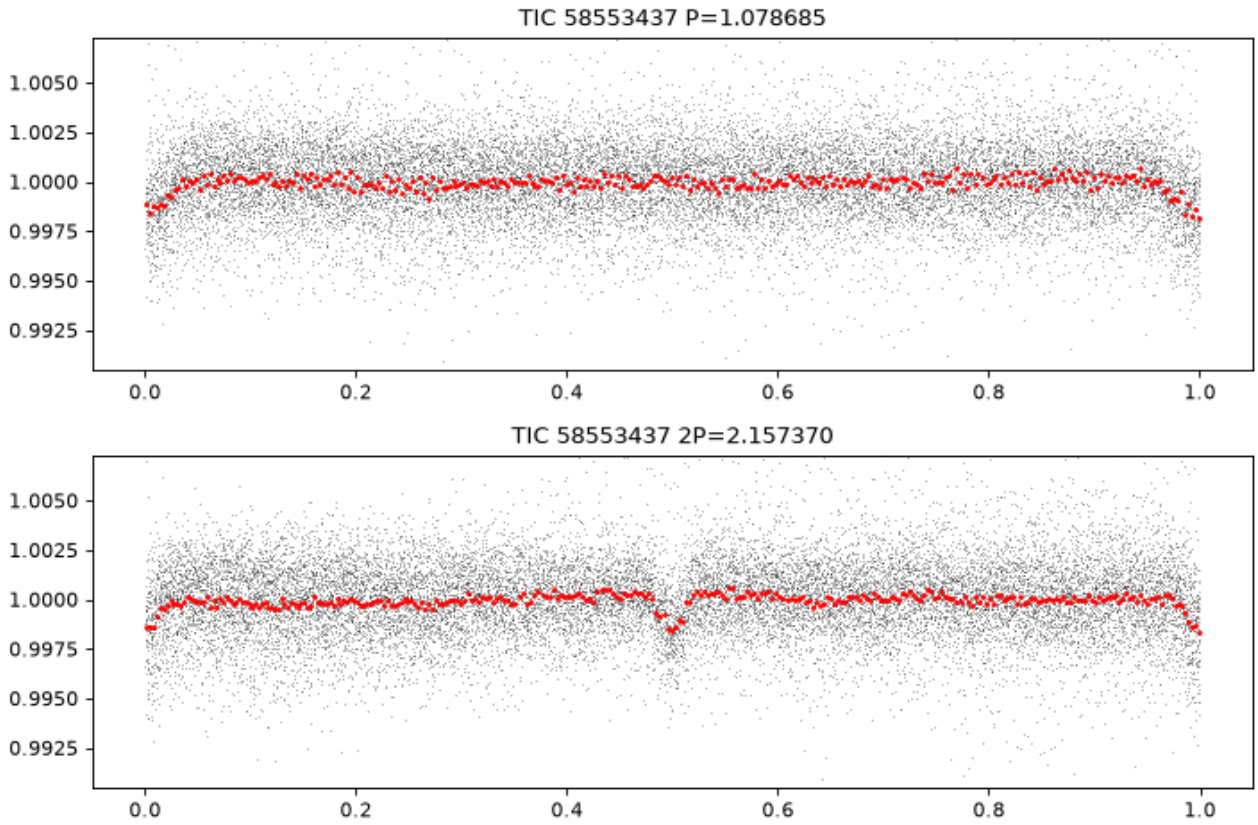


Figure A13. The 1.26-ppt blended signal in the aperture of TIC 58553437, folded on 1.078685 d (top) and on 2P = 2.157370 d (bottom).

Host	One of two G 13.7 stars 8–9″ from the deficit peak: Gaia DR3 2047563024899242368 or … 243264 (parallax 2.78 and 2.75 mas); not split
Period	1.078685 d, or 2.157370 d if the eclipses are twins (odd = even)
Depth	1.26 ppt, blended
Catalogue status	Uncatalogued: no VSX entry within 45″, no TESS-EB entry within 90″

A14 Near TIC 71124276

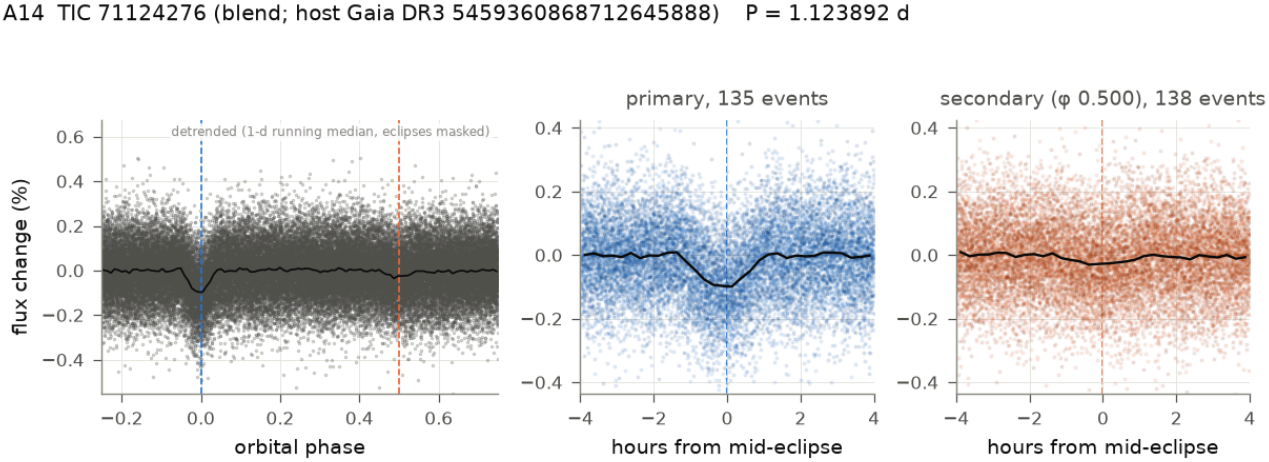


Figure A14. The 0.82-ppt blended signal in the aperture of TIC 71124276, folded on 1.123892 d.

Host	Gaia DR3 5459360868712645888, G 16.78, 14″ from the deficit peak

Period	1.123892 d; secondary 0.22 ppt
Depth	0.82 ppt, blended
Catalogue status	Uncatalogued. The aperture also holds Gaia DR3 5459360662554215168 (*G* 12.14, 15.4″ from the TIC position), a VSX variable of type VAR at P = 0.1286 d; it would need a 0.14% signal to make the blended 0.82 ppt, against 10% for the G 16.78 star named above, and neither VSX period matches 1.1239 d
Open issues	One sector only; weak; the host is not pinned

A15 TIC 261126769 = Gaia DR3 1956328333130771456, near TOI-6255

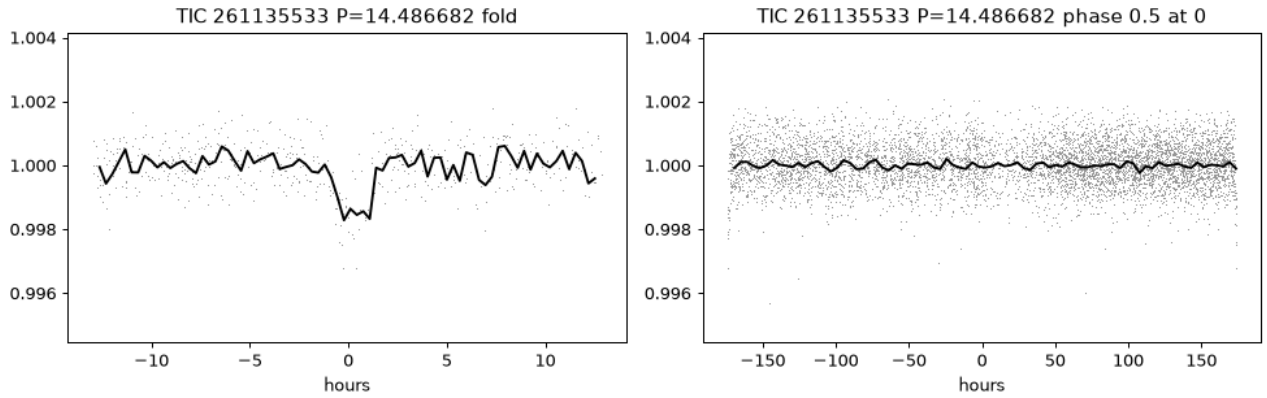


Figure A15. The 1,399-ppm signal in the aperture of TOI-6255 (TIC 261135533), folded on 14.486682 d, centred on the event (left) and on phase 0.5 (right); the pixels place it on a G 15.65 star 29″ south.

Host	TIC 261126769 = Gaia DR3 1956328333130771456: G 15.65, BP−RP 0.82, parallax 0.54 mas, RUWE 1.01, no Gaia variability or NSS flag; 29″ S of TOI-6255
Period	14.486682 d, or 28.97 d if the eclipses are twins (P/2 excluded: odd epochs at 7.24 d give −57 ± 101 ppm)
Depth	1,399 ppm on the M dwarf TOI-6255 (5 events, all positive, odd = even); ≳ 10% on the host, which is ~5 mag fainter
Pixel attribution	Deficit peaks 1.5–2 px off TOI-6255 in the 200-s FFIs of sectors 56 and 83; centroid 9.6″ (S56) and 5.1″ (S83) from the host
Catalogue status	VSX has nothing there, only a rotator 26″ away
Open issues	Probably an uncatalogued eclipsing binary; the centroid rests on two sectors, with the noise unquantified

The signal is not contamination from TOI-6255 b (its events fall at scattered phases of b), not momentum dumps (the nearest is 6–56 h away) and not the TESS orbit.

3.6 Catalogue corrections (B1–B11 and further cases)

These are stars that are already *in* a catalogue, with the wrong type, the wrong period or no period. Each is a revision of an existing entry rather than a new object. Table 2 lists the eleven numbered corrections. Nine are established by the data in their notes. Two are proposed and await a further step, stated in their notes: B1 (the host is not yet separated from two possible unresolved neighbours) and B7 (a contamination ratio of 1.77 leaves a blend possible). Short notes on each follow, and then the further corrections found by the candidate checks (Sections 2.4 and 2.5).

Table 2. Catalogue corrections B1–B11.

#	TIC	name	catalogue says	TESS says
B1	48875187	Gaia DR3 5451130921097245056	VSX type RS (Gaia classifier), no period	detached twin EB, P_{orb} = 0.862264 d, equal 9.1% eclipses (TGLC). **Held**: host not submission-grade

#	TIC	name	catalogue says	TESS says
B2	382523691	ASASSN-V J005242.03-513706.0	VSX ROT 5.3637 d	eclipsing at Gaia's SB1 period, P 4.915053 d, 24 + 12 ppt
B3	397233697	Gaia DR3 522317746033943808	VSX ROT (Gaia); Gao ROT 8.40 d	eclipsing, P 17.13557 d, 100 ppt + 5 ppt at φ 0.3
B4	333792991	HAT-177-0002605	VSX VAR 9.69 d; ASAS-SN ROT:	eclipsing, P 6.21467 d, 45 + 13 ppt
B5	385557214	V1274 Tau	GCVS RS 6.62 d; VSX RS 6.448 d; ASAS-3 ESD/EC 1.727 d, 0.12 mag	eclipsing at Gaia's 5.62323 d, 4.5 + 2 ppt; ASAS-3's own photometry shows only the 6.45-d rotation
B6	464315748	HD 171459	VSX DSCT|GDOR at 3.19721 d	near-twin eclipses 12.7/13.4 ppt; P_{orb} = 6.3955 d
B7	189269584	CD-28 7251	VSX VAR at 0.0146 d	20.7/23.9% twin eclipses, P_{orb} = 6.9610 d
B8	99028319	LP 852-5	TESS-EB (Prša) P 9.4045943 ± 0.00045 d; not in VSX	P = 9.400436 d, 32.9/18.5 ppt
B9	80158129 (blend)	CzeV3749, 9″ from the peak	VSX EA, no period	P 10.898518 d, no secondary
B10	120660110 (blend)	WISE J105540.2-521127, 11″	VSX EW/EA P 0.8152227 d	P 0.817052 d
B11	306658481	NSV 3445 (1.5″)	VSX suspected variable, no type or period	eclipsing, 1.66 d; depth sector-dependent

B1 TIC 48875187 = Gaia DR3 5451130921097245056

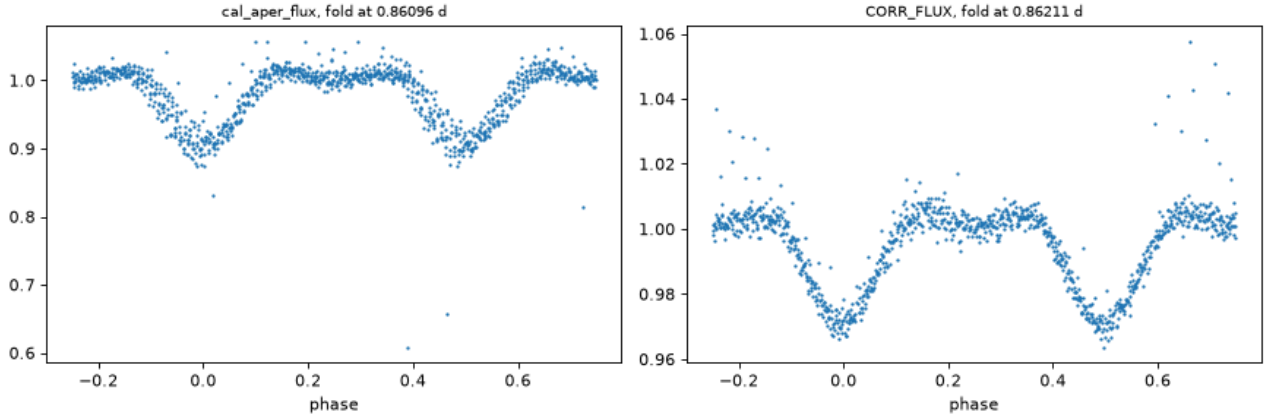


Figure B1. TIC 48875187 (sector 9), folded near P_{orb} = 0.862264 d (left: TGLC cal_aper_flux folded at 0.86096 d; right: GSFC-ELEANOR-LITE flux folded at 0.86211 d): two equal, wide eclipses with flat light between them.

Found as a 2.156-d, 263-ppm signal on the M dwarf TIC 48875199, 61″ away. In the variable's own pixels the 2.156-d signal is the fifth subharmonic of 0.4312–0.4313 d. Folded at twice that period, it shows two equal eclipses, 9.1%/9.1% in TGLC and 2.8%/2.8% in eleanor-lite. The star is therefore a detached near-twin eclipsing binary, not an RS CVn spot wave, at P_{orb} = 0.862264 d (2 × 2.155659/5, from the multi-year baseline), with an eclipse at BTJD 1546.164. A neighbour 4″ away, TIC 942681515, is unresolved by TESS, and the pipelines disagree on which star dips: TGLC's PSF fit gives the deeper dips to the neighbour (11% vs 9%), while eleanor-lite gives them to TIC 48875187 (2.8% vs 0.5%). The centroid and Gaia's variability flag favour TIC 48875187, but the period is real and the host is not submission-grade. The revision is deferred until higher-resolution photometry identifies the host. The star is listed in Kostov et al. (2025), table 2.

B2 TIC 382523691 = ASASSN-V J005242.03-513706.0

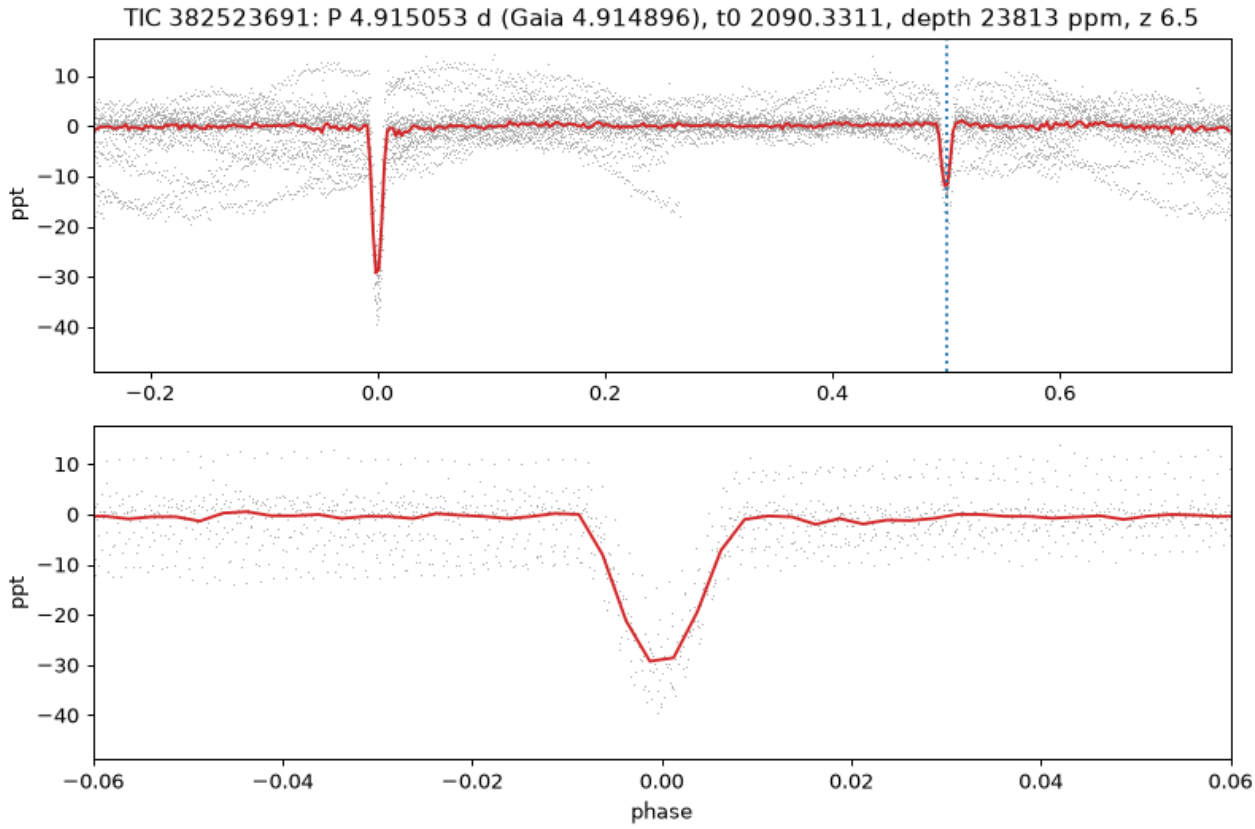


Figure B2. TIC 382523691 folded on its orbital period, 4.915053 d: 24-ppt and 12-ppt eclipses.

Gaia DR3 4922422783421253632 has an SB1 orbit at P 4.914896 ± 0.00029 d. TESS shows eclipses at P 4.915053 d, t_0 = BTJD 2090.3311, primary ~24 ppt, secondary ~12 ppt at phase 0.5, on Gaia's predicted conjunction (Δφ = +0.010). Tmag 11.24, 91 pc, K-type, contamination ratio 0.0007. VSX, Christy et al. (2023) and Gao et al. (2025) all type it a rotator at 5.36 d (Schmitt et al. 2026: 5.353 d). The spot period is real and differs from the orbit by 9%. Earlier studies detected only the spot modulation. The revision is to add an eclipsing type and the orbital period.

B3 TIC 397233697 = Gaia DR3 522317746033943808

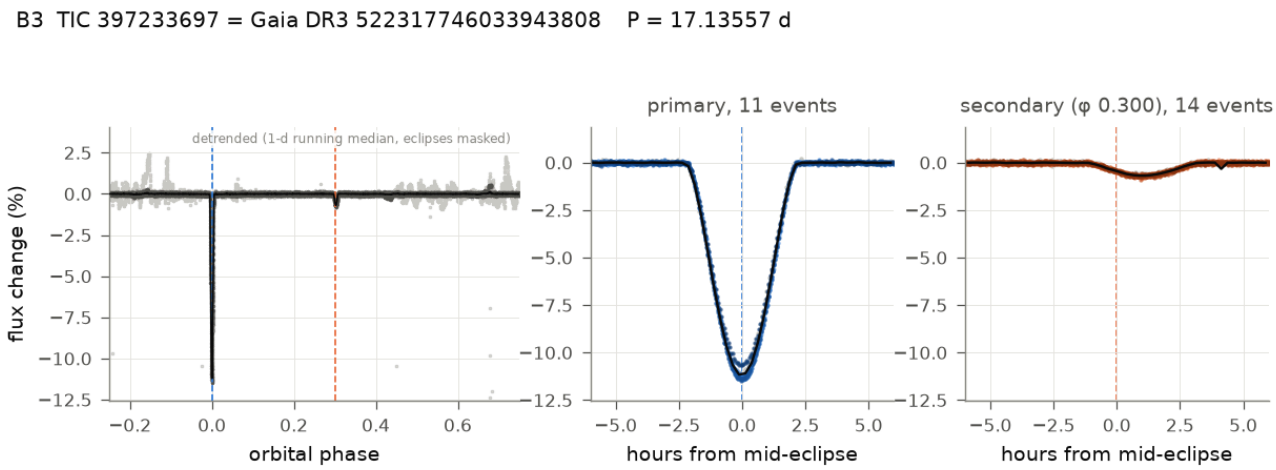


Figure B3. TIC 397233697 folded on 17.13557 d: a 100-ppt primary and a 5-ppt secondary at phase 0.3.

Eclipsing at 17.13557 d, 100 ppt with a 5-ppt secondary at phase 0.3 (an eccentric orbit), on Gaia's conjunction to Δφ = −0.001. VSX carries it as a rotator from the Gaia classifier (Gaia SOLAR_LIKE), and Gao et al. (2025) as a rotator at 8.40 d. Tmag 8.61.

B4 TIC 333792991 = HAT-177-0002605

B4 TIC 333792991 = HAT-177-0002605 P = 6.21467 d

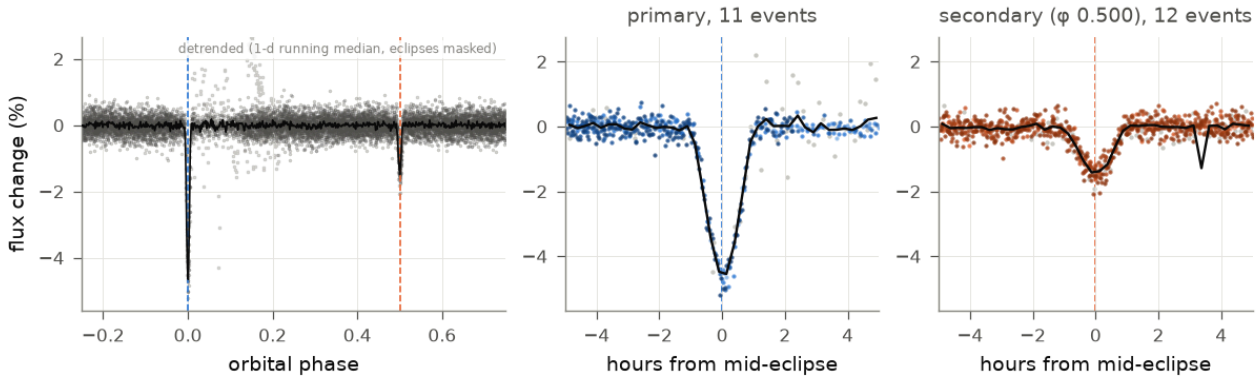


Figure B4. TIC 333792991 folded on 6.21467 d: 45-ppt and 13-ppt eclipses.

Eclipsing at 6.21467 d, 45 + 13 ppt, Δφ = +0.026 from Gaia's conjunction. VSX lists it as VAR at 9.6885 d and ASAS-SN as a probable rotator.

B5 TIC 385557214 = V1274 Tau

B5 TIC 385557214 = V1274 Tau P = 5.62323 d

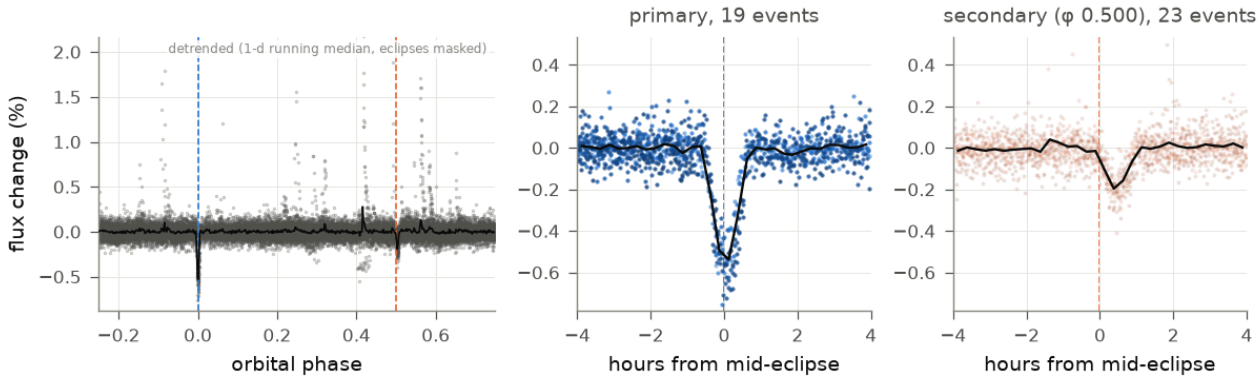


Figure B5. V1274 Tau folded on 5.62323 d: 4.5-ppt and 2-ppt eclipses.

Eclipsing at Gaia's period, 5.62323 d (Gaia 5.6237), 4.5 + 2 ppt, Δφ = −0.020. GCVS types it RS at 6.62 d and VSX RS at 6.448 d. The ASAS-3 catalogue of variable stars (Pojmański 2002; VizieR II/264, queried 2026 October 13) lists it 1.4″ from the TIC position as ASAS 034558+2733.6, class ESD/EC, P = 1.727 d, amplitude 0.12 mag in V. An eclipsing classification is therefore already on record. Its period and amplitude cannot be those of the 4.5-ppt eclipses found here, and the relation of that period to the 6.4–6.6-d rotation period of GCVS and VSX is not established. The ASAS-3 photometry itself (63 grade A–B points over 677 d, fetched 2026 October 13) does not support that entry. Its Lomb–Scargle periodogram peaks at 6.450 d (power 0.54, the GCVS/VSX rotation period, 0.11 mag) and has no power

at 1.727 d (0.009) or at 5.623 d (0.035). The 4.5-ppt eclipses are far below its 0.035-mag errors. The correction is therefore to an existing eclipsing classification whose period is wrong and whose variability is rotational. The eclipses are at Gaia's 5.623 d.

B6 TIC 464315748 = HD 171459

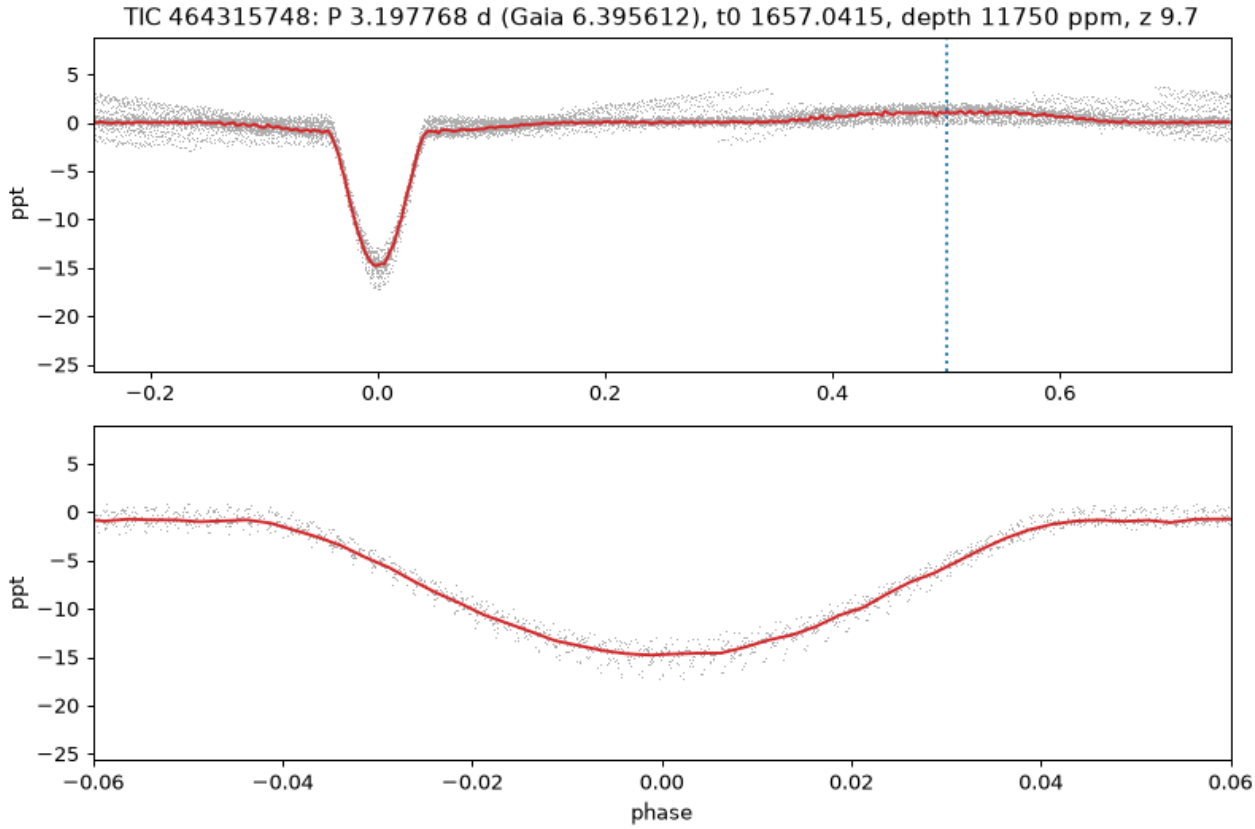


Figure B6. HD 171459, the discovery fold at P/2 = 3.197768 d, half the 6.3955-d orbital period, which stacks the near-twin eclipses of 12.7 and 13.4 ppt into one dip.

A circular double-lined Gaia orbit (SB2C) with near-twin eclipses of 12.7 and 13.4 ppt at P_{orb} = 6.3955 d. VSX has the star as a δ Scuti/γ Doradus pulsator at 3.19721 d, which is the spacing between the two eclipses. The VSX classification is incorrect, as the independent check confirmed.

B7 TIC 189269584 = CD-28 7251

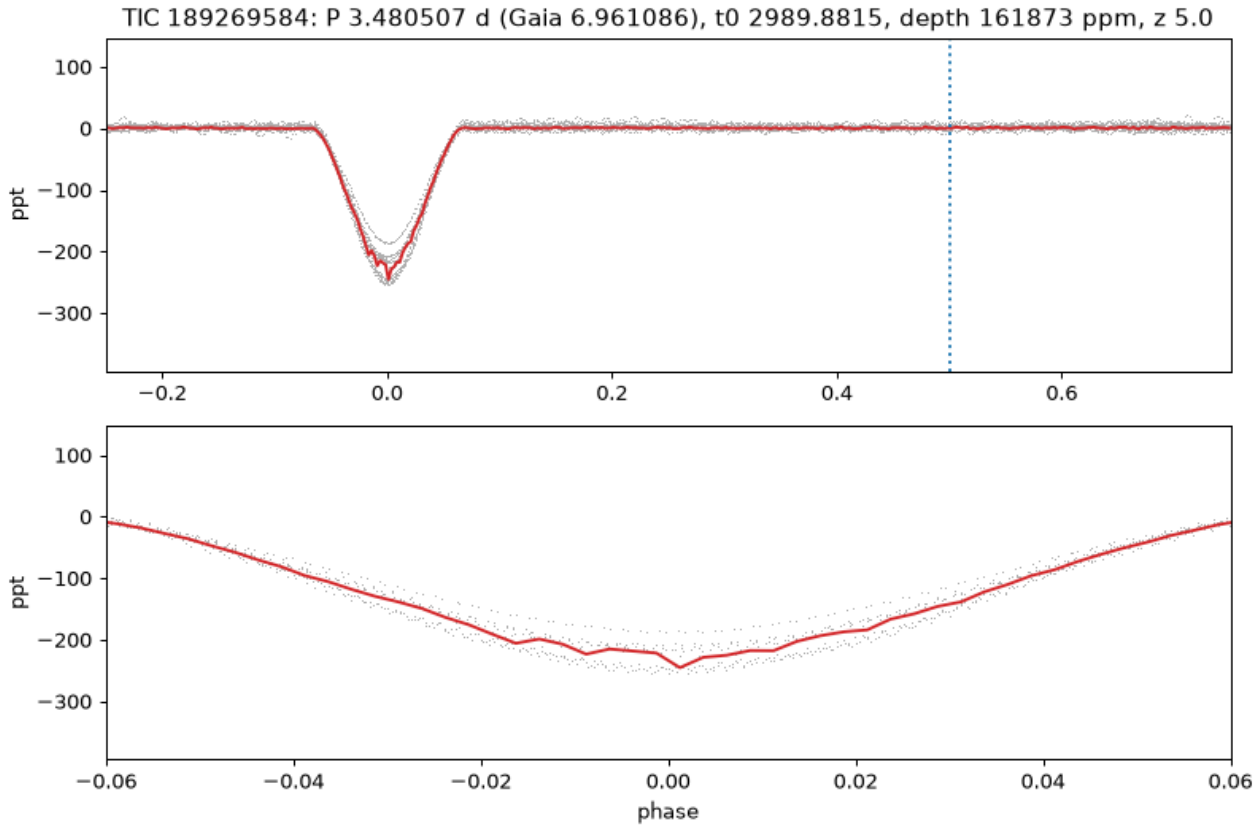


Figure B7. CD-28 7251, the discovery fold at P/2 = 3.480507 d, half the 6.9610-d orbital period, which stacks the twin eclipses of 20.7% and 23.9% into one dip.

Twin eclipses of 20.7% and 23.9% at P_{orb} = 6.9610 d, Δφ = −0.039 from Gaia's conjunction. VSX lists it as "VAR" at a spurious period of 0.0146 d. Caveat: the TIC contamination ratio is 1.77, so a blend is possible, but no catalogued neighbour matches the period (the only eclipsing neighbour, 105″ away, has P 3.21869 d).

B8 TIC 99028319 = LP 852-5

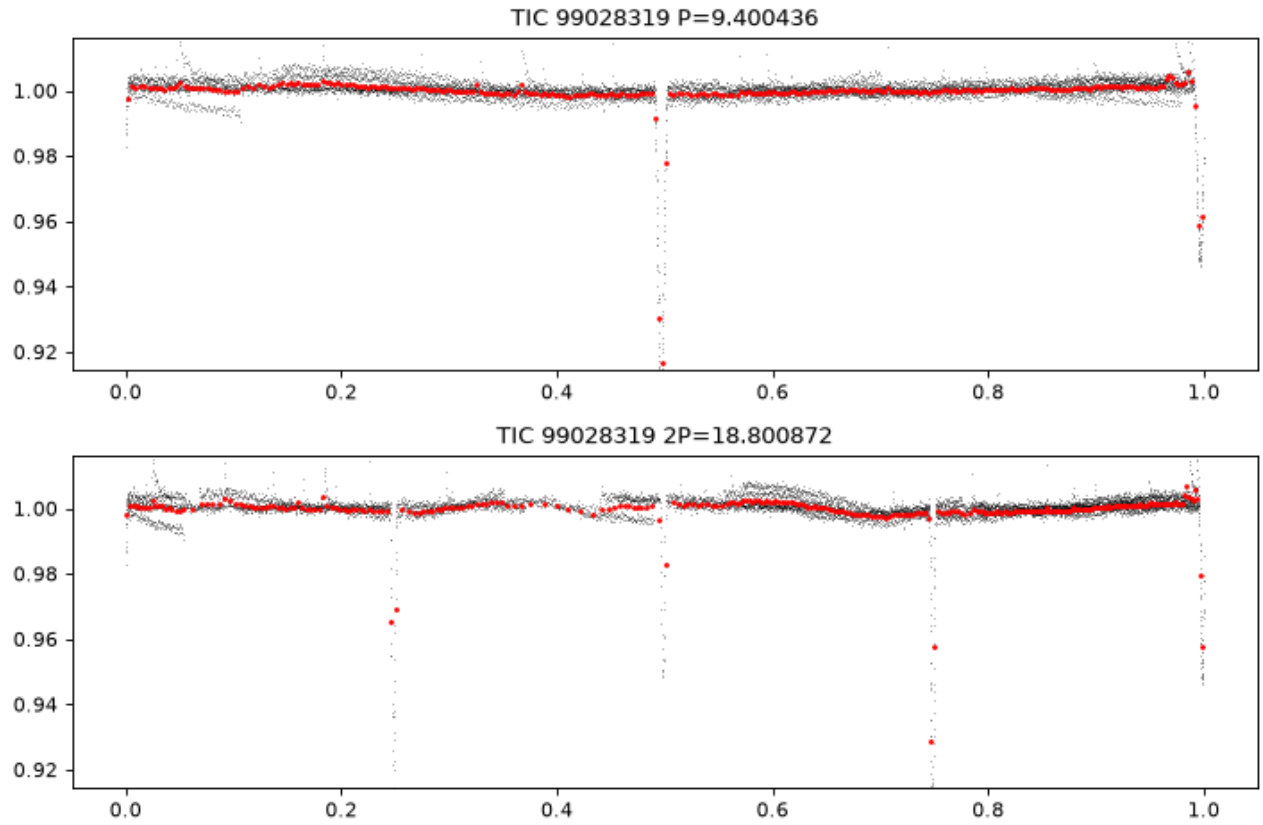


Figure B8. LP 852-5 folded on 9.400436 d (top) and on 2P = 18.800872 d (bottom): detached eclipses whose fitted box depths are 32.9 and 18.5 ppt (the binned minima in the panels are deeper, about 8% and 4.5%, because the boxes average over the eclipse).

An M3.0 dwarf at 39 pc (G 12.01; LDS 4178A, with B = LP 852-6 16.5″ away), with detached eclipses of 32.9 ± 3.6 and 18.5 ± 2.3 ppt half an orbit apart, P = 9.400436 d, primary T_0 ≈ BTJD 1576.925; on target in sector 90. It is already in the TESS eclipsing-binary catalogue at 9.4045943 ± 0.00045 d from sector 10 alone. Folded at that period the eclipses wash out after sector 10 (−0.5 to +1.0 ppt in sectors 36–101). At 9.400436 d they hold at 31.5–34.8 ppt in every sector. Not in VSX. Gaia: RUWE 1.55, RV amplitude 55 km s^{-1}.

B9, B10 CzeV3749 and WISE J105540.2-521127 (blends)

Two signals in the crowded-field search traced to catalogued eclipsing binaries near the targets. **B9:** a 3.2-ppt blended signal on TIC 80158129 at P = 10.898518 d with no secondary, from CzeV3749, 9″ from the peak. VSX types it EA with no period, so this would be its first period determination. **B10:** a 1.3-ppt blended signal on TIC 120660110 at P = 0.817052 d (secondary 0.17 ppt), from WISE J105540.2-521127, 11″ away. VSX gives P 0.8152227 d, which is 0.22% off, far more than the 3,000-d baseline allows.

B11 TIC 306658481 = NSV 3445

An eclipsing signal at 1.66 d (at the half period, 0.829 d, the fold shows a 15σ odd/even difference), with a depth that changes from sector to sector. It is 1.5″ from NSV 3445, a VSX suspected variable with no type or period. Two GDS variables 40″ and 80″ away (0.3 and 0.53 mag ranges) may dilute it differently in each sector. A type and period revision is warranted.

The crowded-field search also matched 25 of 52 strong peaks to VSX entries at simple harmonics of the catalogued period. These are recorded as period-of-record notes rather than corrections.

Further corrections from the candidate checks

TYC 1194-680-1 = TIC 435855781: a SuperWASP period corrected

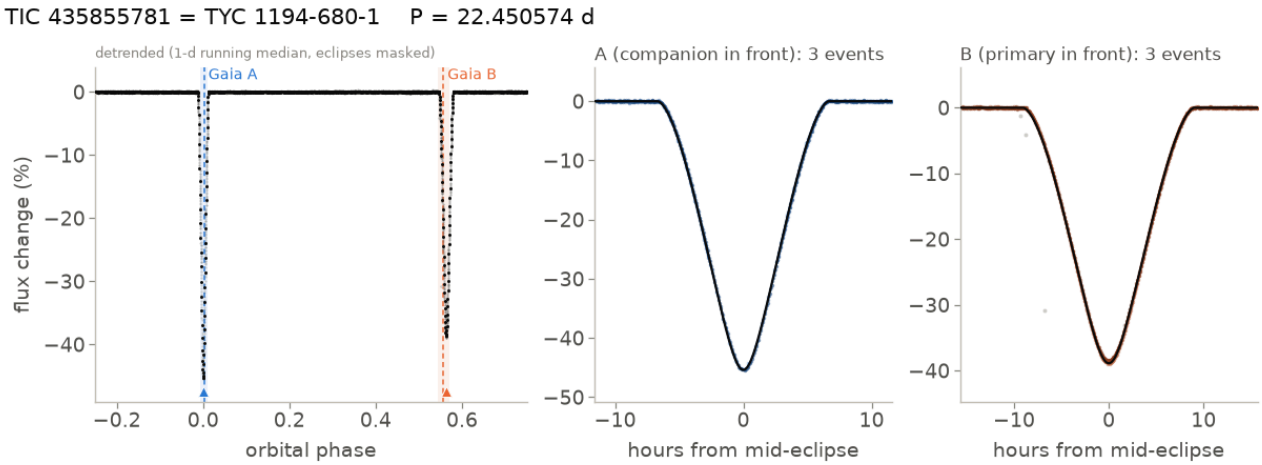


Figure B12. TYC 1194-680-1 folded on P = 22.450574 d: 45% and 39% eclipses of a pair of slightly evolved near-twins.

Identifiers	TIC 435855781; TYC 1194-680-1; 1SWASP J004527.74+210427.9
Catalogue says	EA/EB with P 1.66 d in the SuperWASP variable-star catalogue of Thiemann et al. (2021); not in VSX
Period	22.450574 ± 0.000054 d from the A eclipses (22.450619 from B; 22.450609 from the physical fit). Gaia SB2: 22.44398 ± 0.0033 d, 2.0σ short
Epochs	A: BTJD 2276.97457 ± 0.0016 = BJD_{TDB} 2459276.97457; B: BTJD 3210.09581 ± 0.0017
Eclipses	45% and 39%; durations 11.6 and 15.7 h. TESS-SPOC (43.6%, 36.8%) and TARS agree on the times
Orbit	e = 0.1826, ω = 57.4° (Gaia 0.182 ± 0.008, 62.3 ± 2.8°); i = 89.25°
Stars	Joint fit with Gaia's double-lined K_1 = 51.07, K_2 = 50.31 km s^{-1}: M_1 = 1.144, M_2 = 1.161 $M_\odot$; R_1 = 1.93, R_2 = 2.13 $R_\odot$; a = 44.2 $R_\odot$
Pixel attribution	Difference-image centroid 1.1–1.3″ from the target at S/N 670–960; no Gaia star brighter than G 19 within 60″
Next eclipses	B: BTJD 4332.627 (2026-10-19 03:02 UT); A: BTJD 4342.427 (2026-10-28 22:15 UT); ±8–9 min
Method	Conjunction-window search (Section 2.5); checked by independent re-analysis (Section 2.5)

A confirmed eclipsing binary, but not new: SuperWASP already identified it as an eclipser, with a period wrong by more than a factor of ten. It is a period and type revision for VSX, citing Thiemann et al. (2021) as the prior identification. Because Gaia measured both stars' velocities, the light curve and the published orbit together give masses and radii, in principle to percent-level precision.

Wrong VSX periods on two neighbours of conjunction-window targets. Gaia DR3 3327090097954378752 (G 14.8, 63″ from TIC 206952704) is in VSX as "E" with P 3.23146 d; 22 TESS eclipses with alternating depths give P = 5.725588 ± 0.000018 d, circular. Gaia DR3 5956391969014516736 (G 13.9, 53″ from TIC 34344890) is in VSX as "E" with P 12.3202 d; its TESS eclipses alternate at P = 1.5432 d. Both signals contaminate the QLP apertures of the brighter stars, and were detected through this contamination.

VSX type revisions among the new eclipsers. Five of the appointments objects are in VSX under a non-eclipsing type, so for them the submission is a revision: A20 (HD 170829 = NSV 24480, "VAR:", no type or period), A25 (BD+12 121 = ASAS J005930+1304.9, "MISC" at 91.5 d, which is spot modulation of the giant), A31 (V1192 Cas, SRA at 37.4 d, which is the orbital period), A32 (TYC 97-228-1, ROT at 39.81 d, the giant's near-synchronous rotation) and A52 (TYC 5406-1905-1, "MISC" at 79.3 d). A21's only prior record is an ExoFOP CTOI (403113935.01, a 14.9 $R_\oplus$ "single transit") that is the star's own secondary eclipse.

Gaia DR3 orbit solutions contradicted by the eclipses. Eclipses fix the period, and their spacing and durations fix e cos ω and e sin ω, independently of any velocity measurement. We count a Gaia solution as contradicted when its period is spurious or wrong, or when the period or e cos ω differs from the eclipse value by 3σ or more of Gaia's quoted error, whether Gaia's covariance is propagated in full or its correlations are ignored, and however the ephemeris is carried. The eclipse e cos ω is taken from the spacing of the two eclipses, through Kepler's equation at Gaia's e sin ω. Conjunction offsets alone, which move by several σ between those choices, are recorded as tensions, and so are eccentricities and arguments of periastron from eclipse durations, which depend on the assumed geometry. The rule is applied in the same way to every system in this paper. Among A5–A74, ten systems meet it:

- **Spurious or wrong periods (2). A54, TYC 447-247-1 (TIC 108344775):** a 3.02348-d circular eclipsing K-dwarf pair. Its Gaia SB2 solution (P 39 d, $K_1 + K_2$ 190 km s^{-1}) would need 14 + 13.5 $M_\odot$ components and is not a sampling alias of the true period: spurious. **A9, HD 7601 (TIC 399009076):** Gaia's SB2 period of 13.42 d does not phase the 27.08-d eclipses ($2P_{Gaia}$ differs by 100σ), and its $(M_1+M_2)\sin^3 i$ = 0.49 $M_\odot$ is impossible for this pair: an erroneous period (Section 3.3).
- **Period off (3). A52, TYC 5406-1905-1 (TIC 307170556):** eclipses at 40.46953 d where Gaia's SB2 gives 40.543 ± 0.015 d (4.9σ; Gaia goodness of fit 11.5); one Gaia epoch-photometry point is consistent with the TESS period and not with Gaia's. **A12, HD 217492:** period 4.2σ and conjunction phase 5.5σ from Gaia's SB2 elements (Section 3.3). **A29, BD-01 648:** the eclipses give 29.60666 ± 0.00002 d against Gaia's 29.5888 ± 0.0053 d (3.4σ), and at Gaia's own epoch the eclipse falls 1.67 d (formally 25σ) from Gaia's B conjunction. We judge the SB2 solution distorted (goodness of fit 20).
- **Shape off (5). A65, BD+27 4098 (TIC 88619181):** the transit and occultation fall only 1.95 d apart and both are flat, which needs e = 0.790, ω = 209°. Gaia's SB1 solution (e 0.688 ± 0.017, ω 224 ± 4°) predicts a spacing that is 4.0–5.0σ from the observed one. The period and periastron epoch agree. A third body is likely (RUWE 7.8, eclipse-timing variations). **A28, CD-32 4901:** secondary at phase 0.5000, so the orbit is circular, and the spacing Gaia's e = 0.042 implies is 5.7–6.1σ off (goodness of fit 14.9); the period is also 4.5σ off (51.37333 ± 0.00020 d against 51.311 ± 0.014 d). **A30, BD+29 2845 (TIC 311297982):** the eclipses are 0.7582 apart in phase against Gaia's predicted 0.7711 ± 0.0023, so e cos ω is 5.3–5.6σ off; the period agrees (SB2, goodness of fit 23.7). **A63, TYC 3801-1499-1:** e cos ω 3.8σ (4.1–4.3σ from the eclipse spacing) and e 3.3σ from Gaia's. **A21, TYC 4244-623-1:** e cos ω 3.7σ off, in a solution with RUWE 35 and significance 53.

Several systems fall below the bar or are not counted. A37, CPD-25 5943, is 2.7–2.9σ off in e cos ω by the eclipse spacing (3.0σ in the first-order relation), in a weak orbit of significance 11 whose K_1 is probably diluted. In A48, TYC 3166-453-1, the eclipse phases give e = 0.173 against Gaia's 0.211 ± 0.014 (2.8σ). A55 is 2.5–2.9σ off in e cos ω. In A24 the duration-based e is 3.3σ below Gaia's, while ω and e cos ω agree within 1.2σ; this is the kind of duration-based tension the rule leaves out. In A109, BD+43 1585, Gaia's period is 4.3–6.5σ shorter than the eclipse period on the formal errors, but the eclipse period itself changes by 16 ppm between TESS baselines (the sector-20 eclipses are 26 min early, probably light-travel time about a third body); with that change added to Gaia's error the difference is 1.8–2.7σ. A45, TYC 7897-48-1, is left open: its 31.15-d eclipses do not follow its 21.68-d Gaia SB2 orbit at all. Either Gaia's period is an alias, or the system is a 2+2 quadruple, or the eclipses come from a

blend within 2″; in the last two cases Gaia's orbit is not contradicted. HD 151770 (Table 3) is a further ~7σ case whose eclipses are not attributed to the Gaia pair. For comparison, Rowan et al. (2023) found only about half of Gaia's SB2 solutions for ASAS-SN eclipsing binaries consistent with the eclipses, and Callahan et al. (2025) found 85% of the SB1 and 20% of the SB2 orbits of their heartbeat stars consistent with the TESS light curves. Our rate, 12 of the 91 counted systems whose eclipses were compared with a non-circular Gaia spectroscopic orbit (two of them SB2 periods that are wrong outright, and two among A75–A109, Section 5.3), is in the same range.

3.7 Planet candidates (C1–C3)

Per-object entries (3: C1, C2, C3) are in the Supplement, Section S3.7, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

3.8 Other objects

Table 3 lists other objects of note: variables that are not eclipsing binaries, false positives kept because they are instructive, and rejected candidates. The six new heartbeat stars and one candidate are in Section 3.1.3.

Table 3. Other objects. The last row comes from the short-period search of the edge-on astrometric sample (Section 2.6); independent re-analysis (Section 2.5) rated it plausible pending a third event. Three others from that search are A72, A73 and A74 (Section 3.2).

object	what	status
GJ 4348 = TIC 206327797 (M3.5, 25 pc; B of LDS 819)	Günther et al. (2020) list P_{rot} 0.223861 d. The 2f term holds phase at 31–37° in all eight sectors 2018–2025 while the spot term drifts 186° → 9°; Gaia RV amplitude 387 km s^{-1} (more than any orbit gives, so partly outliers, but 5–7× fast-rotator controls); M_G +2.02 mag (G) and +2.15 mag (K_s) above 306 stars of the same colour, of which an unresolved pair can supply at most 0.753 mag; listed in SPYGLASS I (Kerr et al. 2021). Interpretation: a young star, with a detached near-twin M+M pair at P_{orb} 0.223864 d plausibly on top of it. Independent re-analysis finds the case undetermined but leaning towards a binary, with nothing supporting a contact configuration; the luminosity excess is quantified above	open; one spectrum at quadrature settles it
TYC 3763-577-1 = TIC 322347036 (T_{eff} 7263 K, 511 pc)	Low-amplitude δ Scuti star, frequencies 15.32, 20.53 and 27.14 d^{-1}, ~1 ppt (diluted); per-pixel sine fit puts it 1.7–2.0″ from this star. Detected by the search as a 0.261-d transit-like signal on the M dwarf TIC 322347050, 70″ away. SIMBAD "*", no references; not in VSX	uncatalogued; of minor interest
TIC 234495456 (M dwarf, 38 pc)	Uncatalogued BY Draconis spot rotator at ~0.7623 d; the apparent secondary at 16.7σ was the sinusoid filling the longest trial duration. No Gaia flags; not in VSX	not submission priority
TIC 189882802 (M3V, 36.9 pc)	A 3,444-ppm, 4-h single event at SNR 16.5 that passed isolation, catalogues, Gaia, the difference-image centroid (on target, 0.2σ) and flat depth across apertures. It is two flares 6–10 h apart raising both side windows; in injections a flare pair manufactures a candidate 16–17% of the time, and a cut on a distant baseline removes 71 of 71	dead; kept as the best false positive of the hunt
TIC 96202086 (M3.5V, 20.5 pc)	Strongest candidate of the initial search: 0.63048 d, 307 ± 32 ppm, ~0.7 $R_\oplus$; null 3/100 exceed, so ~40 expected at that level over 1,299 stars	not a detection
TIC 150282725, 29938816, 70442390, 88379993	Planet-like peaks rejected by the centroid test (84″/123″ off at 15–18σ; 11.9″ off at 10.7σ), by dependence on a single sector, or by odd/even depths (an eclipsing binary at twice the period)	dead
Four demoted fold candidates	HD 34757 (TIC 309416862; a 0.9-ppt claim; SPOC sees nothing at three events: dead); HD 44342 (TIC 38022998; 1.6 ppt; raw SAP 0.1 ppt; no SPOC products: unresolvable); HD 196402 (TIC 100614055; 1.6 ppt; a 0.7-ppt SPOC dip at the event but the star varies by 1.5 ppt off-phase: stays demoted); **HD 151770** (TIC 337393383; real 16.3 + 11.2 ppt eclipses, but their separation, 0.593, implies e cos ω 0.146 against Gaia's 0.223 ± 0.011, ~7σ, and both conjunctions miss: open; either Gaia's SB2 geometry is biased or a neighbour with a near-coincident period, an L-type Gaia variable 27″ away, is the source)	3 dead or unresolvable, 1 open

object	what	status
BD−16 52 = EBLM J0021−16 = TIC 406352710	A published eclipsing binary, a G dwarf with a 0.19-$M_\odot$ M-dwarf companion (Triaud et al. 2017; Spejcher et al. 2026), recovered blind by the G 9–10 fold (Section 2.4) and passed by the full re-analysis: 25.7-ppt primary and 1.3-ppt secondary at phase 0.5005, P = 5.9672641 ± 0.0000012 d from QLP sector 29 and SPOC 2-min sector 107, primary epoch BTJD 4189.23001 ± 0.00008. The orbit is circular (published e = 0.0005), yet its eclipse falls 1.8σ from Gaia's conjunction A, because Gaia's e = 0.026 ± 0.012 moves the predicted conjunction by about 0.004 in phase. Our period is 6.7×10^{-6} d (5.6σ on our error) shorter than the published 5.96727082 d	known; a blind recovery
BD+30 9 = TIC 22503559 (F0, slightly evolved; T_{eff} 6823 K, 3.18 $R_\odot$; 305 pc)	New ellipsoidal variable on a Gaia SB1 orbit (P 1.5786487 ± 0.0000408 d, K_1 19.4 km/s), picked up by the G 9–10 fold (Section 2.4). A smooth double wave, 16.1 ppt peak to peak (2nd harmonic 6.98 ppt; minima 16.1 and 13.3 ppt below maximum), at P = 1.5786673 ± 0.0000010 d from QLP sectors 17, 57 and 84 (+0.46σ from Gaia's); TESS-SPOC and TARS agree to 3% in amplitude, and difference images put the variation on the target (≤ 0.7″). No eclipse: 3σ limits 0.2–0.65 ppt at either conjunction for durations of 1–8 h, against 6–14 ppt for a transit of the companion; the amplitude implies $i \approx 28$–$55°$ and $m_2 \approx 0.22$–$0.34\ M_\odot$. The minima fall on Gaia's conjunctions, the deeper on B (+0.0056 ± 0.0086 in phase) and the shallower on A (+0.0027 ± 0.0061), with the TESS period carried to Gaia's epoch. No SPOC TCE; in no variability catalogue checked	not an eclipsing binary; new ellipsoidal variable, consistent with Gaia's orbit
Marginal folds	TIC 66645701, 355686601, 363573162, 371947250, 133853923, 93028775, 436979038, 364422623, 344186882, 159448461, 45239182, 141045993, 340182320, 317170426, 468394784, 118578234, 351454794, 358098040: eclipse-like folds in no catalogue	recorded, not chased
TIC 4382254	A lone 7,328-ppm, 0.72-h dip in sector 70 at BTJD ~2452.2; a possible single eclipse; unconfirmed	not a candidate
Gaia binary suspects among bright late-M dwarfs	TIC 35760711 (RUWE 1.71, RV amplitude 8.2 km s^{-1}), TIC 238865036 (RV amplitude 12.4 km s^{-1}), and five rows with RUWE 1.4–1.9	noted
TYC 3707-251-1 = TIC 316045876 (G 9.9)	4%, ~3-h dips every 5.854 d in raw QLP SAP (sectors 18, 19, 59, 86). The earlier interpretation as an eclipser on a 197-d edge-on Gaia astrometric orbit is withdrawn: the two events are 185 cycles of the 5.855-d series apart. Refit: P = 5.854343 d, circular, 3.8% primary and 0.34% secondary at phase 0.50; eclipse timing variations consistent with light-travel time on Gaia's 198-d orbit (K ≈ 5 min), so a probable hierarchical triple; see the entry below.	new 5.854-d EB; triple probable, host not pixel-verified
TIC 271520202 (G 12.05; edge-on sample)	Two identical eclipses on the target: BTJD 3104.089 (18.5%, 10.35 h) and 3873.652 (17.3%, 10.38 h), 4 × 192.391 d apart; Gaia 193.38 ± 0.81 d (−1.2σ; chance match ~4%), e 0.32. Difference images: 18.6% and 18.4% at the target pixel; the G 11.83 neighbour at 46″, which could host it, loses +1.4% and −0.2%. The sector-94 event is 1.35 d before that sector's known common-mode dip and is not it. TESS allows N = 1, 2, 4 and Gaia selects 4. The secondary falls only in sectors 101–102, where the star's raw flux ramps by several percent, so it is untested at the ~2% level. An 18.5% eclipse needs a companion of ~0.8–0.9 $M_\odot$ supplying ~20% of the light. In no catalogue checked. Next primary BTJD 4450.82 (2027-02-14)	plausible (strong); one more primary would confirm

GJ 4348 = TIC 206327797

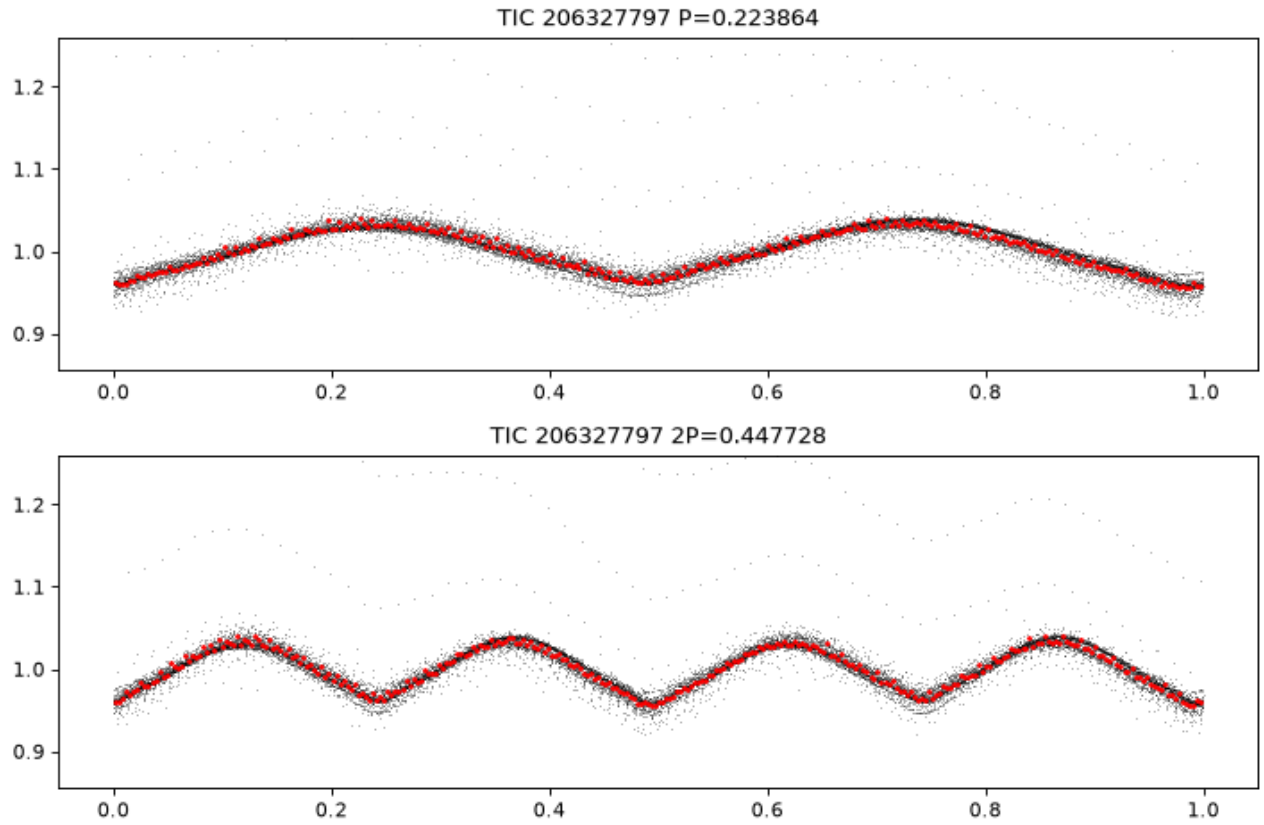


Figure D1. GJ 4348 folded on 0.2238640 d (top) and on 2P = 0.447728 d (bottom): a double wave with unequal minima (58 and 51 ppt in the blended QLP aperture) and flares on top.

A 0.2238640-d double wave (fitted over sectors 1–104, 2,878 d) whose 2f term keeps its phase for ~12,860 cycles while the 1f term drifts. This indicates a fixed orbit with spots migrating on top of it; a single star's spot wave does not hold phase for seven years. Independent re-analysis finds no eclipse and judges the case undetermined but leaning towards a binary; the self-consistent picture is a detached near-contact twin at i ~ 50°. If that is right, LDS 819 is a 2+1+1 quadruple 25 pc away and the star of the flare studies is an unresolved binary. It would not be a record (the shortest verified M-dwarf contact binary is at 0.19856 d), but at 25 pc it would be the nearest known M-dwarf binary at or near contact. One high-resolution spectrum at quadrature would resolve it.

HD 151770 = TIC 337393383

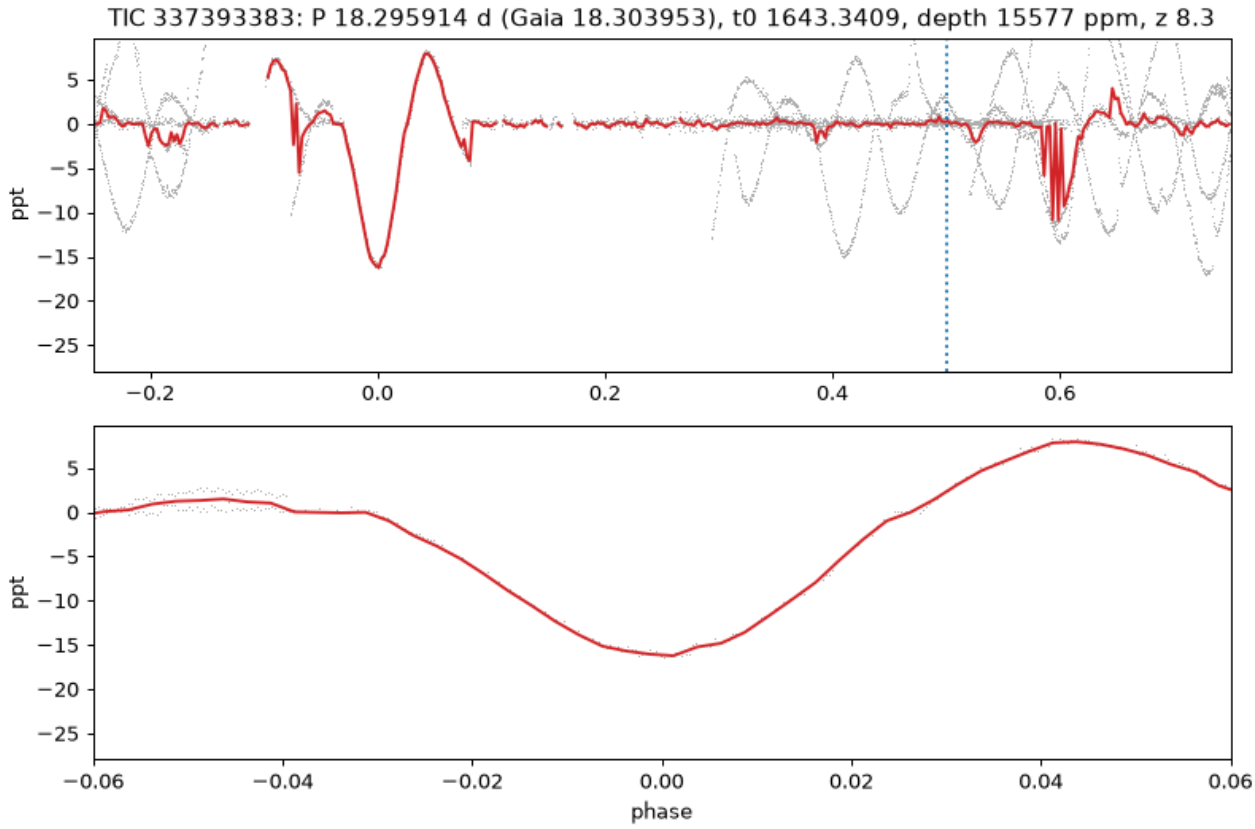


Figure D2. HD 151770 folded on 18.2959 d: a 16.3-ppt primary and an 11.2-ppt secondary at phase 0.593.

The eclipses are real (16.3 and 11.2 ppt; separation 0.5933 over a 2,224-d baseline), but their attribution to the Gaia pair fails. Gaia's SB2 orbit (e 0.2232 ± 0.0103, ω 4.09 ± 2.09°) predicts a separation of 0.641. Both absolute conjunctions miss (+0.165 and −0.475), and the TESS period, 18.2959 d, is 3.4σ from Gaia's 18.3040 d. Resolving it requires a per-pixel eclipse map or re-derived radial-velocity geometry.

Per-object entry (TIC 316045876) are in the Supplement, Section S3.8, with their figures; the entry format and the status words are defined in Section 1 (Conventions).

4. Transit search of nearby M dwarfs: null result and expected yield

4.1 The searches

We searched nearby M dwarfs in public TESS data for new transiting planets. Table 1 lists every search made. The sample throughout is TIC dwarfs with T_{eff} 3200–4300 K, Tmag 8.5–11, distance < 40 pc, and no TOI or CTOI, except where a row says otherwise. None found a planet.

Table 1. The planet searches, their samples and their completeness.

search	stars	result	completeness
Incoherent multi-sector BLS, 0.5–15 d, contamination ratio < 0.05, 3–8 QLP sectors	1,299	no planet; found A1	TIC 4206066 c would be the top peak (SNR 15); b (SNR 10) near the floor

search	stars	result	completeness
Coherent BLS over all sectors, full four-year grid	1,299 (1,101 + the faintest 198)	no planet; found B1, A3	injection-recovery: ≥ 1.5 $R_\oplus$ inside 6 d excluded with high probability; ≥ 1 $R_\oplus$ inside 2 d probably excluded; poorly constrained below 1 $R_\oplus$
Ultra-short periods, 0.17–0.5 d	1,299	no planet; ten SNR ≥ 10 peaks all explained (six by VSX entries); found the δ Scuti star	5/5 known sub-0.5-d planets in the parent sample recovered as the top peak; injections: 95% recovery (53/56) at 1.0–1.5 $R_\oplus$, 74.5% (149/200) over all radii. Expected yield 1–2, and such planets were already TOIs
Faint late-M dwarfs (T_{eff} 2600–3200 K, Tmag 11–13.5)	796	no planet; found A2. SNR at fixed planet radius falls by a factor of 0.58 for the fainter sample (the opposite had been pre-registered)	—
Bright late-M subsample (radius 0.10–0.25 $R_\odot$, Tmag < 11)	53 of 54 (36 with ≥ 3 sectors, 17 with 1–2)	no planet; includes transit nulls on Barnard's Star, CD Cet and Ross 508	84–92% complete to 1.0–1.4 $R_\oplus$ over 0.5–10 d for the 36 (at known ephemeris)
Single transits, 24 d and beyond	1,299	no planet; 16 surviving candidates, all explained, one of them the flare pair of Table 3	56–59% at ≥ 3 $R_\oplus$, limited by the post-gap trim; expected 2.7 detections, 0 found, p = 0.067
Crowded fields (contamination ratio ≥ 0.05)	1,009	no planet candidate; found B8–B10, A13, A14, GJ 4348	insensitive: QLP apertures too contaminated
Additional planets around known hosts (known signals masked; the not-a-TOI cut not applied)	78 hosts	three signals, all in a preprint (C3)	3/3 calibration planets recovered

The injection-recovery grid of the coherent search counts the injected box transits recovered at SNR ≥ 10 at the correct period, or at half or twice it. In the 0.5–2, 2–6 and 6–15 d bins, the counts are 10/32, 6/31 and 0/27 at 0.6–1.0 $R_\oplus$; 25/31, 18/23, 4/14 at 1.0–1.5 $R_\oplus$; 20/22, 12/13, 12/20 at 1.5–2.0 $R_\oplus$; and 32/33, 19/20, 14/16 at 2.0–3.0 $R_\oplus$. The 6–15 d column is partly lost to the 12–14.8 d exclusion band, which also discards real injections.

4.2 Expected yield

The probability calls used to pool the searches were fixed before any search result was examined. There were five calls at first, then eight; four of the first five resolved false, and the estimator used below was built after two earlier versions had failed. Six of the eight searches of Table 1 enter the pool. The search for additional planets around known hosts is excluded because it is not a field-occurrence search over a sample stripped of known candidates. The incoherent BLS is excluded because the coherent search covers the same stars.

Under the estimator's assumptions, set out below, the expected number of detections is about 1.2 from the repeated-transit searches and about 4.5 with the single-transit search included, with a range of about 1 to 12. Every probability in this section is conditional on those assumptions. The escape fraction ε is estimated from planets that other surveys found, and their selection differs from that of planets no survey has found. The faint subsample's recovery is borrowed. The single-transit search's ε → 1 assumes that the TOI list holds no single-transit candidates in this sample, which has not been checked. The

P(0) values, the 95% upper totals and the single-transit shortfall are model results, not measured bounds. At the floor, the probability of no detection, P(0), is 0.30. At E = 4.55, P(0) = 0.011, and at the 95% upper total of 11.8 it is 7×10^{-6}. The mechanism set out below puts the repeated-transit searches at the low end of the range, where the null is unremarkable. The single-transit search, taken on its own, shows a 2.5σ shortfall.

A naive calculation gives E = 75–95, which against zero detections would mean p ~ 10^{-36}. It multiplies the occurrence rate, the transit probability and our measured recovery, and sums over the samples (M-dwarf occurrence from Dressing & Charbonneau 2015). Fifty-four transiting planets are already known around 42 stars that pass exactly our stellar cuts at P 0.5–15 d (NASA Exoplanet Archive, queried October 12). The TOI cross-match used below counts 53 around 41, because one host drops out of the match. With so many already known, an occurrence integral of about 160 transiting planets over these samples is not unreasonable in itself. The discrepancy arises from the sample definition. Every search used the cut *not a TOI or CTOI*. That cut removed only 2.2% of the stars (3,490 dwarfs become 3,412), but it removed all of the known hosts: **41 of 41** stars in this population known to host a detectable 0.5–15 d planet have a TOI. The probability ε that a star hosting a findable planet survived the cut is therefore 0/41 in the window where most of the searching happened (95% upper bound 7.3%).

Table 4. Expected detections by subsample. "Transiting" is the occurrence-integral number of transiting planets; "× recovery" applies our measured recovery; ε is the fraction of known planets in the subsample whose host has no TOI. The single-transit search is entered with ε → 1 (QLP's TOI search needs repeated transits), as in the text; the original calculation entered 0 there. The 95% upper totals are sums of per-subsample upper bounds, which overstate a 95% bound on the sum.

subsample	transiting	× recovery	ε	E	E (95% upper)
coherent BLS 0.5–15 d, parent 1,299	65.7	35.71	0.000	0.00	2.61
crowded fields 0.5–15 d, 1,009	51.5	27.97	0.000	0.00	2.05
faint late-M 0.5–15 d, 796	24.2	13.14	0.091	1.19	2.99
bright late-M 0.5–10 d, 53	1.4	0.76	0.000	0.00	0.06
monotransit 24–200 d, parent 1,299	21.4	3.36	→ 1 (mechanism)	3.36	3.36
USP 0.17–0.5 d, parent 1,299	1.7	1.18	0.000	0.00	0.71
total, repeated-transit searches (a floor)	**145**	**78.8**		**1.19**	**8.4**
total with the single-transit search	**166**	**82.1**		**4.55**	**11.8**

Two factors suppress the yield: the not-a-TOI cut, ×0.015 (1/69), and our measured recovery, ×0.495 (1/2.0). The TOI cut suppresses the yield 35 times more strongly than detection efficiency does. The error in the naive version is to multiply our recovery by (1 – TESS's recovery) as if the two were independent. They are not. Both measure whether a box search can find a signal of a given SNR in the star's TESS photometry, and the planets TESS misses are the low-SNR tail that our searches miss too.

The only measured incompleteness of the TOI list is in the faint tail (T_{eff} 2600–3200 K, Tmag 11–13.5), where ε = 2/22. Both exceptions are K2 discoveries whose hosts have no TOI: K2-239 (three planets, 1.0–1.1 $R_\oplus$, P 5.2, 7.8 and 10.1 d, Tmag 12.24) and K2-129 b (1.04 $R_\oplus$, P 8.24 d, Tmag 11.23). All of the repeated-transit E = 1.19 comes from that subsample, and that term is itself uncertain in both directions. The subsample is a different population from the parent sample, yet it is credited with the parent sample's recovery of 0.543, as every subsample is, because its own completeness was not measured (Table 1). And under the noise model the two exceptions would reach only SNR 4–6 in four sectors, at or below our threshold. The subsample contributes to the yield not because its stars are fainter but because TOI membership is a poor proxy for prior detection there: those planets were found by a different survey, and their hosts have no TOI.

Estimated this way, from known planets whose hosts have no TOI, ε is a lower bound. It can only count planets that the TOI list missed and another survey found; planets that no survey has found are invisible to it. E = 1.19 is therefore a floor under these assumptions rather than a measured lower bound, and the 95% upper value is the other end of the range. The data cannot choose between them. A further caveat applies: the mechanism argument below is much weaker for crowded fields and for single transits. It could be tightened by measuring QLP's TOI completeness directly against injections, which has not been done.

The repeated-transit searches are placed at the low end of the range by a mechanism, not by the arithmetic. A second test, with its criteria written down before it was run (`PREREG2.md` in the deposit), applied our actual search criteria and noise model to the 53 known planets in our stellar sample. The median known planet would have SNR 26.8; 51 of 53 (96%) clear SNR 10, and **48 of 53 (91%)** pass every criterion. The injection grid, applied planet by planet to the same 53, predicts 68.5% where the noise model gives 96%. The injections were run end to end through the search code and are the more reliable of the two. Our pipeline is at best comparable to the one whose output defined the cut. That is why ε = 0/41 matters: the cut removed the stars where our searches would have succeeded.

The estimator sees only planets in the confirmed-planet table. Detectable signals that are not there are invisible to it: TIC 4206066, which passes every parent-sample cut and whose two candidates this search finds at SNR 15 and 10 (C1); the two TOI-6284 signals and the TOI-1467 threshold-crossing event (C3); and the SPOC events of Section 2.7 that never became TOIs. One such exception among the 42 bright hosts (ε = 1/42) would raise E by 1.5 and lower P(0) to about 0.07. The single-transit search is different in kind. QLP's TOI search needs repeated transits, so for single transits ε → 1, giving E = 3.36 and P(0) = 0.035. That agrees to within 25% with the single-transit estimate in Table 1 (2.7 expected, p = 0.067), which uses a different occurrence source. Table 4 enters the search on that basis, and the pooled total including it is 4.55.

The quantity to estimate is the incompleteness of the TOI list, not the abundance of planets. A search of an under-searched parameter range is productive where the inherited cut is a poor proxy for prior detection (K2 fields, single transits, crowded apertures), not where the stars are merely fainter or the periods merely longer. No new planet is to be expected from this sample. For the repeated-transit searches the null is not for want of a working pipeline, and the single-transit search is the only one in which a real shortfall remains. The search for additional planets around known hosts, the only search in which the not-a-TOI cut was not applied, recovered three transit signals (C3).

4.3 The cost of excluding the TESS-orbit band

Every search here discarded peaks at P = 12–14.8 d, where the 13.7-d TESS orbit produces systematics. Widening the band to ~10.4 d, where spurious peaks appear to begin, was considered (Section 2.8). Measured against the 53 known planets in our stellar sample, the band discards real planets:

Table 5. Known planets in our stellar sample discarded by a hard exclusion band.

band	real planets discarded
12.0–14.8 d (used)	3 / 53 = 5.7%: LTT 3780 c, LHS 1903 d, Gliese 12 b
11.0–14.8 d	4 / 53 = 7.5% (adds TOI-270 d)
10.4–14.8 d (proposed)	5 / 53 = 9.4% (adds TOI-1801 b)
10.0–14.8 d	6 / 53 = 11.3% (adds TOI-198 b)

The two planets that the widening would newly discard are not marginal: TOI-1801 b and TOI-270 d come out at SNR 29 and 31 in this model. The widening improves purity at the cost of 3.7 percentage points of planets. A lossless alternative is to make the band a ranking penalty rather than an exclusion.

That is also the only version that would have let these searches see Gliese 12 b, a 0.93 $R_\oplus$ planet at 12.76 d on a Tmag 10.2 star 12 pc away.

5. Discussion

5.1 Why the appointments method works

The Gaia elements fix the conjunction times. A long-period eclipse lasts hours to a day once every few weeks, and a TESS sector catches at most one of each kind. Without a period, a few isolated dips in years of data cannot be phased. With Gaia's elements, a single dip either lands in a window of a few days that was fixed in advance or it does not. This reduces the number of trials by orders of magnitude, makes a single event meaningful, and attributes the eclipse to the star with the orbit in the same step. The control sample validates the predicted windows: 32 of 40 Gaia-flagged eclipsers showed their eclipse at the predicted conjunction, mostly within 0.03 in phase. Ephemerides from the re-analyses predicted events in sectors the search had not used, before those data were examined, and the events were found: three eclipses of A47 in sector 106 within 1.5 min, the sixth transit of A34 in sector 105 to 1.3 min, the third eclipse of A52 in sector 108 to 2 min, and the transit ingress of A57 at the end of sector 88.

Effect of pipeline detrending. Eclipses in this period range last from about 3 h to 46 h (A41 to A36). QLP's detrended flux is built to flatten variability on those time-scales, and in post-sector-56 files it removes or halves such eclipses (Section 2.8). Searching the raw aperture flux, with the per-star random-window null and the far ratio in place of a detrend, keeps them. Many of these stars had a SPOC or TESS-SPOC threshold-crossing event that never became a TOI: 31 of the 54 appointment systems and 75 of all 109 A-objects (Section 2.7, Table A1). Some of these events are at the true period (A61, A73), some at an alias of it (A16 at 13.6 d, A41 at 2P, A43 at 15P, A54 and A70 at P/2, A47 at the B-to-A spacing of 9.27 d), and some are unrelated. One system had a CTOI that was its own secondary eclipse (A21). These events were detected by the pipelines but never vetted.

Eclipses in flagged cadences. Some eclipses fall in cadences that QLP flags, usually for scattered light near orbit edges. Five of the ten eclipses of A30 are in such cadences, all five under scattered-light flags. So are the third eclipse of A19, one eclipse of A53, one each of A58 and A59, and one of A40's two events. A search restricted to QUALITY = 0 sees fewer events or none. These were re-measured from the raw flux and confirmed against independent pipelines, the target's pixels, and the absence of the same feature in other stars on the detector.

5.2 Yields

The appointments search found 13 new eclipsing systems among 1,422 stars at G < 10, a rate of 0.9 per hundred stars searched, and 41 (four of them probable) among 2,327 stars at G 10–11, a rate of 1.8 per hundred. Over the whole sample the rate is 1.4 per hundred. The first Gaia-orbit fold at P < 20 d found 3 new eclipsers, all probable, and 4 mistyped ones among 736 nearby orbits. The second found 5 new (one probable) and 2 mistyped among 1,338 science targets at G < 9. The third, at G 9–10, found 34 new eclipsing systems, all confirmed, among 2,559 science targets, a rate of 1.3 per hundred; it also recovered one published system blind and gave the first ephemeris for one flagged by SuperWASP (A93). The edge-on astrometric search found 4 among 177 stars with TESS coverage, plus one plausible two-event case. A19 (probable) came from the window search, and A72, A73 and A74 (the last probable) from the short-period search of the same light curves. A further candidate, TIC 316045876, was withdrawn as a long-period eclipser. The M-dwarf transit searches, over about 3,100 distinct stars, found 4 eclipsing systems (one probable), three of them on a star other than the one searched first.

These rates come from searches built differently and are not directly comparable. Together they indicate that the Gaia catalogue of orbits holds many more uncatalogued eclipsers than have been examined here. Two parts of it have not been searched: the orbits with G 11–12 (part of the 11,446 at P 20–60 d), and the about 52,400 P < 20 d orbits fainter than G 10 with parallax ≤ 10 mas or missing. The fold's rate rose from 0.4 per hundred at G < 9 to 1.3 per hundred at G 9–10. The share of eclipse-like stars absent from the vetted catalogues changed little, from 68% to 76%; the rise is in the later inspection and vetting steps, which differed between the two runs, so the two funnels are not directly comparable. The G 9–10 search recovered 199 of its 218 known eclipsers (91%), which anchors its completeness. Applied like for like to the fainter orbits (about 85% of them science targets, as at G 9–10), the G 9–10 rate gives about 590 ± 100 uncatalogued eclipsers and the G < 9 rate about 170. The fainter orbits therefore hold of order 200–600, fewer if the falling photometric precision lowers the rate.

5.3 Constraints on the Gaia orbits

A spectroscopic orbit gives the period, eccentricity, argument of periastron and the mass function, but not the inclination or the sizes of the stars. Eclipses give the inclination (close to 90°) and the radii relative to the orbit. Where Gaia measured both stars' velocities (SB2), the two together give absolute masses and radii. For A47 (HIP 1060) one model gives both masses (1.10 and 1.22 $M_\odot$) and radii (1.26 and 1.83 $R_\odot$); for TYC 1194-680-1 the joint fit gives 1.144 and 1.161 $M_\odot$ and 1.93 and 2.13 $R_\odot$. Some of the new systems contain stars whose radius is otherwise poorly constrained: a G8 subgiant with an asteroseismic mass (A20), an oscillating red giant with a seismic SB1 orbit (A60), and a δ Scuti/γ Doradus hybrid pulsator (A51).

The eclipses are also an independent test of the Gaia orbits. The eclipse period is usually good to seconds to minutes, and the spacing of the two eclipses measures $e \cos \omega$ directly. Of the systems with a Gaia orbit at the eclipse period, most agree with Gaia within about 2σ in period and in $e \cos \omega$, so for them Gaia's solution has passed an independent test. Twelve do not, by the rule of Section 3.6 (period or $e \cos \omega$ 3σ or more off by Gaia's quoted errors under every treatment of its covariance); they are 12 of the 91 counted systems compared with a non-circular Gaia orbit. Ten are in Section 3.6: two Gaia periods are spurious or wrong (A54, A9), three are 3.4–4.9σ off (A52, A12, A29), and five have $e \cos \omega$ 3.7σ or more off (A65, A28, A30, A63, A21). Two are among A75–A109, where Gaia's period is 4.1σ short for A86 and 5.0σ short for A102. For A86 the eclipse spacing gives 0.7694 against Gaia's 0.748 ± 0.008 (2.7σ), and the conjunction offset depends on the treatment: −0.0037 ± 0.0039 in phase when Gaia's covariance is conditioned on the eclipse period, and −0.0214 ± 0.0064 (3.3σ) when the correlations are ignored. A102's period, 1.9385707 ± 0.0000005 d, rests on its ASAS-3 photometry of 2000–2009: the TESS period alone (±0.000029 d) carries the ephemeris back the 3,700 cycles with an uncertainty of 0.055 in phase, so the cycle count is secure. Conditioning Gaia's orbit on that period reduces the tension in Gaia's e = 0.07 but does not remove it: e becomes 0.021 ± 0.006, still 3.5σ from circular, $e \cos \omega$ is 2.9σ off, and the conjunction offsets are 2.3σ and 2.7σ.

Three conjunction offsets among A75–A109 are in tension by Gaia's quoted errors without being counted, because they move between about 2σ and about 8σ with the treatment. For A89 the offset is 6.7σ when the full covariance of a poorly fitting double-lined solution (goodness of fit 4.5) is propagated and 2.2σ when the correlations are ignored. For A104 it is 8.2σ when Gaia's epoch is carried with the TESS period and 0.2σ with Gaia's own period; the eclipse spacing gives $e \cos \omega = 0.408$ against Gaia's 0.434 ± 0.009 (2.9σ), in a solution with goodness of fit 17. For A87 it is 4.3σ at Gaia's epoch and 1.7σ at the TESS epoch carried with Gaia's period. Below these, A106 and A108 have secondaries 2.6–2.7σ from Gaia's conjunctions. A109's period is 4.3–6.5σ from Gaia's on the formal errors and 1.8–2.7σ once its

observed 16-ppm period change is included (Section 3.6). Carried to Gaia's epoch with the TESS period, its primary falls on conjunction A (−0.011 ± 0.024 in phase); carried with Gaia's own period, it falls 4.7σ late.

Taken together, the offsets indicate that Gaia's quoted errors for these orbits are too small. Across A75–A109 the 92 offsets in period and conjunction phase quoted in the entries have an rms of 1.9 in units of their errors, and 1.5 without the three above 5σ. Thirteen of the 31 systems with a non-circular orbit have a largest offset of 2.3σ or more, where about two are expected. For the 17 orbits with $e < 0.1$, the median e/σ_e is 1.9, against 1.18 for the Rayleigh distribution a truly circular orbit would give, and seven exceed 2.45, where 0.8 are expected. Small fitted eccentricities are biased high because the eccentricity cannot be negative (Lucy & Sweeney 1971), but values of e/σ_e of 2.6–2.8 (A76, A87, A96, A106) pass Lucy & Sweeney's own 2.45 test, so that bias alone does not explain them. Gaia's quoted errors for these orbits appear to be underestimated by a factor of about 1.5; with that factor, the number of systems with a largest offset of 2.3σ or more (13, against about 10 expected) and the eccentricity distribution (median e/σ_e 1.3) are close to what chance gives. With the errors inflated by 1.5, A102's period (3.4σ) and the conjunction offsets of A89 (4.5σ) and A104 (5.5σ, each in the treatment that gives the largest value) remain above 3σ; A86's period (2.7σ) and A87's offset (2.8σ) do not, and nor does A109's period once its observed period change is included. Among the short-period orbits that tides should have circularised (A87, A96, A106, A108 and the published EBLM J0021–16 of Table 3), Gaia's eccentricities of 0.01–0.08 move the predicted conjunctions; for A87, after that shift, about 2.8σ remains unexplained.

The eclipse timings also reveal what the Gaia orbit alone does not. In A95 the eclipse times vary with an amplitude of 3.4 ± 0.3 min on the 541-d period of Gaia's astrometric solution for the star, which is the outer orbit of the eclipsing pair about a third body. The timing is consistent with the light-travel time predicted by Gaia's astrometric orbit (chance coincidence ~10%), which is at least 3.25 min if the third body's share of the light is smaller than its share of the mass. In A82 the secondary eclipse has moved 49 min earlier relative to the primary over four years, apsidal motion more than a hundred times faster than general relativity can drive, which with a RUWE of 6.7 points to a third body. A89's primary timings scatter by 3.1 min rms, about four times their errors; a third body is possible, but A87 shows the same scatter and there it is attributed to systematics, so it remains an open possibility. Separately, A45's 31.15-d eclipses do not follow its 21.68-d Gaia orbit at all, and which of the two is wrong is open. Several of these systems, and A41's eclipse-timing variations, point to third bodies. A73's hours-long timing variations and fading eclipses need one. El-Badry et al. (2026) report that about half of their Gaia SB1 solutions for compact-object candidates are spurious. The eclipsers here are a different population, selected for having a real binary, but the contradicted solutions show that a Gaia orbit requires independent confirmation.

5.4 Limits

- **TESS pixels are 21″.** Where a neighbour sits inside one pixel, difference images cannot always separate the two. A68 is probable for that reason: a G 14.5 star 4.4″ away could host a 33% eclipse, and the attribution rests on the period and phase match. A35's co-moving companion 1.65″ away and A42's visual companion 1.3″ away cannot be separated either. Those two are confirmed with that caveat, A42 probably on component B. A46's host component is unresolved, and it is not counted. B1 is held for the same reason, and A13–A15 are probable, not confirmed. All depths are aperture depths, diluted by whatever else is in the aperture unless the pipeline corrects for it (Section 2.8).
- **Some objects rest on one or two events.** A40 has one transit and one occultation in a single sector. A37 has one transit and two occultations in one sector, with no second pipeline. A53 has one clean event and one in flagged data. A25 has two events of one kind. A19 has three events in four

years, two of them only in flagged or QLP-only data. A74 has two TESS events and a third seen only as one night of ZTF photometry. A37, A53, A19 and A74 are probable for these reasons. BD+16 2639 and Wolf 73 are plausible single events and are not counted among the confirmed systems.

- **No spectroscopy.** Everything here is photometry plus public Gaia results. For A3 the companion's period is not known, so a planet is strongly disfavoured but not formally excluded. For A19 the decisive test is a spectroscopic orbit; for GJ 4348, one spectrum at quadrature.
- **Catalogue checks are not a literature search.** The ADS full-text checks of 490 identifiers (Section 1) found one prior mention of eclipses, A7, in the text of a planet-vetting paper (Bryant et al. 2023). The query for the 242 identifiers of the G 9–10 systems skipped survey designations (2MASS, WISE, UCAC, GSC, PPM, SAO and Gaia DR2). The index does not reach every table, so a star listed only under a name we did not query could still be missed. Several objects have prior records of some kind: C3 in a preprint, TYC 1194-680-1 in SuperWASP, A21 as a CTOI, the VSX non-eclipsing types of Section 3.6, and the unvetted pipeline threshold-crossing events of 72 counted systems (Section 2.7, Table A1). A live cone search of A1–A109 in 14 VizieR catalogue tables and SIMBAD (Section 2.7) found no eclipsing-type entry for any of them except A93, which SIMBAD types as an eclipsing binary and which is not counted.
- **The conjunction test inherits Gaia's period.** The test that an eclipse falls at a Gaia conjunction is made at Gaia's period, so when that period is erroneous the test can pass by coincidence. A9 did: its eclipses passed the phase test at Gaia's 13.42 d and its true period is 27.08 d. Where two or more cycles are in the data, the eclipse timings fix the period on their own and the agreement with Gaia is a check on Gaia, not on the eclipses. Where only one cycle is covered (A37, A40, A53, A25), the period rests on Gaia's and the eclipses only show that it is consistent.
- **The search's completeness is not measured.** Section 2.5 reports what the windows contain and how candidates were admitted. But no eclipses were injected into the 3,749 light curves and recovered. The fraction of real eclipsers at P = 20–60 d that the appointments search would find is therefore not known, and neither is its false-alarm rate as a function of depth and duration. The injection–recovery tests of Section 2.3 cover the M-dwarf search only. Injecting and recovering eclipses in the appointments search is the next step, and a comparison on the same stars against a blind period search would show what the orbit prediction buys. Pipelines are not independent observations: QLP, SPOC and TESS-SPOC reduce the same pixels, so their agreement tests the reduction, not the photons. An independent observation of any system in this paper is an eclipse seen from the ground at the time Appendix A predicts.
- **Some physics is unresolved.** A27's eclipse is real, and the difference images favour the target over its 6.2″ neighbour at 9–16σ. But the fitted position is itself 2–5σ from the target, and the 1.07% depth lies at the top of the 0.62–1.13% range that an occultation of the mass-function companion allows, so no simple geometry fits comfortably. A26's expected occultation is absent. A37, A58 and A59 have radius or depth tensions recorded in their entries.

5.5 Predicted eclipses

Every confirmed and probable system except A3 has a predicted next eclipse (Appendix A). Most are good to a few minutes; A31, A32, A35–A38, A40 and A41 are good to 0.02–0.10 d. For the systems found by the search of Section 2.5 (A16–A18, A20–A71), the first predicted primary eclipses after 2026 October 20 (geocentric UTC) are: A54 on October 21 at 03:50, A34 on October 21 at 10:57, A61 on October 21 at 21:25, A56 on October 22 at 00:03, A25 on October 22 at 04:04, A37 on October 22 at 11:30, A43 on October 23 at 07:05, A54 again on October 24 at 04:24, A28 on October 24 at 06:19, A29 on October 25 at 14:51, A22 on October 25 at 21:23 and A53 on October 26 at 05:24; A47's annular eclipse falls on November 2 at 06:37. The shorter-period systems (A1, A2, A4–A12, A75–A109) eclipse

every few days to weeks; Appendix A lists their next two. Two of them have short TESS baselines and wider predictions: A80's are uncertain by about 2.5 h and A101's by about half an hour (3σ of the ephemeris). A109's eclipse times vary by about 30 min (Section 5.3). These are falsifiable predictions made before the events. The predictions are computed from the ephemerides of Appendix A and are current as of 2026 October. Most of these stars are G 9–11, and their eclipses are 1–40% deep and last hours. Many are therefore within reach of small telescopes, including amateur ones, and they make a natural target list for the AAVSO community. A missed eclipse would show an error in an ephemeris; an eclipse that arrives early or late would measure the eclipse-timing variations that several systems already hint at (A41, A47, A65).

5.6 The eclipses on the Harvard plates

The ephemerides of Appendix A can also be run backwards. The Harvard plate collection, digitised by DASCH (Grindlay et al. 2012; data release 7), holds photometry of most of the sky from 1890 to 1990 at a scatter of about 0.13 mag per plate for stars of G 10–11, which cannot show a 2% eclipse but can show a 10–40% one if enough plates fall inside the window. After the manuscript was complete, every object among A1–A74 and B1–B11 with a primary or secondary eclipse of 8% or more, G ≤ 12.5 and a period and epoch, 31 systems (A73 was examined separately; Section 3.2), was fetched from DASCH through its public service, cleaned with fixed cuts (no calibrated magnitude, DASCH's rejection flag, its bad-measurement flags, or within 0.5 mag of the plate's local limit; blends kept, since a blend dilutes an eclipse without moving it), detrended with a running median of ±365 d per plate series, and folded on the period and epoch of Appendix A, extrapolated back 600–12,000 cycles. The statistic was the mean residual inside a window one eclipse duration wide, as a z value against the plate scatter, maximised over ±5 period errors; the null was 200 permutations of the residuals over the real plate dates with the same scan; a detection is a false-alarm probability below 1%. For every system the TESS eclipse was also injected into the real plates, ten times at shifted epochs, to see whether the plates could have shown it. The cuts, the statistic and five predictions were fixed and committed before the first fold; four came true, and the fifth, which asked whether the plates' best period fell within two errors of the TESS period, asked something the plates cannot resolve (below) and did not test what it was meant to. A75–A109 were added to the paper after this test and were not run; five of them (A82, A83, A86, A98 and A108) meet its depth cut.

Seventeen of the 31 show the eclipse on the plates at the paper's ephemeris (Table 6). The deep ones are unmistakable: A30's 42% eclipse is a 0.6-mag dip on 27 plates from 1900 to 1960 (z = 19); A47 (HIP 1060), the system submitted to VSX, gives z = 17 on 22 plates; A28 z = 17, A9 z = 13, A48 z = 12. For 15 of the 17, the TESS period unchanged puts the plate eclipses inside the window (z at the TESS period within 1.5 of the best over the scan). The two exceptions carry information. A37, probable here because its period rests on one sector and Gaia, has z = −2.5 at the TESS period and z = 11.9 at P = 22.6131 ± 0.0002 d, inside the TESS error but twenty times tighter. A54, a 3.02-d by-catch over 12,000 cycles, is seen only at P = 3.023482 ± 0.000003 d, 3.4σ from the single-sector TESS value, on a marginal detection (FAP 0.5%, in a default 8-h window). The ASAS-SN V photometry of 2015–2018, 700–1,100 cycles before sector 80, gives 3.023466 independently (bootstrap 16–84%: 3.023454–3.023478; none of 300 resamples reaches 3.02355), and the reductions of sector 80 disagree with each other by more than its quoted error, so the period adopted here is the archives', 3.02348 ± 0.00001 d (Table A1). A48 is seen at the TESS period but more strongly 6σ below it, a 3.8-h shift of eclipse centre over 830 cycles: an underestimated period error or apsidal motion in an eccentric pair. Two systems are untestable (A19 and A72, at 215–291 d, never put five plates in a window). Of the twelve not detected, six are too shallow for their plates (injections recovered 2–4 times in 10); five have plate depths consistent with TESS within 1–3σ and are simply below threshold; and one, A63, is a real absence: nine plates in its window

give 0.066 ± 0.036 mag against the 0.21 mag its 18% TESS eclipse predicts, with injections recovered 10 times in 10 and nothing within ±40 period errors. Like A73, A63 is an eclipse that was shallower or absent in the past, or a period wrong by more than its error.

The plate depths are means over a box the length of the TESS eclipse, so for a V-shaped eclipse they are half to two-thirds of the maximum; and the plates are blue-sensitive, so the occultation of a hot companion (A33 and A38, where the 9% occultations are the eclipses folded) is deeper than in TESS and the transit of a cool one shallower. Turning the box depths into depths per decade, which would ask A73's question of all 17, is the next step. The plates cannot improve a long period already known to 10^{-5} d, because a 7-h eclipse over 1,500 cycles resolves the period only to about 5×10^{-5} d; they refine it where the TESS baseline is one or two sectors (A37) or the cycle count is very large (A54, A11, A4). What they do show, for 17 systems, is that eclipses reported here from 2018–2026 were photographed a century earlier at the times this paper's ephemerides give.

Table 6. The 31 deepest systems among A1–A74 and B1–B11 on the Harvard plates (DASCH DR7, 1890–1990), folded on the ephemerides of Appendix A (A54 on its sector-80 period, 3.02355 d; Appendix A gives the archive period, Section 5.6). TESS depth is the depth of the eclipse folded (the primary of Table A1; for A33 and A38 the 9% occultation of the hot companion), with its box-equivalent magnitude in brackets. Plates: the number kept after the cuts, and their years. n: plates inside the eclipse window at the TESS period. Box depth: the mean in-window residual in DASCH's calibrated magnitude (blue-sensitive plates), which for a V-shaped eclipse is half to two-thirds of the maximum and is a lower bound where the DASCH catalogue flags a blend. $z(P_0)$: the window statistic at the TESS period unchanged; z_{best}: its maximum over ±5 period errors, with the offset in units of the error in brackets. FAP: the fraction of 200 permutations of the residuals over the real plate dates that reach z_{best}. Inj.: TESS-depth eclipses injected into the real plates at shifted epochs, recovered out of 10 (not run where the folded depth is below 8%). Detected = FAP < 1%.

Label	Name	P (d)	TESS depth % (mag)	Plates	Years	n	Box depth (mag)	$z(P_0)$	z_{best} ($\Delta P/\sigma_P$)	FAP	Inj.	Verdict
A4	UCAC4 537-050325	3.465754	30.2 (0.39)	245	1902–1989	8	0.252 ± 0.044	2.4	5.7 (+1.0)	0.000	10/10	detected
A6	BD-13 1046	5.0782783	10.97 (0.13)	1362	1889–1990	19	0.122 ± 0.019	5.4	6.4 (+0.7)	0.000	10/10	detected
A7	UCAC4 130-138342	13.91066352	8.0 (0.09)	652	1897–1990	7	−0.015 ± 0.043	−0.6	−0.4 (−2.5)	0.815	2/10	not detected; too shallow for the plates
A9	HD 7601	27.0822013	39.8 (0.55)	1158	1899–1990	24	0.311 ± 0.023	12.8	13.3 (−1.3)	0.000	10/10	detected
A11	BD+40 3254	4.075190394	10.11 (0.12)	3275	1894–1990	139	0.050 ± 0.008	4.3	6.0 (−2.5)	0.000	10/10	detected
A16	TYC 2173-965-1	23.802223	10.0 (0.11)	1818	1891–1990	63	0.114 ± 0.017	6.6	6.9 (−4.5)	0.000	10/10	detected
A19	Gaia DR3 144024581875447	290.9037	12.5	521	1891–1990	<5						untestable (fewer than five plates in any window)
A22	HD 309050	23.034195	8.3 (0.09)	976	1889–1985	19	0.069 ± 0.025	1.8	2.8 (−5.0)	0.040	9/10	not detected; injections recovered
A28	CD-32 4901	51.37333	20.0 (0.24)	1195	1890–1989	37	0.271 ± 0.016	16.9	17.3 (+0.7)	0.000	10/10	detected
A29	BD-01 648	29.60666	9.5 (0.11)	2119	1892–1990	12	0.074 ± 0.026	2.2	2.9 (+1.0)	0.140	10/10	not detected; injections recovered
A30	BD+29 2845	27.94676	42.0 (0.59)	2073	1891–1990	27	0.384 ± 0.020	18.4	19.0 (−3.8)	0.000	10/10	detected

Label	Name	P (d)	TESS depth % (mag)	Plates	Years	n	Box depth (mag)	z(P_0)	z_{best} ($\Delta P/\sigma_P$)	FAP	Inj.	Verdict
A33	TYC 7691-3893-1	45.59795	9.0 (0.10)	1060	1890–1989	27	0.103 ± 0.019	4.6	5.5 (+5.0)	0.000	–	detected
A37	CPD-25 5943	22.6115	18.7 (0.22)	1768	1890–1990	47	0.159 ± 0.014	−2.5	11.7 (+0.5)	0.000	10/10	detected
A38	TYC 3715-870-1	37.1165	8.7 (0.10)	1680	1890–1990	49	0.127 ± 0.013	7.4	9.4 (−1.3)	0.000	–	detected
A47	CD-31 59 = HIP 1060	24.10512	34.0 (0.45)	1101	1890–1990	22	0.327 ± 0.019	17.3	17.4 (−2.8)	0.000	10/10	detected
A48	TYC 3166-453-1	43.19607	27.0 (0.34)	1768	1890–1990	15	0.287 ± 0.024	9.4	11.9 (−5.0)	0.000	10/10	detected
A50	TYC 3729-162-1	24.17247	11.5 (0.13)	1869	1890–1990	30	0.073 ± 0.018	3.3	4.1 (−1.8)	0.010	10/10	not detected; injections recovered
A54	TYC 447-247-1	3.02355	18.8 (0.23)	1320	1888–1990	142	0.032 ± 0.007	−0.4	4.4 (−3.5)	0.005	10/10	detected
A55	BD+60 397	21.25484	19.5 (0.24)	2329	1899–1990	37	0.123 ± 0.023	5.2	5.5 (+0.3)	0.000	10/10	detected
A56	HD 279736	32.28659	17.5 (0.21)	1618	1890–1990	31	0.061 ± 0.019	2.8	3.2 (+4.5)	0.005	10/10	detected
A58	BD-14 1559	35.97605	9.8 (0.11)	1420	1888–1990	10	0.016 ± 0.028	−1.1	0.6 (−5.0)	0.785	2/10	not detected; too shallow for the plates
A59	TYC 3187-1132-1	42.64893	12.3 (0.14)	1890	1892–1990	7	0.102 ± 0.036	2.2	2.9 (+1.0)	0.115	2/10	not detected; too shallow for the plates
A61	TYC 4308-545-1	21.552334	14.6 (0.17)	2829	1890–1990	21	0.129 ± 0.029	3.7	4.5 (−4.2)	0.000	10/10	detected
A63	TYC 3801-1499-1	57.24	18.0 (0.22)	1819	1893–1989	9	0.066 ± 0.036	1.9	1.9 (+0.0)	0.505	10/10	not detected; injections recovered
A65	BD+27 4098	26.76832	10.0 (0.11)	2067	1891–1990	21	0.073 ± 0.022	2.4	3.3 (+3.0)	0.010	10/10	not detected; injections recovered
A69	TYC 243-1377-1	42.46396	14.6 (0.17)	790	1894–1989	8	0.173 ± 0.040	3.7	4.3 (+3.8)	0.010	4/10	not detected; too shallow for the plates
A70	TYC 4094-1571-1	22.73503	8.2 (0.09)	2469	1890–1990	45	0.080 ± 0.021	3.2	3.8 (+4.2)	0.020	9/10	not detected; injections recovered
A71	TYC 3747-2187-1	24.44534	9.7 (0.11)	2175	1891–1990	15	0.109 ± 0.032	2.5	3.4 (+2.3)	0.000	8/10	detected
A72	Gaia DR3 472037370920235	215.0116	38.0	949	1891–1990	<5						untestable (fewer than five plates in any window)
B3	Gaia DR3 522317746033943	17.13557	10.0 (0.11)	1212	1903–1952	19	0.071 ± 0.036	0.6	2.0 (−4.8)	0.110	4/10	not detected; too shallow for the plates

Label	Name	P (d)	TESS depth % (mag)	Plates	Years	n	Box depth (mag)	z(P_0)	z_{best} ($\Delta P/\sigma_P$)	FAP	Inj.	Verdict
B7	CD-28 7251	6.961	20.7 (0.25)	88	1896–1951	8	0.202 ± 0.025	5.9	8.0 (−3.3)	0.115	2/10	not detected; too shallow for the plates

6. Summary

1. We report 104 eclipsing systems, 93 confirmed and 11 probable, whose eclipses are in no vetted variable-star or eclipsing-binary catalogue, plus three probable blends with the host unresolved; all were found in public TESS photometry. A full-text search of NASA ADS found one prior mention (A7), and 72 systems carry unvetted pipeline threshold-crossing events that were not followed up.
2. Fifty-four came from *eclipses by appointment*: searching TESS only inside the conjunction windows predicted by Gaia DR3 spectroscopic orbits with P = 20–60 d. Among 3,749 stars with G < 11 it found about 1.4 new eclipsers per hundred stars. Each of the 54 was re-analysed independently with separately written code; 50 are confirmed and 4 probable.
3. Forty-six came from M-dwarf transit searches and Gaia-orbit period folds at P < 20 d, 34 of them, all confirmed, from 2,559 science targets among 3,022 orbits at G 9–10, and four from the edge-on Gaia astrometric orbits. One of the four (A73) is a 17.5-d eclipsing binary whose eclipses have faded from 33% to 13% in four years and whose timings need a third body. A further edge-on candidate (TIC 316045876) was withdrawn because its dips recur every 5.854 d.
4. Six eccentric binaries are new heartbeat stars, one with tidally excited oscillations, and a seventh is a candidate. Four of them were found at P < 10 d, inside the period range of the published heartbeat searches.
5. Seventeen of the 31 deep systems among A1–A74 and B1–B11 that were examined show their eclipse on the Harvard plates of 1890–1990 at the ephemerides given here; one (A63) does not, although the plates could have shown it.
6. Eleven catalogue entries have the wrong type or period or none, and twelve Gaia DR3 orbit solutions are contradicted by the eclipses in period or e cos ω; Gaia's quoted errors for the G 9–10 orbits appear to be underestimated by a factor of about 1.5. In one system (A95) the eclipse times are consistent with the light-travel time predicted by Gaia's astrometric orbit (chance coincidence ~10%), which is the outer orbit of the eclipsing pair about a third body.
7. No new planet was found. Pooled over the repeated-transit searches, about 1.2 detections were expected under the estimator's assumptions, with P(0) = 0.30 and a range of about 1 to 12 allowed by those assumptions. The expectation is low because removing known candidates from the sample removed most of the stars where a detectable planet is known. We independently recovered both TIC 4206066 candidates and showed TOI-5394.01 to be the occultation of an M-dwarf companion.
8. Appendix A gives ephemerides and next-eclipse predictions for the confirmed and probable systems. The first of these systems, HIP 1060 (A47), has been submitted to the AAVSO VSX and is pending; submissions for the others are in preparation.

7. Contributions and AI use

The author posed the question and directed the work. The author also commissioned and read the internal adversarial review of the draft (Appendix B), decided what is claimed and at what status, and is the author of record and the submitter, responsible for the content of this paper. The searches, measurements, checks and text

were carried out by six AI language-model agents running on the author's computer. Each is credited once here, where its contribution is introduced, and not again in the running text; the per-object record is kept with the paper's sources (Appendix B).

Claude Opus 5.5 (Anthropic 2026): the M-dwarf transit-search design with its injection–recovery and pixel tools (Section 2.3) and the first searches (A1, A13–A15, B1, B8–B11, GJ 4348, TYC 3763-577-1); the first Gaia-orbit period fold (Section 2.4; A5–A7, B2–B5) and the explanation of TOI-5394.01 (C2); the blind recovery of the TIC 4206066 candidates (C1) and of the TOI-6284 and TOI-1467 signals (C3); the by-eye classification and the independent re-analysis, with separately written code, of every appointments candidate and of A19 and TIC 316045876 (Section 2.5); the short-period sweep of the edge-on sample that found A72, A73, A74 and the plausible case of Table 3, A73's pre-TESS photometry, and the independent re-analysis of the four heartbeat systems of Section 3.1.3 and of the 37 candidates found by the fold's continuation (A75–A109, EBLM J0021–16 and the ellipsoidal variable BD+30 9). **Claude Opus 5**: the faint and bright late-M searches (A2), the flare-pair false positive, the photometric excess of GJ 4348, the expected-yield analysis of Section 4, and the live check of all A-objects against 14 VizieR catalogues and SIMBAD (Section 2.7). **Claude Fable 5**: the vetting that produced A3, the resolution of A4's primary eclipse, the rest of the bright late-M subsample, and the scaled-up Gaia-orbit fold with its independent check (A8–A12, B6, B7) and its continuation to G 9–10, which produced the eclipsing systems A75–A109 and the four heartbeat systems of Section 3.1.3 (three heartbeat stars and one candidate). **Claude Fable 5.1**: the appointments method of Section 2.5 and its run (A16–A18, A20–A71 and the three heartbeat stars), the working catalogue, the assembly of this paper, the response to the review, the archival (ZTF) check that raised A74 to probable, and the Harvard-plate folds of Section 5.6. **GPT-5.6 "Sol"** (OpenAI 2026): the edge-on astrometric-orbit search (Section 2.6; A19, and TIC 316045876 as first found), the look-elsewhere null and the grid-spacing criterion for TIC 4206066 (C1), and the VSX submission worksheet. **GPT-6 "Astra"** read and cross-checked the work and ran no searches. The adversarial review of Appendix B was run by a separate Claude Fable 5.1 session with nine Claude Opus 5.5 agents, briefed to referee the draft and given no access to the writing session. It was internal quality control with independently written code, not journal peer review. Systems A75–A109 were added after that review. Each was re-analysed independently to the same protocol, and the new material was then read by five further agents, briefed as independent referees and given no access to the writing session; this version answers their findings.

The agents used only public data; no telescope time, spectroscopy or proprietary data were used. Every number in this paper traces to an analysis file recorded in the paper's source (Appendix B). The adversarial review re-derived about 2,500 of these numbers from those files and re-reduced eight systems from the raw light curves; for A75–A108 the second review re-measured eight weak systems from the raw data and repeated the catalogue checks. The author takes responsibility for the content; the use of AI systems is disclosed here and in the Acknowledgements, as the journal requires.

8. Acknowledgements

This paper includes data collected by the TESS mission, which are publicly available from the Mikulski Archive for Space Telescopes (MAST); funding for the TESS mission is provided by NASA's Science Mission Directorate. We used the TESS light curves and full-frame-image products of the MIT Quick-Look Pipeline (QLP), the Science Processing Operations Center (SPOC) at NASA Ames Research Center and its TESS-SPOC full-frame-image products, TGLC, eleanor and GSFC-ELEANOR-LITE, the TESS Asteroseismic Science Operations Center (TASOC), CDIPS, TARS, T16 and TEQUILA, and the TESScut service; we thank the teams that built and maintain them. This work has made use of data from the European Space Agency (ESA) mission Gaia, processed by the Gaia Data Processing and Analysis Consortium (DPAC); funding for the DPAC has been provided by national institutions, in particular the institutions participating in the Gaia Multilateral Agreement. This research has made use of the VizieR catalogue access tool and the SIMBAD database, operated at CDS, Strasbourg, France; of the Exoplanet Follow-up Observation Program (ExoFOP) website and the NASA Exoplanet Archive, operated by the

California Institute of Technology under contract with NASA; and of the International Variable Star Index (VSX) database, operated at the AAVSO, Cambridge, Massachusetts, USA. The analysis used NumPy (Harris et al. 2020), Astropy (Astropy Collaboration et al. 2022) and Matplotlib (Hunter 2007).

The analysis, the checks and the text of this paper were produced by AI language-model agents (Claude Opus 5.5, Claude Opus 5, Claude Fable 5 and Claude Fable 5.1, Anthropic 2026; GPT-5.6 and GPT-6, OpenAI 2026) directed by the author, as set out in the Contributions section, which also records the internal adversarial review that preceded this version.

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Appendix A. Ephemerides and next eclipses

The table covers every confirmed and probable eclipsing system, the by-catch variable A46 and the previously typed A93 (both marked). For each it gives the period, an epoch, the next predicted eclipses after 2026 October 14 (BTJD 4327.5) and the status tier (Section 1; probable: A3, A5, A6, A7, A10, A19, A37, A40, A53, A68, A74). It also gives any prior records of the object: a Kostov et al. 2025 table 2 flag, a VSX entry of a non-eclipsing type, an ExoFOP CTOI, or an unvetted SPOC/TESS-SPOC threshold-crossing event (TCE) from the scan of Section 2.7, with the TCE's period where the scan matched it to the eclipses. For A16–A71 the ephemerides are the fitted ephemerides from the independent re-analyses. For A5–A12 and A75–A109 they are timing fits made for this paper on the raw QLP SAP flux (trapezoids per event, weighted linear fit; for A75–A109, in the independent re-analyses). A1 and A2 carry the timing ephemerides of their discovery notes, and A4 a timing fit to its 11 primary and 10 secondary eclipses. A3 carries the fold period and box epoch of its discovery; it has no timing fit, so no prediction. A19 carries the three-event ephemeris of its independent re-analysis. A72–A74 carry the ephemerides of their independent re-analyses (Section 2.5): for A72 three primaries over 12 cycles, for A73 the mean period of its primaries (the period varies; Section 3.2), and for A74 the two TESS eclipses with the ZTF night that fixes the cycle count between them. Rows marked * were propagated for this paper from those discovery ephemerides with their formal errors; none of the three has a published ephemeris. "prim." and "sec." are the deeper and shallower eclipse; A and B are the two conjunctions (Section 1, Conventions). Dates are geocentric UTC calendar dates and times. They are BJD_{TDB} with the light-travel time to the solar-system barycentre in the star's direction removed (up to ±8.3 min) and with TT − UTC = 69.184 s applied. An observer's local correction is at most 21 ms. BJD_{TDB} = BTJD + 2457000.

Table A1. Ephemerides, next eclipses, status and prior records of the 104 eclipsing systems (93 confirmed, 11 probable), the by-catch variable A46, and A93, an eclipsing binary already typed in SIMBAD. The ± on the next eclipses of A80 and A101 is 3σ of the ephemeris; on the rows marked * it is the formal error of the discovery ephemeris. Neither includes event-to-event timing scatter, which the entries give where it is larger, nor A109's eclipse-timing variations of about 30 min. A status in brackets of the form “Gaia P off” or “Gaia e cos ω off” marks a Gaia orbit contradicted by the rule of Section 3.6.

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A1	311958594	UCAC4 083-054218	1.2443947 ± 0.0000016	1629.45849 (shallower)	4327.928 (deeper), 4328.551 (shallower) ± 0.004*	2026-10-14 10:17, 2026-10-15 01:13	confirmed; Kostov t2
A2	183494856	UCAC4 070-000968	6.129392 ± 0.000006	3.74608 (deeper)	4331.097 ± 0.004*	2026-10-17	confirmed; Kostov t2
A3	85379511	HIP 83929	9.474664 (fold)	1993.379 (box epoch)	none: no timing fit		probable; TCE 9.47 d (MES 8)
A4	70665829	UCAC4 537-050325	3.4657540 ± 0.0000093	prim. 2528.2147 ± 0.002 sec. 2529.8114 (phase 0.461)	prim. 4330.407, 4333.873, 4337.338 sec. 4328.538, 4332.004, 4335.469	prim. 2026-10-16 21:49, 2026-10-20 08:59, 2026-10-23 20:10 sec. 2026-10-15 00:58, 2026-10-18 12:09, 2026-10-21 23:18	confirmed; TCE 3.4658 d (P)
A5	22388285	HD 106770	19.454722 ± 0.000010	prim. 3494.45294 ± 0.00039	prim. 4331.006, 4350.461	prim. 2026-10-17 12:13, 2026-11-05 23:08	probable; TCE 19.454 d (P)
A6	348773952	BD−13 1046	5.0782783 ± 0.0000010	prim. 3609.56921 ± 0.00016 sec. 3657.81301 ± 0.00016	prim. 4330.685, 4335.763 sec. 4328.146, 4333.224	prim. 2026-10-17 04:20, 2026-10-22 06:12 sec. 2026-10-14 15:24, 2026-10-19 17:16	probable; TCE 5.0783 d

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A7	271321097	UCAC4 130-138342	13.9106635 ± 0.0000071	prim. 3685.73790 ± 0.00034 sec. 3887.68375 ± 0.0007	prim. 4339.539, 4353.450 sec. 4332.825, 4346.736	prim. 2026-10-26 00:59, 2026-11-08 22:52 sec. 2026-10-19 07:51, 2026-11-02 05:43	probable; TCE 13.9105 d; Gaia DR2 variability flag; Bryant et al. (2023) saw the eclipses
A8	402800288	HD 131614	10.4462098 ± 0.0000098	prim. 3391.45705 ± 0.00071	prim. 4331.616, 4342.062	prim. 2026-10-18 02:49, 2026-10-28 13:32	confirmed; TCE 10.446 d (P)
A9	399009076	HD 7601	27.0822013 ± 0.0000180	prim. 3097.60593 ± 0.00023 ‡ sec. 3253.43045 ± 0.00028	prim. 4343.387, 4370.469 sec. 4336.718, 4363.800	prim. 2026-10-29 21:08, 2026-11-25 23:07 sec. 2026-10-23 05:04, 2026-11-19 07:03	confirmed (Gaia P wrong); TCE 13.554 d (P/2)
A10	300395201	HD 99010	7.8127421 ± 0.0000013	prim. 3384.50567 ± 0.0001 sec. 3427.43069 ± 0.00016	prim. 4329.847, 4337.660 sec. 4333.709, 4341.521	prim. 2026-10-16 08:23, 2026-10-24 03:54 sec. 2026-10-20 05:05, 2026-10-28 00:34	probable; TCE 3.906 d (P/2)
A11	159892657	BD+40 3254	4.0751904 ± 0.0000009	prim. 3133.82003 ± 0.00011 sec. 3184.76020 ± 0.00014	prim. 4327.851, 4331.926 sec. 4329.889, 4333.964	prim. 2026-10-14 08:25, 2026-10-18 10:13 sec. 2026-10-16 09:20, 2026-10-20 11:08	confirmed; Kostov t2; TCE 4.0752 d
A12	436758044	HD 217492	14.0343984 ± 0.0000110	prim. 3203.47040 ± 0.00028	prim. 4340.257, 4354.291	prim. 2026-10-26 18:03, 2026-11-09 18:53	confirmed (Gaia P 4.2σ off); TCE 14.0346 d
A16	257352286	TYC 2173-965-1	23.802223 ± 0.000022	prim. (A) 2804.28519 ± 0.00089 sec. (B) 2817.93973 ± 0.00142	prim. (A) 4327.627, 4351.430 sec. (B) 4341.282, 4365.084	prim. 2026-10-14 02:57, 2026-11-06 22:16 sec. 2026-10-27 18:42, 2026-11-20 13:59	confirmed; TCE 13.6 d (A-to-B spacing)
A17	219230041	HD 26958	40.45701 ± 0.00002	prim. (B) 386.08460	prim. (B) 4350.872, 4391.329	prim. 2026-11-06 08:51, 2026-12-16 19:50	confirmed; none
A18	189668511	TYC 2819-1027-1	31.80877 ± 0.00001	prim. (B) 1766.58610 sec. (A) 1783.48000 ± 0.0015	prim. (B) 4343.096, 4374.905 sec. (A) 4328.170, 4359.978	prim. 2026-10-29 14:09, 2026-11-30 09:35 sec. 2026-10-14 15:56, 2026-11-15 11:20	confirmed (P_A, P_B 4.7σ apart); none
A19	219766411	Gaia DR3 1440245818754478976	290.9037 ± 0.0006	1936.3573 ± 0.0028	4554.491 ± 0.005*	2027-05-28	probable; Kostov t2
A20	235666667	HD 170829 = HR 6950 = NSV 24480	26.384493 ± 0.000040	prim. (A) 2403.97980 sec. (B) 2414.45020	prim. (A) 4330.048, 4356.432 sec. (B) 4340.518, 4366.903	prim. 2026-10-16 13:09, 2026-11-11 22:24 sec. 2026-10-27 00:27, 2026-11-22 09:43	confirmed; VSX VAR:; TCEs 26.3845 d (P) and its secondary
A21	403113935	TYC 4244-623-1	30.763077 ± 0.000013	prim. (A) 2849.89772 ± 0.00041 sec. (B) 2830.31820 ± 0.00041	prim. (A) 4357.288, 4388.052 sec. (B) 4337.709, 4368.472	prim. 2026-11-12 18:52, 2026-12-13 13:12 sec. 2026-10-24 04:57, 2026-11-23 23:17	confirmed (Gaia e cos ω 3.7σ off; no difference image; RUWE 35); Kostov t2; TESS-SPOC TCE (secondary); CTOI
A22	324515186	HD 309050	23.034195 ± 0.000010	prim. (A) 2657.89250 ± 0.00031 sec. (B) 2670.06297 ± 0.00085	prim. (A) 4339.389, 4362.423 sec. (B) 4328.525, 4351.559	prim. 2026-10-25 21:23, 2026-11-17 22:12 sec. 2026-10-15 00:39, 2026-11-07 01:28	confirmed; TCE 23.034 d (P)

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A23	142935468	TYC 4160-760-1	22.244792 ± 0.000017	prim. (A) 2228.68805 ± 0.00056 sec. (B) 2243.31708 ± 0.00107	prim. (A) 4341.943, 4364.188 sec. (B) 4334.328, 4356.572	prim. 2026-10-28 10:37, 2026-11-19 16:28 sec. 2026-10-20 19:52, 2026-11-12 01:42	confirmed; TCE 22.245 d (P)
A24	331507968	TYC 3669-262-1	24.690127 ± 0.000031	prim. (A) 1797.42249 ± 0.00146 sec. (B) 1813.60985 ± 0.00219	prim. (A) 4340.506, 4365.196 sec. (B) 4332.003, 4356.693	prim. 2026-10-27 00:01, 2026-11-20 16:35 sec. 2026-10-18 11:57, 2026-11-12 04:30	confirmed; TCE 8.51 d (unrelated)
A25	258780392	BD+12 121	43.8230 ± 0.0025	prim. (B) 2451.28700 ± 0.0013	prim. (B) 4335.676, 4379.499	prim. 2026-10-22 04:04, 2026-12-04 23:52	confirmed; VSX MISC
A26	89031985	TYC 3091-707-1	42.14749 ± 0.00117	prim. (A) 2645.40345 ± 0.01221	prim. (A) 4331.303, 4373.451	prim. 2026-10-17 19:17, 2026-11-28 22:51	confirmed; none
A27	143198335	HD 63923	41.30385 ± 0.00030	prim. (B) 2963.04837 ± 0.00438	prim. (B) 4367.379, 4408.683	prim. 2026-11-22 21:04, 2027-01-03 04:19	confirmed (blend within 2″ not excluded; physics open); none
A28	145034339	CD-32 4901	51.37333 ± 0.00020	prim. (A) 1512.23071 ± 0.002 sec. (B) 1537.91852 ± 0.00193	prim. (A) 4337.764, 4389.137 sec. (B) 4363.450, 4414.824	prim. 2026-10-24 06:20, 2026-12-14 15:13 sec. 2026-11-18 22:45, 2027-01-09 07:40	confirmed (Gaia P 4.5σ and e cos ω 6σ off); none
A29	248652521	BD-01 648	29.606659 ± 0.000010	prim. (B) 2177.83838 ± 0.0005	prim. (B) 4339.124, 4368.731	prim. 2026-10-25 14:51, 2026-11-24 05:23	confirmed (Gaia P 3.4σ off); none
A30	311297982	BD+29 2845	27.94676 ± 0.00001	prim. (A) 1994.50441 ± 0.0018 sec. (B) 1987.74538 ± 0.00042	prim. (A) 4342.028, 4369.975 sec. (B) 4335.273, 4363.220	prim. 2026-10-28 12:43, 2026-11-25 11:28 sec. 2026-10-21 18:36, 2026-11-18 17:20	confirmed (Gaia e cos ω 5σ off); TCEs 3.6, 15.5 and 22.4 d (the eclipses misfolded)
A31	2048935928	V1192 Cas	37.4133 ± 0.0005	prim. (B) 1764.90610 ± 0.018 sec. (A) 1783.58515 ± 0.011	prim. (B) 4346.423, 4383.836 sec. (A) 4327.691, 4365.104	prim. 2026-11-01 22:03, 2026-12-09 07:59 sec. 2026-10-14 04:29, 2026-11-20 14:24	confirmed; Kostov t2; VSX SRA
A32	399758284	TYC 97-228-1	40.51032 ± 0.00011	prim. (B) 1463.92360 ± 0.024 sec. (A) 1443.67330 ± 0.017	prim. (B) 4340.156, 4380.667 sec. (A) 4360.416, 4400.927	prim. 2026-10-26 15:37, 2026-12-06 03:51 sec. 2026-11-15 21:50, 2026-12-26 10:06	confirmed; VSX ROT
A33	4341778	TYC 7691-3893-1	45.59795 ± 0.00008	prim. (B) 1573.67953 ± 0.0042 sec. (A) 1550.87600 ± 0.003	prim. (B) 4355.154, 4400.752 sec. (A) 4332.351, 4377.949	prim. 2026-11-10 15:42, 2026-12-26 05:59 sec. 2026-10-18 20:28, 2026-12-03 10:45	confirmed; Kostov t2
A34	139285862	CD-45 14957	45.161458 ± 0.000109	prim. (A) 1354.30334 ± 0.00477	prim. (A) 4334.960, 4380.121	prim. 2026-10-21 10:57, 2026-12-05 14:54	confirmed; TCE 15.89 d (unrelated)
A35	11488700	CD-29 366	20.289728 ± 0.000160	prim. (A) 1383.84197 ± 0.00342	prim. (A) 4346.142, 4366.432	prim. 2026-11-01 15:17, 2026-11-21 22:17	confirmed (1.65″ companion unresolved); none
A36	43473641	TYC 5326-1527-1	54.851934 ± 0.000191	prim. (A) 2192.78258 ± 0.0053 sec. (B) 1448.54445 ± 0.00418	prim. (A) 4332.008, 4386.860 sec. (B) 4355.697, 4410.549	prim. 2026-10-18 12:05, 2026-12-12 08:30 sec. 2026-11-11 04:36, 2027-01-05 01:03	confirmed; none

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A37	24755724	CPD-25 5943	22.611537 ± 0.002850	prim. (A) 3793.29985 ± 0.002 sec. (B) 3778.13171 ± 0.00203	prim. (A) 4335.977, 4358.588 sec. (B) 4343.420, 4366.032	prim. 2026-10-22 11:30, 2026-11-14 02:12 sec. 2026-10-29 22:09, 2026-11-21 12:52	probable; none
A38	316326405	TYC 3715-870-1	37.116502 ± 0.000308	prim. (B) 1839.85111 ± 0.00926 sec. (A) 1821.29322 ± 0.01104	prim. (B) 4363.773, 4400.890 sec. (A) 4345.215, 4382.332	prim. 2026-11-19 06:26, 2026-12-26 09:14 sec. 2026-10-31 17:03, 2026-12-07 19:50	confirmed; TCE 18.575 d (P/2)
A39	22620802	TYC 3396-1373-1	35.692330 ± 0.000062	prim. (A) 2602.95779 ± 0.0007 sec. (B) 2582.96490 ± 0.00106	prim. (A) 4351.882, 4387.574 sec. (B) 4331.889, 4367.581	prim. 2026-11-07 09:05, 2026-12-13 01:38 sec. 2026-10-18 09:17, 2026-11-23 01:50	confirmed; TCE 19.99 d (unrelated)
A40	165541068	TYC 986-148-1	24.7458 ± 0.0025 (Gaia; one event of each kind)	prim. (A) 3463.59430 ± 0.001 sec. (B) 3476.19840 ± 0.003	prim. (A) 4329.696, 4354.442 sec. (B) 4342.300, 4367.046	prim. 2026-10-16 04:44, 2026-11-09 22:41 sec. 2026-10-28 19:15, 2026-11-22 13:11	probable (one sector, one event of each kind; P is Gaia's); none
A41	156182608	TYC 6568-1036-1	23.53654 ± 0.00017	prim. (B) 1501.94000 ± 0.011	prim. (B) 4349.861, 4373.398	prim. 2026-11-05 08:39, 2026-11-28 21:30	confirmed; Kostov t2; TCE 47.069 d (2P)
A42	832117928	CD-25 5638B	22.029385 ± 0.000018	prim. 1508.34974 ± 0.00098	prim. 4328.111, 4350.140	prim. 2026-10-14 14:40, 2026-11-05 15:20	confirmed (host probably B); Kostov t2
A43	241445833	TYC 4685-2083-1	23.205492 ± 0.000007	prim. (B) 1389.70430 ± 0.00052	prim. (B) 4336.802, 4360.007	prim. 2026-10-23 07:05, 2026-11-15 12:01	confirmed; TCE 348.08 d (15P)
A44	355477435	TYC 8461-136-1	21.106026 ± 0.000006	prim. (B) 1325.49469 ± 0.00051	prim. (B) 4343.656, 4364.762	prim. 2026-10-30 03:41, 2026-11-20 06:16	confirmed; none
A45	34344890	TYC 7897-48-1	31.152794 ± 0.000072	prim. 2382.75960 ± 0.0037	prim. 4345.386, 4376.538	prim. 2026-10-31 21:19, 2026-12-02 01:00	confirmed (by-catch; relation to Gaia orbit open); none
A46	383529972	BD+30 2277 (contact/ellipsoidal variable, not counted)	0.381845 ± 0.000002	prim. 2645.01110 ± 0.0007	prim. 4327.802, 4328.184	prim. 2026-10-14 07:20, 2026-10-14 16:29	by-catch variable, not counted; none
A47	70858500	CD-31 59 = HIP 1060	24.105122 ± 0.000006	prim. (B) 1357.74508 ± 0.00079 sec. (A) 1367.01018 ± 0.00054	prim. (B) 4346.780, 4370.885 sec. (A) 4331.940, 4356.045	prim. 2026-11-02 06:37, 2026-11-26 09:11 sec. 2026-10-18 10:26, 2026-11-11 12:59	confirmed; TCE 9.266 d (B-to-A spacing)
A48	66900858	TYC 3166-453-1	43.196065 ± 0.000034	prim. (A) 2804.38481 ± 0.00045 sec. (B) 2441.94311 ± 0.00079	prim. (A) 4359.443, 4402.639 sec. (B) 4342.570, 4385.766	prim. 2026-11-14 22:36, 2026-12-28 03:21 sec. 2026-10-29 01:37, 2026-12-11 06:23	confirmed; TCE 14.36 d (P/3)
A49	286926100	TYC 3870-1689-1	28.098880 ± 0.000013	prim. (A) 2645.01560 ± 0.00026 sec. (B) 2688.69960 ± 0.00076	prim. (A) 4330.948, 4359.047 sec. (B) 4346.534, 4374.632	prim. 2026-10-17 10:46, 2026-11-14 13:09 sec. 2026-11-02 00:50, 2026-11-30 03:11	confirmed; TCE 15.58 d (A-to-B spacing)
A50	450218810	TYC 3729-162-1	24.172470 ± 0.000015	prim. (A) 2926.66270 ± 0.00041 sec. (B) 3274.97930 ± 0.00112	prim. (A) 4328.666, 4352.838 sec. (B) 4338.568, 4362.740	prim. 2026-10-15 03:53, 2026-11-08 07:59 sec. 2026-10-25 01:31, 2026-11-18 05:38	confirmed; Kostov t2; TCE 14.29 d (unrelated)

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A51	18000700	BD+28 4244	29.32848 ± 0.00033	prim. (A) 3558.15104 ± 0.060 sec. (B) 3571.18560 ± 0.0044	prim. (A) 4350.020, 4379.348 sec. (B) 4333.726, 4363.055	prim. 2026-11-05 12:24, 2026-12-04 20:19 sec. 2026-10-20 05:19, 2026-11-18 13:16	confirmed; TCE 15.45 d (unrelated)
A52	307170556	TYC 5406-1905-1	40.46953 ± 0.00005	prim. (A) 2206.58750 ± 0.0019	prim. (A) 4351.473, 4391.942	prim. 2026-11-06 23:17, 2026-12-17 10:29	confirmed (Gaia P 4.9σ off); VSX MISC
A53	96844936	TYC 8355-1052-1	36.7755 ± 0.006 (one interval)	prim. (B) 4155.84550 ± 0.0007	prim. (B) 4339.723, 4376.498	prim. 2026-10-26 05:24, 2026-12-02 00:02	probable; none
A54	108344775	TYC 447-247-1	3.02348 ± 0.00001	prim. 3482.03790 ± 0.0001 sec. 3483.54967 ± 0.0001	prim. 4328.612, 4331.636 sec. 4330.124, 4333.148	prim. 2026-10-15 02:41, 2026-10-18 03:16 sec. 2026-10-16 14:59, 2026-10-19 15:33	confirmed (Gaia P spurious); TCE 1.5114 d (P/2)
A55	11156440	BD+60 397	21.254843 ± 0.000012	prim. (A) 1801.37455 ± 0.00065 sec. (B) 1813.59825 ± 0.00078	prim. (A) 4330.701, 4351.956 sec. (B) 4342.925, 4364.179	prim. 2026-10-17 04:43, 2026-11-07 10:49 sec. 2026-10-29 10:05, 2026-11-19 16:10	confirmed; TCE 21.255 d (P)
A56	367706305	HD 279736	32.286594 ± 0.000012	prim. (A) 1817.15278 ± 0.00047 sec. (B) 1826.96486 ± 0.00044	prim. (A) 4335.507, 4367.794 sec. (B) 4345.319, 4377.606	prim. 2026-10-22 00:02, 2026-11-23 06:54 sec. 2026-10-31 19:31, 2026-12-03 02:23	confirmed; TCE 7.74 d (unrelated)
A57	145469097	TYC 7121-1393-1	43.324287 ± 0.000075	prim. (B) 1530.31198 ± 0.00384 sec. (A) 1508.64804 ± 0.00138	prim. (B) 4346.391, 4389.715 sec. (A) 4368.051, 4411.375	prim. 2026-11-01 21:22, 2026-12-15 05:05 sec. 2026-11-23 13:10, 2027-01-05 20:54	confirmed; none
A58	421900559	BD-14 1559	35.976045 ± 0.000018	prim. (B) 2203.38951	prim. (B) 4361.952, 4397.928	prim. 2026-11-17 10:45, 2026-12-23 10:08	confirmed; none
A59	372183259	TYC 3187-1132-1	42.648930 ± 0.000055	prim. (B) 3537.57169	prim. (B) 4347.901, 4390.550	prim. 2026-11-03 09:33, 2026-12-16 01:11	confirmed; TCEs 17.37 and 7.33 d (unrelated)
A60	142084127	TYC 9189-1130-1	45.49880 ± 0.00004	prim. (B) 2553.26450 ± 0.00083 sec. (A) 2615.83808 ± 0.00145	prim. (B) 4327.718, 4373.217 sec. (A) 4344.791, 4390.290	prim. 2026-10-14 05:13, 2026-11-28 17:12 sec. 2026-10-31 06:58, 2026-12-15 18:57	confirmed; TCE 45.50 d
A61	468529113	TYC 4308-545-1	21.552334 ± 0.000007	prim. (A) 2481.89522 ± 0.00022	prim. (A) 4335.396, 4356.948	prim. 2026-10-21 21:25, 2026-11-12 10:39	confirmed; TCE 21.55 d
A62	464830297	TYC 110-583-1	22.34089 ± 0.00001	prim. (B) 1440.02660 ± 0.0004 sec. (A) 1456.45160	prim. (B) 4344.342, 4366.683 sec. (A) 4338.426, 4360.767	prim. 2026-10-30 20:05, 2026-11-22 04:14 sec. 2026-10-24 22:06, 2026-11-16 06:16	confirmed; none
A63	467158573	TYC 3801-1499-1	57.2400 ± 0.0003	prim. (A) 1868.88650 ± 0.005 sec. (B) 3319.52800	prim. (A) 4330.207, 4387.447 sec. (B) 4349.848, 4407.088	prim. 2026-10-16 16:57, 2026-12-12 22:37 sec. 2026-11-05 08:18, 2027-01-01 13:59	confirmed (Gaia e cos ω 4σ off; no complete eclipse; cycle count from Gaia); TCEs 16.87 and 14.65 d (unrelated)

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A64	249895319	BD+50 578	29.27232 ± 0.00005	prim. (A) 1793.94630 ± 0.0007 sec. (B) 1811.97330	prim. (A) 4340.638, 4369.910 sec. (B) 4329.393, 4358.665	prim. 2026-10-27 03:11, 2026-11-25 09:42 sec. 2026-10-15 21:18, 2026-11-14 03:49	confirmed; TCE 11.19 d (unrelated)
A65	88619181	BD+27 4098	26.768321 ± 0.000009	prim. (A) 2819.55960 ± 0.00015 sec. (B) 2821.51261 ± 0.00061	prim. (A) 4345.354, 4372.122 sec. (B) 4347.307, 4374.075	prim. 2026-10-31 20:25, 2026-11-27 14:54 sec. 2026-11-02 19:18, 2026-11-29 13:46	confirmed (Gaia e cos ω 4σ off); TCEs 15.93 and 9.28 d (unrelated)
A66	260563222	CD-41 5677	38.791921 ± 0.000024	prim. (A) 1553.49853 ± 0.00089 sec. (B) 3744.23157 ± 0.00226	prim. (A) 4346.517, 4385.309 sec. (B) 4364.902, 4403.694	prim. 2026-11-02 00:27, 2026-12-10 19:24 sec. 2026-11-20 09:40, 2026-12-29 04:37	confirmed; TCE 13.72 d (unrelated)
A67	173468180	TYC 3623-443-1	29.182072 ± 0.000005	prim. (B) 1756.76116 ± 0.00042 sec. (A) 1740.76463 ± 0.00027	prim. (B) 4353.966, 4383.148 sec. (A) 4337.969, 4367.151	prim. 2026-11-09 11:05, 2026-12-08 15:29 sec. 2026-10-24 11:08, 2026-11-22 15:32	confirmed; TCEs 9.09 and 13.19 d (unrelated)
A68	80107967	TYC 8136-2645-1	34.184472 ± 0.000076	prim. (A) 1521.30741 ± 0.00302	prim. (A) 4358.619, 4392.803	prim. 2026-11-14 02:50, 2026-12-18 07:13	probable; none
A69	377060838	TYC 243-1377-1	42.463964 ± 0.000016	prim. (B) 1517.53332 ± 0.00059 sec. (A) 1538.63354 ± 0.00042	prim. (B) 4362.619, 4405.083 sec. (A) 4341.255, 4383.719	prim. 2026-11-18 02:50, 2026-12-30 13:53 sec. 2026-10-27 18:09, 2026-12-09 05:11	confirmed; TCE 8.507 d (P/5)
A70	70789999	TYC 4094-1571-1	22.735028 ± 0.000008	prim. (A) 1823.53837 ± 0.00039 sec. (B) 1836.88172 ± 0.00042	prim. (A) 4347.126, 4369.862 sec. (B) 4337.735, 4360.470	prim. 2026-11-02 14:56, 2026-11-25 08:34 sec. 2026-10-24 05:33, 2026-11-15 23:10	confirmed; Kostov t2; TCE 11.3675 d (P/2)
A71	87303050	TYC 3747-2187-1	24.445338 ± 0.000011	prim. (A) 1833.11980 ± 0.00047 sec. (B) 1826.93956 ± 0.00066	prim. (A) 4350.990, 4375.435 sec. (B) 4344.809, 4369.255	prim. 2026-11-06 11:38, 2026-11-30 22:18 sec. 2026-10-31 07:19, 2026-11-24 17:59	confirmed; Kostov t2; TCEs at eight unrelated periods
A72	279636729	Gaia DR3 4720373709202359808	215.0116 ± 0.0002	prim. 1341.4321 ± 0.0009 sec. 2075.6456 ± 0.0012 (phase 0.415)	prim. 4351.595 ± 0.002, 4566.607 sec. 4440.773 ± 0.003	prim. 2026-11-07, 2027-06-10 sec. 2027-02-04	confirmed; Kostov t2; TCE 17.07 d (unrelated)
A73	315721826	TYC 4590-364-1	17.538 (varying; single cycles 17.39–17.85)	prim. 2940.800, 2958.651 (two clean, consecutive) sec. phase 0.47 (2024)	prim. 4344.09, 4361.62, 4379.16 (±0.3) sec. 4334.60, 4352.13	prim. 2026-10-30, 2026-11-17, 2026-12-04 sec. 2026-10-21, 2026-11-07	confirmed; Kostov t2; TCEs 17.537 d (P), 17.472 d, 8.09 d
A74	427345302	TIC 427345302	173.8755 ± 0.0005	prim. 2884.9860 ± 0.0013 (TESS); 3232.72 (ZTF night) sec. phase 0.438 (ZTF)	prim. 4449.87, 4623.74 sec. 4352.11, 4525.98 (±0.05)	prim. 2027-02-13, 2027-08-06 sec. 2026-11-07, 2027-04-30	probable (third eclipse one ZTF night, comparison stars unchecked); Kostov t2; TCE 1.143 d (unrelated)
A75	202282515	TYC 3255-1824-1	6.929866 ± 0.000015	prim. 3111.42326 ± 0.00065 sec. 3114.85767	prim. 4331.080, 4338.010 sec. 4327.584, 4334.514	prim. 2026-10-17 13:47, 2026-10-24 12:06 sec. 2026-10-14 01:54, 2026-10-21 00:13	confirmed; TCE 6.9299 d (P)

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A76	202710940	TYC 3256-1533-1	3.3063718 ± 0.0000018	prim. 3299.60466 ± 0.00020 sec. 3294.64483	prim. 4327.886, 4331.193 sec. 4329.539, 4332.846	prim. 2026-10-14 09:09, 2026-10-17 16:30 sec. 2026-10-16 00:49, 2026-10-19 08:10	confirmed; TCE 1.6531 d (P/2)
A77	196475343	BD+44 219	6.370415 ± 0.000005	prim. 3407.33044 ± 0.00027 sec. 3321.33200	prim. 4331.041, 4337.411 sec. 4327.858, 4334.228	prim. 2026-10-17 12:50, 2026-10-23 21:44 sec. 2026-10-14 08:27, 2026-10-20 17:20	confirmed
A78	268587594	HD 6217	9.140474 ± 0.000004	prim. 1764.02178 ± 0.00018	prim. 4332.495, 4341.635	prim. 2026-10-18 23:44, 2026-10-28 03:07	confirmed; TCE 9.1405 d (P)
A79	416277351	BD+64 520A	6.607651 ± 0.000012	prim. 3149.97660 ± 0.00045 sec. 3153.28375	prim. 4332.746, 4339.354 sec. 4329.446, 4336.053	prim. 2026-10-19 05:50, 2026-10-25 20:24 sec. 2026-10-15 22:37, 2026-10-22 13:12	confirmed; TCE 6.6076 d (P)
A80	19702288	TYC 1310-1984-1	17.02161 ± 0.00032	prim. 2516.54908 ± 0.00042 sec. 2524.0685	prim. 4337.861, 4354.883 sec. 4328.359, 4345.381 (±148 min)	prim. 2026-10-24 08:34, 2026-11-10 09:03 sec. 2026-10-14 20:32, 2026-10-31 21:01	confirmed; TCE 17.035 d (one event)
A81	109061795	CD−24 5530	13.2271715 ± 0.0000023	prim. 3040.42781 ± 0.00013 sec. 3047.00989	prim. 4336.691, 4349.918 sec. 4330.046, 4343.273	prim. 2026-10-23 04:33, 2026-11-05 09:59 sec. 2026-10-16 13:05, 2026-10-29 18:30	confirmed; Kostov t2; TCE 6.6138 d (P/2)
A82	317250623	CD−83 126	19.605278 ± 0.000014	prim. 3060.10199 ± 0.00051 sec. 3071.72505	prim. 4334.445, 4354.050 sec. 4346.043, 4365.648	prim. 2026-10-20 22:41, 2026-11-09 13:13 sec. 2026-11-01 13:02, 2026-11-21 03:34	confirmed; TCEs 19.6042 d (MES 1472) and 19.6053 d (MES 302, secondary)
A83	7060775	BD+13 1919	6.422180 ± 0.000001	prim. 2741.55165 ± 0.00012 sec. 2841.74306	prim. 4327.830, 4334.252 sec. 4331.689, 4338.111	prim. 2026-10-14 07:56, 2026-10-20 18:03 sec. 2026-10-18 04:32, 2026-10-24 14:39	confirmed; TCEs 6.4222 d (MES 2600) and 6.4222 d (MES 1017, secondary)
A84	456339196	TYC 3805-1342-1	7.1061147 ± 0.0000043	prim. 2530.89350 ± 0.00012 sec. 2534.48917	prim. 4328.741, 4335.847 sec. 4332.336, 4339.442	prim. 2026-10-15 05:46, 2026-10-22 08:18 sec. 2026-10-18 20:03, 2026-10-25 22:35	confirmed; TCE 3.5530 d (P/2)
A85	38388435	CD−27 6336	2.6164745 ± 0.0000005	prim. 3518.67725 ± 0.00008 sec. 3519.97302	prim. 4329.784, 4332.401 sec. 4328.464, 4331.080	prim. 2026-10-16 06:52, 2026-10-18 21:39 sec. 2026-10-14 23:10, 2026-10-17 13:57	confirmed; TCE 2.6165 d (MES 504)
A86	170891257	BD−01 2238	15.798442 ± 0.000003	prim. 2811.62273 ± 0.00024 sec. 2744.78595	prim. 4328.273, 4344.072 sec. 4340.429, 4356.227	prim. 2026-10-14 18:36, 2026-10-30 13:44 sec. 2026-10-26 22:18, 2026-11-11 17:26	confirmed (Gaia P 4.1σ and eclipse spacing 2.7σ off); TCE 15.798 d (MES 632)
A87	463896609	HD 89725	2.2366733 ± 0.0000008	prim. 3195.87831 ± 0.00025 sec. 3196.9962	prim. 4327.635, 4329.872 sec. 4328.753, 4330.990	prim. 2026-10-14 03:17, 2026-10-16 08:58 sec. 2026-10-15 06:07, 2026-10-17 11:47	confirmed; TCE 2.2366 d (MES 43)

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A88	302351416	TYC 3827-170-1	13.9574485 ± 0.0000066	prim. 2928.45698 ± 0.00023	prim. 4338.159, 4352.117	prim. 2026-10-24 15:50, 2026-11-07 14:47	confirmed
A89	388287047	CPD−85 289	17.638544 ± 0.000019	prim. 3032.65988 ± 0.00069	prim. 4337.912, 4355.551	prim. 2026-10-24 09:54, 2026-11-11 01:14	confirmed; TCE 17.638 d (MES 232)
A90	454747287	CD−73 685	5.6347635 ± 0.0000015	prim. 2691.10727 ± 0.00013 sec. 2693.92630	prim. 4330.823, 4336.458 sec. 4328.008, 4333.642	prim. 2026-10-17 07:47, 2026-10-22 23:02 sec. 2026-10-14 12:13, 2026-10-20 03:27	confirmed; TCE 5.6348 d (MES 217)
A91	311635507	HD 311112	18.5428744 ± 0.0000049	prim. 3479.18152 ± 0.00017 sec. 3490.52516	prim. 4332.154, 4350.697 sec. 4343.497, 4362.040	prim. 2026-10-18 15:44, 2026-11-06 04:46 sec. 2026-10-29 23:59, 2026-11-17 13:01	confirmed; TCE 18.543 d (MES 750)
A92	148877735	BD+69 653	3.6074499 ± 0.0000008	prim. 3110.82736 ± 0.00008 sec. 3101.80861	prim. 4330.145, 4333.753 sec. 4328.342, 4331.949	prim. 2026-10-16 15:29, 2026-10-20 06:04 sec. 2026-10-14 20:12, 2026-10-18 10:46	confirmed; TCE 1.8037 d (P/2, MES 314)
A93	175734936	HD 112601	4.337953 ± 0.000001	prim. 3380.72231 ± 0.00018 sec. 3382.89142	prim. 4330.734, 4335.072 sec. 4328.565, 4332.903	prim. 2026-10-17 05:43, 2026-10-21 13:50 sec. 2026-10-15 01:40, 2026-10-19 09:47	known (SIMBAD EB*, from a SuperWASP flag without period; ephemeris new here, not counted); TCE 2.1690 d (P/2)
A94	245194662	HD 114997	17.2795467 ± 0.0000052	prim. 3353.15051 ± 0.00022 sec. 3364.96274	prim. 4338.085, 4355.364 sec. 4332.617, 4349.897	prim. 2026-10-24 14:06, 2026-11-10 20:49 sec. 2026-10-19 02:53, 2026-11-05 09:36	confirmed; Kostov t2; TCE 17.280 d (MES 320)
A95	154346187	HD 234179	3.549541 ± 0.000002	prim. 2489.65835 ± 0.00031 sec. 2491.43279	prim. 4328.321, 4331.870 sec. 4330.095, 4333.645	prim. 2026-10-14 19:44, 2026-10-18 08:55 sec. 2026-10-16 14:19, 2026-10-20 03:30	confirmed; TCE 1.7748 d (P/2)
A96	426877400	HD 140021	2.7560089 ± 0.0000006	prim. 3574.15160 ± 0.00014 sec. 3575.5325	prim. 4329.298, 4332.054 sec. 4327.923, 4330.679	prim. 2026-10-15 19:11, 2026-10-18 13:20 sec. 2026-10-14 10:10, 2026-10-17 04:19	confirmed; TCE 2.756 d (MES 110)
A97	351595900	TYC 1529-387-1	6.647424 ± 0.000005	prim. 2826.14200 ± 0.00052	prim. 4328.460, 4335.107	prim. 2026-10-14 23:05, 2026-10-21 14:37	confirmed; TCE 6.649 d (MES 38)
A98	157469799	BD+45 2574 A	3.5063244 ± 0.0000002	prim. 3172.71372 ± 0.00002 sec. 3174.46688	prim. 4329.801, 4333.307 sec. 4328.048, 4331.554	prim. 2026-10-16 07:13, 2026-10-19 19:23 sec. 2026-10-14 13:09, 2026-10-18 01:18	confirmed; TCE 1.7532 d (P/2; MES 4313)
A99	352236055	TYC 3925-1852-1	18.821610 ± 0.000008	prim. 3160.83034 ± 0.00023	prim. 4327.770, 4346.591	prim. 2026-10-14 06:25, 2026-11-02 02:09	confirmed
A100	425304484	HD 353655	16.260285 ± 0.000024	prim. 3531.06071 ± 0.00039 sec. 3231.16518	prim. 4327.815, 4344.075 sec. 4336.865, 4353.125	prim. 2026-10-14 07:30, 2026-10-30 13:47 sec. 2026-10-23 08:43, 2026-11-08 15:00	confirmed; Kostov t2; TCE 16.260 d (MES 181)

label	TIC	name	P (d)	epoch (BTJD)	next eclipses (BTJD)	date (geocentric UTC)	status; prior records
A101	14596588	HD 192721	4.478718 ± 0.000065	prim. 3818.15412 ± 0.00013 sec. 3820.41690	prim. 4328.728, 4333.207 sec. 4330.991, 4335.469 (±32 min)	prim. 2026-10-15 05:25, 2026-10-19 16:55 sec. 2026-10-17 11:44, 2026-10-21 23:14	confirmed (one sector; no SPOC product)
A102	72160748	HD 194269	1.9385707 ± 0.0000005	prim. 3815.72949 ± 0.00012 sec. 3814.76054	prim. 4327.512, 4329.451 sec. 4328.482, 4330.420	prim. 2026-10-14 00:14, 2026-10-15 22:46 sec. 2026-10-14 23:30, 2026-10-16 22:02	confirmed (Gaia P 5.0σ short); TCE 1.93857 d (MES 721) and its secondary
A103	188258837	BD+41 3794	5.9855010 ± 0.0000159	prim. 3144.94717 ± 0.00108 sec. 3135.20653	prim. 4330.076, 4336.062 sec. 4332.307, 4338.292	prim. 2026-10-16 13:46, 2026-10-22 13:25 sec. 2026-10-18 19:18, 2026-10-24 18:57	confirmed
A104	353453805	BD+42 3880	14.3509487 ± 0.0000051	prim. 3309.02608 ± 0.00010 sec. 3319.88908	prim. 4327.943, 4342.294 sec. 4338.806, 4353.157	prim. 2026-10-14 10:34, 2026-10-28 19:00 sec. 2026-10-25 07:17, 2026-11-08 15:43	confirmed; TCE 14.351 d (P)
A105	321718006	BD+67 1313	3.9253291 ± 0.0000007	prim. 3144.07847 ± 0.00008 sec. 3146.08177	prim. 4329.528, 4333.453 sec. 4327.606, 4331.531	prim. 2026-10-16 00:36, 2026-10-19 22:48 sec. 2026-10-14 02:28, 2026-10-18 00:40	confirmed; Kostov t2; TCE 3.9253 d (P)
A106	352935659	HD 205273	1.3854241 ± 0.0000004	prim. 3402.27646 ± 0.00009 sec. 3402.96917	prim. 4327.740, 4329.125 sec. 4328.432, 4329.818	prim. 2026-10-14 05:39, 2026-10-15 14:54 sec. 2026-10-14 22:16, 2026-10-16 07:32	confirmed; TCE 0.6927 d (P/2)
A107	397680054	BD+32 4452	2.976903 ± 0.000001	prim. 2681.70466 ± 0.00012 sec. 2739.75430	prim. 4327.932, 4330.909 sec. 4329.421, 4332.397	prim. 2026-10-14 10:15, 2026-10-17 09:42 sec. 2026-10-15 21:58, 2026-10-18 21:25	confirmed; TCE 1.4885 d (P/2)
A108	239649487	TYC 2747-436-1	6.4161826 ± 0.0000024	prim. 3233.51625 ± 0.00014 sec. 3236.72891	prim. 4330.683, 4337.100 sec. 4333.896, 4340.312	prim. 2026-10-17 04:17, 2026-10-23 14:17 sec. 2026-10-20 09:23, 2026-10-26 19:23	confirmed
A109	274331697	BD+43 1585	1.0394878 ± 0.0000005	prim. 3134.11958 ± 0.00009 sec. 3073.30944	prim. 4328.491, 4329.531 sec. 4327.971, 4329.011	prim. 2026-10-14 23:44, 2026-10-16 00:40 sec. 2026-10-14 11:15, 2026-10-15 12:12	confirmed; TCE 1.0395 d (MES 76) and its secondary

‡ A9: the period first catalogued, 13.4166 d, was a fold alias; the timing fit's 27.0822 d is confirmed by the raw sectors and by ASAS-3 photometry from 2003–2009 (see the entry). A40's period is Gaia's, not photometric. A53's period is the single interval between its two events. TCE periods are from the scan of Section 2.7, matched to each ephemeris; “unrelated” means that no TCE period lies within 0.6% of P, P/2, 2P, a small multiple or the A-to-B spacing; such a TCE is either the eclipses folded at a wrong period or a different signal.

Appendix B. Data and code availability

All input data are public: TESS light curves, target-pixel files and full-frame-image cutouts from MAST; Gaia DR3 from the ESA Gaia archive; catalogues from VizieR, SIMBAD, ExoFOP, the NASA Exoplanet Archive and the AAVSO VSX. All light curves used here can be re-fetched from MAST. The analysis code (Python, with NumPy, Astropy, Matplotlib and Requests), the per-object re-analysis reports (one per candidate of Section 2.5), the fitted ephemerides behind Appendix A, the figure sources and the object table are kept with the paper's sources and are in a public data deposit, available from the author. The record is released under CC BY 4.0, and the code in it also under the MIT licence. The code was written for this work and has not been packaged for reuse. The QLP light curves of the P = 20–60 d sample (14 GB) are kept and can be re-fetched from MAST. The analysis file behind each entry is recorded in the source of this document and in the deposit's `tables/objects.csv`.

Labels. The block A20–A71 was renumbered when TIC 316045876 (then A20) was withdrawn. Deposit files written before that use the old labels, in which this paper's A20–A71 are A21–A72 (HIP 1060, this paper's A47, is their A48). **Review.** An internal adversarial review of the draft, by a separate session with its own code, preceded this version; its report, the tone notes and the response plan are kept with the paper's sources. A75–A109 were added after that review. They were re-analysed independently to the protocol of Section 2.5 and then read by five further agents briefed as independent referees, whose findings this version answers; A109 was re-analysed in answer to them.